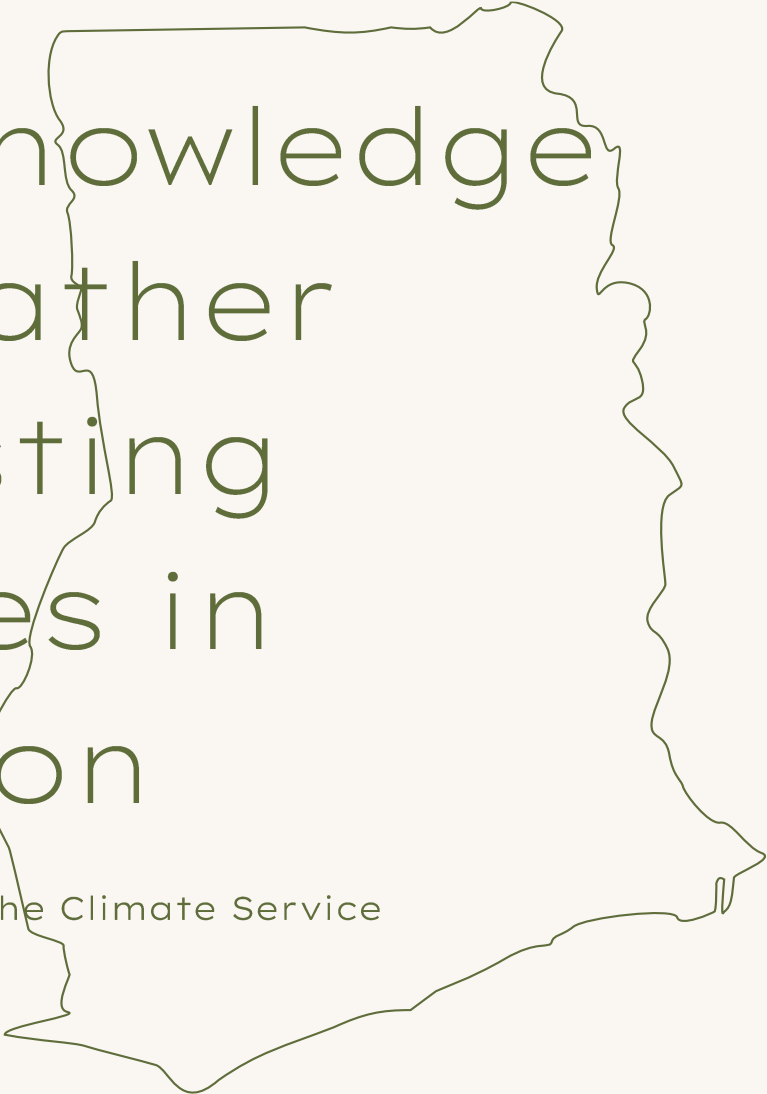


Local Knowledge and Weather Forecasting Practices in Transition



An Exploration through the Climate Service
DROP app in Ghana

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Abstract

In recent years, the emphasis on local knowledge integration within the development of climate services for climate adaptation has grown. However, research on its practical implementation remains limited. Further, climate services have been primarily investigated from a technical perspective without accounting for the social and cultural impacts on communities and knowledge practices. Thus, this thesis analyzes the representation and transformation of local knowledge and community dynamics through the case of an existing climate service and its most visible user touchpoint, the DROP app. The analysis focuses on its current design and deployment in the country of Ghana, where the app was initially developed. The DROP app provides weather forecast information for smallholder farmers and represents one of the first practical examples that integrate scientific and local knowledge into digital climate service tools.

The research is grounded in service design and design research lenses, such as the multi-level service design framework, systems thinking, and the concept of soft impacts of technological innovation. Further, the thesis adopts a qualitative, exploratory methodology with a participatory research approach, including community-based fieldwork in Ghana.

Findings illustrate local knowledge as highly localized, nuanced, contextual, culturally embodied, applied for different purposes, primarily orally transmitted, impacted by climate change, and held unevenly across roles, gender, and generations. Based on these dimensions, the thesis identifies aspects of local knowledge that are represented and those that are simplified when integrated into the DROP app interface. Further, findings indicate that digital climate service tools create social and cultural shifts in norms, meaning, values and, thus, influence relationships and dynamics between community members, such as social positions within community forecasting networks, or intergenerational processes around local knowledge transmission.

Subsequently, the thesis discusses the implications of these findings for the design of the DROP app, and more broadly for the future practice of climate services and service design. Thereby, it surfaces benefits and limitations of both practices and argues for the bridging of these disciplines. Further, the research presents design principles for the future development of climate services. Overall, the research invites practitioners to collectively learn and reflect more rigorously on the meaning of local knowledge integration in practice and illustrates the complexities of designing with various knowledge systems.

Keywords Climate Service, Local Knowledge, Service Design, Participatory Research, Design Research, Systems Thinking, Climate Adaptation

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Additionally, I am grateful to everyone from the SAFE4ALL project and the University for Development Studies who supported me during the fieldwork in Ghana, whether through logistical coordination, translation, facilitation support, or simply by welcoming me warmly.

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Glossary

SAFE4ALL:

Safeguarding African Foodsheds and Ecosystems for all Actors across Local, Regional, and International Levels to Manage Migration

- EU-funded research project, which develops climate services, including the DROP app.

UDS:

University of Development Studies in Ghana, Tamale

- Ghanaian partner of the SAFE4ALL project and host organization during fieldwork.

DROP app:

Weather forecasting app for farmers in Africa developed under the SAFE4All project

- the climate service at the center of this thesis.

SIW app:

Smart Indigenous Weather app for farmers in Ghana

- app selected for benchmarking analysis.

IPCC:

Intergovernmental Panel on Climate Change

- body of the United Nations that evaluates climate change dynamics to inform policy-making

CS:

Climate Service, in literature also referred to as Climate Information Service (CIS) or Weather and Climate Information Service (WCIS)

- the provision of relevant climate and weather information to targeted user groups to help them anticipate, prepare for, and respond to environmental changes.

LK:

Local Knowledge, in literature also referred to as Indigenous Knowledge (IK), Indigenous and Local Knowledge (ILK/IKLK), Traditional Knowledge (TK), Local Ecological Knowledge (LEK), or Traditional Ecological Knowledge (TEK)

- the skills, traditions, and practices developed by communities through long-term engagement with their local and natural environments.

LF:

Local Forecast

- the weather forecast, which is based on local forecasting knowledge.

LFK:

Local Forecasting Knowledge

- one specific type of local knowledge that refers to the practice of forecasting the weather.

LFI:

Local Forecast Indicators

- environmental cues connected to internalized frameworks that connect specific changes in the environment to short- or medium-term weather predictions, such as the onset of rainfall.

SF:

Scientific Forecast

- the weather forecast, which is based on scientific, Western knowledge and on data analysis and prediction models.

SK:

Scientific Knowledge

- knowledge generated from modern technologies, such as satellites or weather stations.

MSD:

Multi-Level Service Design

- framework by Patrício et al. (2011), which outlines the different levels of a service system.

Preface

Viewpoint on Climate Change and Adaptation

Climate change constitutes a global crisis. In Europe, where this thesis is written, climate change manifests itself through milder winters, heavy rains, and wildfires. Despite Europe's historically large contribution to greenhouse gas emissions, the negative effects of climate change are felt less viscerally by its population. They generate awareness, but they fail to create urgency for systemic change. Meanwhile, individuals and communities, that reside in places with less contribution to climate change, experience its impact more strongly, and their daily lives are shaped by the ongoing need to cope and adapt to the consequences. One proposed strategy represents the offering of services that provide relevant climate and weather information. Thereby, these climate services are expected to support local action and the building of resilience. While this strategy is essential, I want to note that it only addresses the symptoms of climate change, and the deeper structural issues remain.

Background and Interest

While deciding on a thesis topic, I encountered the term climate services for the first time. It caught my attention as it appeared to combine my studies in service design with my interest in climate change related topics. I came across the SAFE4ALL EU Horizon-funded research project, which develops climate services for stakeholders in Africa. While African regions face greater vulnerability to climate impacts, this dynamic between Europe and Africa made me question: Why does Europe support climate adaptation in such geographically distant countries? What motivates Europe, and what implications emerge from these motivations and their subsequent support? For instance, does climate adaptation function as a means to manage climate-related migration, or are other strategic interests involved? To what extent does it reflect responsibility for climate change impacts, or overshadow Europe's significant and ongoing contribution to global greenhouse gas emissions, which intensifies climate-related effects in, for example, Africa? Further, the development process of these climate services raises questions: How can European institutions adequately understand the needs of African stakeholders? How are decisions made? And how are local perspectives meaningfully included? Hence, investigating these deeper dynamics of climate services, their interaction with local knowledge, and their

influences on local practices and interactions represents the motivation that drives my research. As a social and service design student, I was particularly interested in exploring what a design-oriented inquiry can contribute to understanding and possibly informing how these services are developed. I contacted the SAFE4ALL research team and decided to orient my thesis around their work. While they provide access to people and practical support, the thesis is conducted independently.

Positioning

I remember looking at my research proposal and being confronted with paradoxes: I say to value local knowledge while my European education taught me to prioritize science-based, systematic, generalizable knowledge. I am questioning the underlying power structures of European climate service funding while studying those services with European funding. I want to center local voices, but I am the one deciding which voices to include and how to interpret them. I want to understand the lived experiences of people, while I do not speak their language to truly listen to their words. I am aware of these contradictions, I reflect on them, but I cannot change them. They make me question whether I, coming from outside the community, the region, and the country, even the continent, can generate meaningful insights. With the lack of local understanding or contextual knowledge that might be needed, will I just reproduce my own assumptions?

This doubt led me to a research paper on indigenous knowledge in South Africa, where one researcher was an Indigenous insider and the other an outsider (Magaya & Fitchett, 2022). Interestingly, the insider researcher, who grew up in the community, encountered instances in which community members did not mention certain information because they assumed she already knew. This made me realize that it is not positionality itself that affects interactions and research, but how the researcher and participant interpret each other's positions. My outsider status might make people explain things that they would expect a local researcher to know. But it might also make them perform a specific narrative or hide information. I will never fully know how people from the communities will see me and how this affects what they tell me. Similarly, I will never fully know how my own background, identity, and perspectives shape what I ask, what I notice, what I miss, and what I interpret. I will never know which questions I should have asked instead, or which framing would have opened up a different conversation.

There is a lot that I don't know, cannot know, and will never know. I do know that all of this will influence my research. However, I also know that asking questions about the representation of local knowledge in climate services and their influence on social and cultural dynamics matters. I take responsibility for the choice to ask them. I want to learn,

and I want to listen, not to facts, but to the stories and contexts. I acknowledge that research can never be objective, but ethically and academically rigorous (Phillips & Johns, 2024). Therefore, I commit to making my research and decision-making process as transparent as possible. I will be conscious about my methods, their implementation, and their limitations. I will reflect on my interpretations, be mindful of my framings, and stay open to discomfort when my assumptions are challenged. I view research participants as co-producers of knowledge (Phillips & Johns, 2024) and, by this, I believe to contribute meaningfully to the discussions on climate services and local knowledge. But most importantly, I want my research to bring benefit to the communities, advocate for their issues and priorities, communicate them, and translate them into implications for the design of the DROP app and the practice of climate services and service design.

Financial Support

I am receiving the DAAD (Deutscher Akademiker Austauschdienst) scholarship from Germany for my master's degree at Aalto University, and a research grant from the Aalto Department of Design that supported my field trip to Ghana.

Declaration for the Use of AI

In this thesis, I utilized Grammarly for spell-checking, auto-correction, and recommendations for improved grammar. Further, I used DeepL for translation and Claude for academic rephrasing.

1 Introduction

Climate change represents a pressing global challenge characterized by rising temperatures, altered rainfall patterns, intense winds, shifted seasons, and other extreme weather events (Danso, 2025; Jabik et al., 2022). While systemic change remains essential, immediate adaptation constitutes an integral strategy for responding to these climate impacts. Adaptation refers to the process of adjusting to current or foreseeable climate effects imposed by climate change (Jiri et al., 2016). It encompasses multiple practices and strategies, with the development of climate services (CS) playing an essential role as they aim to provide climate and weather information relevant for adaptation (Jiri et al., 2016). The European Commission acts as a major funder of global adaptation support and climate service development (European Union, 2025). In its COP30 climate finance factsheet, the Commission expresses a funding objective with a focus on supporting “the poorest and most vulnerable countries and communities” in adapting to climate change and its impacts (European Union, 2025, para. 2). In 2024, the European Union’s climate finance contribution allocated 28.5% to adaptation, 51% to mitigation, and 20.5% to dual-purpose projects. Of this total, 43% of the funds were invested in initiatives supporting African countries. This European-led financing of CS for local stakeholders in Africa raises critical questions about the development process of CS, the interaction with local knowledge, and the influence on the everyday practices of the intended recipients.

Use Case – DROP app

The thesis investigates these dynamics through the case of an existing climate service and its most visible user touchpoint, the DROP app. The analysis focuses on its current design and deployment in the country of Ghana, where the app was initially developed. The development of the DROP app forms part of the EU-funded Horizon Europe project SAFE4ALL Africa (Safeguarding African Foodsheds and Ecosystems for all Actors across Local, Regional, and International Levels to Manage Migration). The DROP app provides weather forecast information to smallholder farmers (SAFE4ALL, 2025). In detail, it displays a scientific weather forecast, based on data generated from modern technologies, and a local weather forecast, aggregated from local inputs of farmers. These local inputs of farmers derive from the local practice of reading environmental signs, such as insect behaviors, to

predict the weather. Therefore, the DROP app represents a practical example of the integration of scientific and local knowledge in digital climate service tools. In the future, the project plans to combine both weather forecast types into a hybrid one to further increase accuracy (Sutanto et al., 2025a, 2025b).

While bodies such as the Intergovernmental Panel on Climate Change (IPCC, 2023) or academic scholars (Nyadzi et al., 2021a; Warner et al., 2022; Kom et al., 2022; Filho et al., 2022a; Makate, 2020) consistently emphasize the importance of local knowledge for climate adaptation, practical implementation remains limited (Filho et al., 2025; Gbangou, 2021; Nyadzi et al., 2021a). In this context, the DROP app constitutes one of the first attempts to translate this principle into a real-world setting. As a result, scientists, researchers, developers, designers, institutions, local knowledge holders, and other involved actors currently navigate uncertainties, implicit assumptions, and power dynamics, and, at the same time, collectively learn from this ongoing process. Overall, the case invites for more rigorous and critical reflection on the meaning and practice of local knowledge representation in climate services. Especially for the field of service design, it illustrates the complexities of designing with various knowledge systems and working in interdisciplinary and intercultural contexts.

Research Objective and Questions

The research is grounded in the theoretical understanding that a service operates across multiple interconnected levels (Patrício et al., 2011) and is embedded within wider design and non-design processes (Vink et al., 2021). Furthermore, the thesis applies a systems thinking approach (Meadows, 2008) to analyze the interconnection between these levels and processes. Lastly, it draws on the concept of soft impacts (Swierstra, 2015), which highlights the subtle influence of design and technologies on social values, norms, and everyday practices of individuals who interact with them. These theoretical concepts are further elaborated in Chapter 3.

From these perspectives, the DROP app forms a multi-level service system. Thus, it functions as a digital interface that represents weather knowledge but simultaneously influences other interconnected processes, such as community practices or institutional structures. Therefore, the thesis explores how the integration of local and scientific weather knowledge in climate services, such as the DROP app, reshapes the form, meaning and social organization around forecasting practices in Ghanaian farming communities. Within this research objective, the thesis investigates the following research questions:

RQ1: How is local forecasting knowledge represented in the climate service DROP app, and what aspects of this knowledge are not reflected in the integration?

This question focuses mainly on the DROP app itself. Specifically, it examines the transformation of local forecasting knowledge when integrated into the app interface. The thesis identifies aspects of local knowledge that are represented, as well as those that are not fully captured or simplified in terms of meaning, interpretation, or contextual richness.

RQ 2: How may the introduction of the climate service DROP app reconfigure the system of community forecasting practices?

This question positions the DROP app within other processes and systems. In detail, it explores the connection between the DROP app and community norms, institutions, and existing weather forecasting practices by investigating potential shifts in these relationships.

The thesis adopts a qualitative, exploratory methodology with a participatory research component to answer the research questions. The data collection process consists of secondary research, benchmarking through service safari of current climate service weather apps, semi-structured interviews with the DROP app research and development team in combination with generative tools, a participatory workshop with farmers, and community-based fieldwork in Ghana. Thereafter, the collected data is analyzed through benchmarking analysis, reflexive thematic analysis, and system mapping.

Research Gap and Contribution

The research contributions occur across two dimensions. Firstly, the thesis contributes to the research field of climate services, which intersects science, policy and practice. Currently, scholars argue for the integration of local knowledge to support climate adaptation (Filho et al., 2022a; Kom et al., 2022; Makate, 2020; Nyadzi et al., 2021a; Warner et al., 2022). While previous studies analyzed hindering factors (Filho et al., 2025) or discussed benefits (Makate, 2020), research on the practical implementation of local knowledge integration remains limited (Filho et al., 2025; Gbangou, 2021; Nyadzi et al., 2021a; Sutanto et al., 2025a). Therefore, the thesis draws upon the DROP app, which exemplifies a possible integration of local forecasting knowledge within CS, to analyze the representation and transformation of local knowledge in practice (Figure 1). Further, existing climate service research focused primarily on the usability of CS

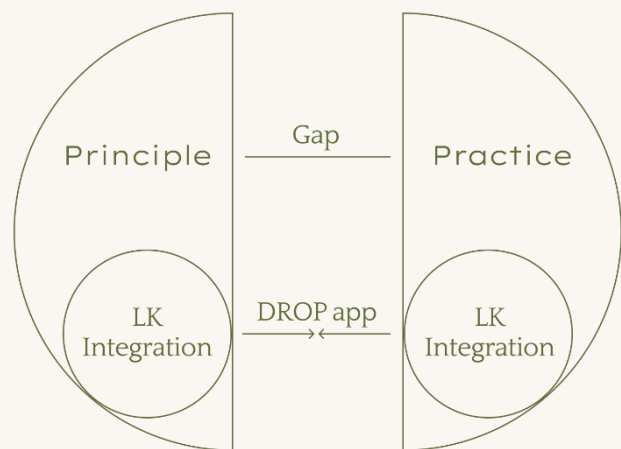


Figure 1: The DROP app as a use case to bridge the gap between principle and practice of local knowledge (LK) integration within climate services (CS) (own elaboration)

(Christel et al., 2018), on factors that shape the uptake and adoption of CS (Chiputwa et al., 2020; Warner et al., 2022), or on the assessment of the value of CS (Tall et al., 2018). In contrast, this thesis contributes to discussions on the social and cultural dimensions of climate services in relation to local knowledge and community practices.

Secondly, the thesis contributes to the field of service design and design research. So far, service design and design research have investigated multi-level service systems (MSD) and their interconnections with broader processes (Patrício et al., 2011; Vink et al., 2021), engaged with systems thinking (ST) to understand complex dynamics (Jones & Kijima, 2018; Vink et al., 2021), and explored the soft impacts (SI) of technology on social values and norms (Swierstra, 2015). Such approaches have been limited in the context of climate services and local knowledge practices. Similarly, the field of climate service development largely evolved without the contribution of service design. This mutual absence creates a gap that this thesis addresses in two directions (Figure 2): (1) The thesis applies service design and design research frameworks to the domain of climate services. Thereby, it examines the social and cultural systems, such as community norms, institutions, and local weather forecasting practices, in which climate services like the DROP app are embedded and which they potentially influence. (2) The thesis reflects on the application of these frameworks within the intercultural and interdisciplinary contexts of climate service development. Thus, it surfaces some of the complexities that arise when service design engages with various knowledge systems and culturally embedded community practices.

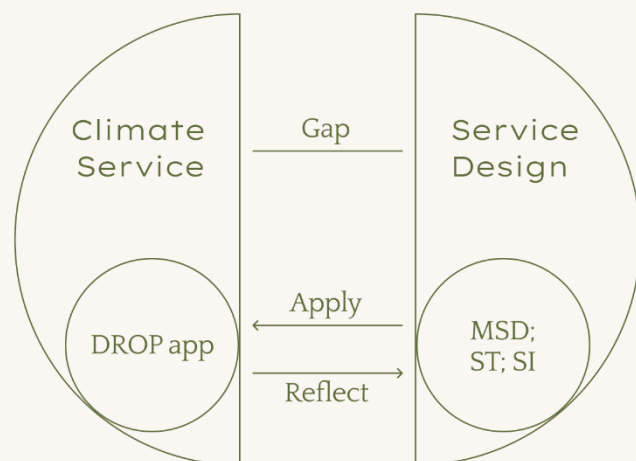


Figure 2: Application of and reflection on service design and design research frameworks (MSD, ST, SI) in the context of CS development through the use case of the DROP app (own elaboration)

Structure of the Thesis

While the introductory chapter offers an overview of the research topic, objective, and contribution of the research, the second chapter presents the contextual background of the thesis. It discusses the current state of climate change and adaptation in Ghana and defines the terms climate services, local knowledge, and knowledge practices. Further, it describes the role of local knowledge integration for climate adaptation. The third chapter introduces the theoretical frameworks on which the thesis is based: the multi-level service design concept by Patrício et al. (2011), the systems thinking perspectives by Meadows (2008), and

the concept of soft impacts of design and technology by Swierstra (2015). The fourth chapter outlines the methodology and methods employed during the thesis process and consists of two sections. The first section presents the methods for data collection, whereas the second section presents the methods for the data analysis. Thereafter, the fifth chapter explains the findings of this thesis in relation to the research questions. Further, the sixth chapter contextualizes the findings by discussing implications for the design of the DROP app and the broader practice of climate services and service design. Additionally, it presents design principles for the future development of climate services. Finally, the seventh chapter summarizes the research, answers the research questions, reflects on limitations and avenues for future research, and outlines research contributions.

2 Background

This chapter establishes the contextual background by discussing the central themes of the research. First, it presents the effects of climate change in Ghana, highlights the need for climate adaptation, and discusses the role of climate services. Second, the chapter provides a literature review on local knowledge, local forecasting knowledge, and the effects of climate change on it. Further, it presents local knowledge as a practice and its position in the social and cultural life of communities. Finally, the chapter examines the advantages and challenges of combining local knowledge and scientific knowledge in weather forecasting for climate adaptation.

2.1 Climate Change and Adaptation in Ghana

The Ghana Meteorological Agency (2025) reported that in Ghana, over the past decade, climate change resulted in unpredictable climate and weather patterns. These include, for example, the erratic onset and intensity of the rainy season or an increase in floods, and prolonged dry spells. In 2024, Ghana reported its highest average temperature of 28.5 °C since records began in the 1980s, and Tamale (Figure 3) recorded the highest maximum temperature in the country at 35.5°C. At the same time, the total annual rainfall declined to the lowest level since 2020. Further, Ghana experiences fluctuating and shortened seasonal patterns. For example, the Northern region around Tamale registered below-average rainfall during the April–May–June period and above-



Figure 3: Map of Ghana (own elaboration)

average rainfall during September–October–November, with severe flooding during June and October 2024.

These climate effects disrupt planting and harvesting cycles and reduce crop yields (Danso, 2025). Consequently, they negatively impact the agricultural sector and food security of Ghana (Danso, 2025). This highlights the need for climate adaptation (Ghana Meteorological Agency, 2025), which refers to the process of adjusting to current or foreseeable climate effects imposed by climate change (Jiri et al., 2016). Adaptation encompasses multiple practices and strategies, with the development of climate services (CS) playing an essential role (Jiri et al., 2016).

2.1.1 Climate Services as Adaptation Strategy

In literature and public discourse, authors frequently substitute the term *climate service* (CS) with *climate information service* (CIS) or *weather and climate information service* (WCIS), as no commonly agreed-upon definition exists. For example, the World Meteorological Organization (n.d.) deploys the term CS. On their website, they describe CS as “the provision and use of climate data, information and knowledge to assist decision-making” and thereby highlight the support for informed climate action (World Meteorological Organization, n.d., *What are climate services?* section, para. 1). In comparison, the United Nations Economic Commission for Africa (n.d.) employs the term CIS on its website. They define CIS as “the packaging and dissemination of climate information to specific users” and therefore emphasize the tailoring of information to specific recipients (United Nations Economic Commission for Africa, n.d., *Climate Information Services* section, para. 1). Further, in academic literature, the definition for WCIS is presented as “the systems and processes that provide data, forecasts, advisories, and warnings about weather conditions and climate patterns to various users, including farmers, policymakers, and the public” (Mabhaudh et al., 2025, p. 2). Their definition focuses on the form and process of information delivery. Nevertheless, the comparison of the different definitions indicates that the concepts CS, CIS, and WCIS largely overlap. They all consistently include the provision of relevant climate and weather information to targeted user groups to help them anticipate, prepare for, and respond to environmental changes. This thesis adopts this understanding and applies the term climate services (CS) for its conciseness.

The CS at the center of this thesis is the DROP app, which provides localized weather forecast information to farmers (SAFE4ALL, 2025). It serves as a tool that supports decision-making regarding farming activities such as sowing, weeding, fertilizer application, or harvesting

(SAFE4ALL, 2025). Early evidence suggests that farmers started adapting their agricultural practices through the use of the DROP app (SAFE4ALL, 2025).

2.2 Local Knowledge

Despite extensive use in the literature, no universally agreed-upon definition for the term *local knowledge* (LK) exists (Filho et al., 2022a; Makate, 2020; Paparrizos et al., 2023). Moreover, the terminology LK frequently overlaps conceptually with other terms, such as *indigenous knowledge* (IK), *indigenous and local knowledge* (ILK/IKLK), *traditional knowledge* (TK), *local ecological knowledge* (LEK) or *traditional ecological knowledge* (TEK) (IPCC, 2023; Kom et al., 2022; Paparrizos et al., 2023). Subsequently, different organizations and research studies propose their own working definitions to reflect specific contextual emphases (Paparrizos et al., 2023). For example, the Intergovernmental Panel on Climate Change (IPCC) defines IK as “the understandings, skills and philosophies developed by societies with long histories of interaction with their natural surroundings” (IPCC, 2018, p. 552). This definition emphasizes the deep cultural and long-standing historical connection between communities and their knowledge systems. In contrast, the IPCC describes LK as “the understandings and skills developed by individuals and populations, specific to the places where they live” (IPCC, 2018, p. 553). This conceptualization highlights the strong location-specific understanding of knowledge, which is rooted in the lived experiences of communities. While the IPCC distinguishes between the terms IK and LK, it predominantly applies the combined version ILK in its report (IPCC, 2023). Similarly, Filho et al. (2022a) adopt ILK to reflect the interconnectedness between local and indigenous knowledge. Nevertheless, this thesis adopts the term local knowledge (LK) as it appears most frequently in academic literature (Paparrizos et al., 2023).

Although terms and definitions differ, several core elements consistently emerge when LK is described. Firstly, LK draws on learning experiences that were accumulated over a long historical timespan and through a deep understanding of place-specific contexts (Kom et al., 2022; Makate, 2020). Secondly, communities transmit LK mainly through oral storytelling or hands-on demonstration, and pass it down through generations (IPCC, 2023; Fernández-Llamazares et al., 2023; Makate, 2020). Thirdly, LK embodies a cultural practice (Dei, 2024) and serves as a basis for everyday decision-making (IPCC, 2023; Makate, 2020; Paparrizos et al., 2023). Building on this, the thesis defines LK as the skills, traditions, and practices developed by communities through long-term engagement with their local and natural environments.

2.2.1 Local Forecasting Knowledge

One specific type of local knowledge (LK) represents *local forecasting knowledge* (LFK), which refers to the practice of forecasting the weather through *local forecast indicators* (LFI) (Filho et al., 2022b; Nyadzi et al., 2022; Paparrizos et al., 2023). More precisely, communities apply internalized frameworks that connect specific changes in their environment to short- or medium-term weather predictions, such as the onset of rainfall (Filho et al., 2022b; Nyadzi et al., 2022). These predictions, based on environmental observations and specific cues, represent a learned skill that individuals acquire through long-term experiences (Nyadzi et al., 2021b). Therefore, elder farmers are considered more skillful in producing accurate forecasts. At the same time, alternative forms of forecasting, which are based on spiritually grounded knowledge rather than systematic environmental indicators, exist (Nyadzi et al., 2021b). These include, for example, rainmaking or other ritualistic ceremonies.

Local Forecast Indicators

Paparrizos et al. (2023) conducted a systematic literature review to identify different types of local forecast indicators (LFI) employed by farmers and reported a total of 984 such indicators across Africa. The range of local indicators might be even larger if documentation of LFI in other languages than English were included in the literature review. In their study, they classify the indicators into five main categories: animal, meteorological, astronomical, plant, and other indicators

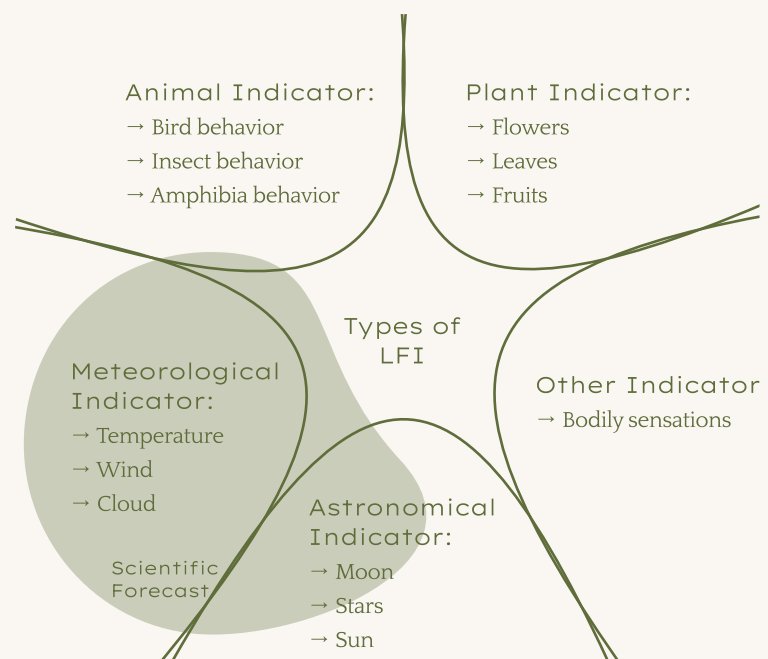


Figure 4: Types of local forecast indicators (LFI) (own elaboration based on a literature review by Paparrizos et al., 2023)

(Figure 4). *Animal indicators* emerge as the most present signal in rain-fed agricultural communities and entail the observation of bird, insect, and amphibian behavior. Secondly, *meteorological indicators* encompass multiple subcategories, such as wind direction, cloud formation, lightning, or temperature changes. Thirdly, *astronomical signs* include observations of halos around the moon or sun, star brightness and constellations. Additionally, *plant indicators* involve, for instance, the timing, abundance, and color of

flowers and leaves. Lastly, *other indicators* relate to human physiology, which include joint pain or body temperature changes. Together, these indicators form a knowledge practice that is based on a deeply sensory, local, and environmental understanding.

The highly localized understanding of these LFI becomes apparent when comparing animal-based indicators, such as the observations of ant behaviors across communities in northern Ghana. A study by Gyampoh et al. (2011) illustrates that, despite the shared general type of an indicator (ant), different communities observe different ant species, apply different cues, and predict rainfall over different time frames or intensities (Figure 5).

Ant Indicator	Local Name	Location	Prediction
Small black ant	Sinsaase	Karemenga Community, Northern Ghana	Seeing them carrying their eggs indicates rainfall starting shortly.
Big black ant	Guri	Karemenga Community, Northern Ghana	Seeing them during the dry season on farm indicates rain in the following day.
Black ants	Ksachi sabliga	Pwalugu Community, Northern Ghana	Seeing them carrying their eggs indicates start of rain season in 4-7 days.
Small, red ants	Salinsar	Yenduri Community, Northern Ghana	Seeing them carrying eggs indicates heavy rainfall.

Figure 5: Comparison of indicator (ant) across communities in Northern Ghana to illustrate the highly localized nature of LFI (own elaboration based on a study by Gyampoh et al., 2011)

Furthermore, a study by Nyadzi et al. (2021b) indicates that LFI applied in northern Ghana vary in its predictive purpose. They documented LFI for daily predictions and LFI for seasonal predictions with highly nuanced indicator-specific interpretations (Figure 6). Indicators for daily predictions support nuanced interpretations of expected rainfall intensity, timing, and probability. Seasonal indicators are utilized once per season to predict the rain level, onset, and cessation of the upcoming rainy season.

Daily Prediction

- rainfall intensity (low/medium/high)
- time frame
- probability

Seasonal Prediction

- above, normal, below rainfall for season
- onset of rainy season
- cessation of rainy season
- probability

Figure 6: Comparison of predictive dimensions in daily and seasonal forecasting (own elaboration based on a study by Nyadzi et al., 2021b)

Impact of Climate Change on LFI

To determine the specific effects of climate change on LFI, Jabik et al. (2022) studied the experiences of farmers in the Garu-Tempene District in Ghana. For example, they found:

Local-level indicators such as the behavior and movement of migratory birds (like the cattle egrets, the African grey hornbill birds, and the fulvous duck), celestial phenomena (like the southwest to northeast winds) and plants phenology (like the flowering of the shea nut and dawadawa trees) are currently not reliable ... (Jabik et al., 2022, p.98)

This description of increased unreliability of indicators results from climatic changes such as rising temperatures. Climatic changes caused certain bird species, for example, the cattle egret, to reside in the region the whole year instead of migrating. Jabik et al. (2022) conclude that these findings allow no generalization to other regions because of their highly environmental and local-specific context. However, since the Ghana Meteorological Agency (2025) reports altered climate and weather patterns throughout Ghana, it can be assumed that LFI lose their reliability in other regions as well. The practice of weather forecasting guides agricultural decision-making and resource allocation and therefore remains especially essential in rural areas (Kom et al., 2022; Paparrizos et al., 2023). Consequently, the loss of certain local indicators poses daily challenges to farmers and their livelihoods.

2.2.2 Community Knowledge Practices

LK encompasses traditions, institutions, beliefs, experiences, norms, and rituals that community members actively perform in interaction with their surrounding natural environment (Dei, 2024; Fernández-Llamazares et al., 2023). Hence, LK represents a community practice. It reflects a deep environmental understanding (Fernández-Llamazares et al., 2023) and operates within social processes (Dei, 2024). For instance, elders and authority figures are regarded as knowledge experts and play an essential role in the transmission of LK (Filho et al., 2022b; Dei, 2024; Kom et al., 2022). They pass down their experiences and skills to the new generations through storytelling, hands-on demonstrations, rituals, or other communal events (Boafo et al., 2016; Kom et al., 2022; Fernández-Llamazares et al., 2023). For example, Kom et al. (2022) reported in their study that one respondent learned to interpret natural cues for weather prediction from their great-grandfather. This intergenerational process of knowledge transmission ensures that LK is owned by every community member. Further, communal activities, such as ceremonies, social or farming events, serve as spaces to actively share and practice LK (Boafo et al., 2016). Nevertheless, research identified a generational trend of younger individuals increasingly devaluing LK (Boafo et al., 2016).

Based on a case study of four communities in the north of Ghana, Boafo et al. (2016) classify knowledge practices into four main categories: *customs and rituals*, *taboos and totems*, *rules and regulations*, and *knowledge about protected areas* (Figure 7). Within this framework, the practice of weather forecasting is placed within customs and rituals. Additionally, this category includes, for example, rainmaking ceremonies to anticipate rainfall. Further, taboos and totems manifest, for instance, in the prohibition of farming activities on Fridays. Rules and regulations structure collective actions, such as requiring approval from the chief before

beginning to harvest. Finally, knowledge about protected areas entails the protection and conservation of riverbanks, groves, and other specific local natural sites. While these particular examples reflect the localized knowledge practices of the specific communities studied by Boafo et al. (2016), they indicate that LK and specifically LFK represent a cultural, collective practice embodied in traditions and the social life of communities.



Figure 7: Classification of LK practices with examples relevant to weather forecasting and farming (own elaboration based on a case study by Boafo et al., 2016)

2.3 Integrating Local Knowledge for Climate Adaptation

Over the past years, the integration of LK into CS for climate adaptation faced several challenges. The highly context- and location-specific nature of LK limits documentation and prevents standardization, both of which prove essential for the upscaling of CS (Filho et al., 2025; Nyadzi et al., 2021a). Further, scientific institutions and external donors of climate adaptation initiatives predominantly regarded local and indigenous perspectives as less credible than scientific approaches (Filho et al., 2025). Scientific approaches refer to scientific, Western knowledge generated from modern technologies, such as satellites or weather stations, and based on data analysis and prediction models (Gbangou, 2021; Paparrizos et al., 2023). Nevertheless, the integration of LK in climate adaptation services offers multiple advantages. Firstly, it contributes to the preservation of the local knowledge system and its cultural practice (Fernández-Llamazares et al. 2023; Filho et al., 2025). Secondly, the inclusion recognizes the knowledge as valuable, creates ownership among communities (Makate, 2020), and thereby increases the uptake of the service (Chiputwa et al., 2020; Warner et al., 2022). Lastly, it increases the accuracy of CS, specifically in the context of weather forecasting (Gbangou, 2021; Nyadzi et al., 2021a; Sutanto et al., 2025a, 2025b). While scientific forecasts remain spatially imprecise (Kom et al., 2022; Gbangou, 2021; Paparrizos et al., 2023) and local knowledge-based forecasts are increasingly disrupted by

climate change (Filho et al., 2022b; Jabik et al., 2022), the integration of LFK with SK aims to address both limitations. (Gbangou, 2021; Nyadzi et al., 2021a; Sutanto et al., 2025a, 2025b). In this context, Sutanto et al. (2025a, 2025b) developed a hybrid forecast that combines both forecast types to increase the accuracy of weather forecast services (Figure 8).

Overall, research regarding the practical implementation of the combination of LK and SK to support climate adaptation remains limited (Filho et al., 2025; Gbangou, 2021; Nyadzi et al., 2021a). The successful integration requires new approaches and methodologies, and researchers emphasize the importance of co-creation with local communities (Chiputwa, 2020; Nyadzi et al., 2022; Filho et al., 2025). The DROP app project aims to integrate this principle by collaborating with local farming communities, agricultural extension advisors, meteorological institutions, and local universities (SAFE4ALL, 2025).

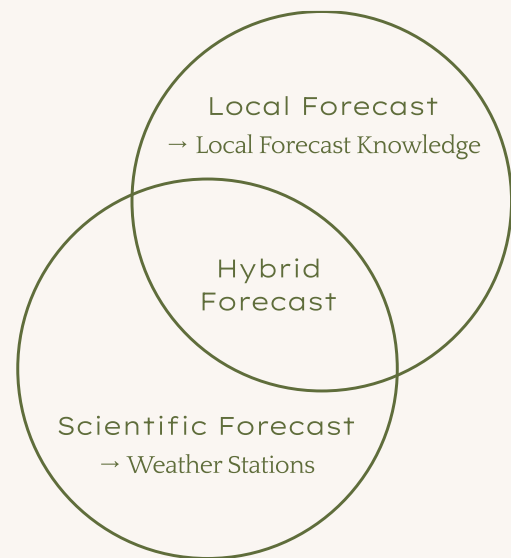


Figure 8: Types of weather forecasts (own elaboration)

3 Theoretical Framework

This chapter introduces three theoretical frameworks that form the basis of this thesis. First, it presents the multi-level service design framework by Patrício et al. (2011), which outlines the different levels of a service system. Secondly, the chapter introduces the systems thinking perspective by Meadows (2008) that serves as a holistic approach to analyze the interconnections and behaviors within complex systems. Thirdly, it discusses the concept of soft impacts of design and technology by Swierstra (2015), which emphasizes the social shifts initiated by new interventions. Each section summarizes the relevant theoretical framework and subsequently contextualizes it in relation to the thesis topic and the DROP app use case. Finally, the chapter concludes by reflecting on the similarities and differences between the frameworks.

3.1 Multi-Level Service Design

This research is conceptually grounded in Patrício et al.'s (2011) multi-level service design (MSD) framework. It describes three different levels at which a service operates (Figure 9 – left side). The first level, *the service encounter level*, refers to the specific service interface or touchpoint and its direct user interactions. The second level, *the service system level*, considers the service as a whole, which includes multiple touchpoints, such as digital, social, or institutional ones. The third level, *the service concept level*, situates the service within a wider ecosystem of interconnected systems. This aligns with Vink et al.'s (2021) perspective, which recognizes a service as embedded in other processes and shaped by underlying institutions. Institutions refer to beliefs, norms, and rules that guide actions within a system (Vargo & Lusch, 2016). In conclusion, the MSD framework provides a structural lens for designing and analyzing a service across multiple scales.

The research questions of this thesis address the different levels of the MSD framework (Figure 9 – right side). In fact, the first research question operates on the service encounter

level and therefore mainly focuses on the DROP app and its interface itself by examining the representation of LK. Further, the second research question spans across the service system level and service concept level of the MSD framework. The service system level, in relation to the DROP app, refers to the overall service of providing weather forecast information to farmers and the mediation of knowledge between community practices and technology. The service concept level highlights the embedding of the DROP app within other processes, such as community dynamics or social structures. Together, the different levels allow for an analytical movement between scales, from the micro-level analysis of the user interface to broader connections between the service, community practices, and social structures. Hence, from an MSD perspective, the climate service DROP app constitutes a multi-layered system that mediates broader processes of knowledge transformation.

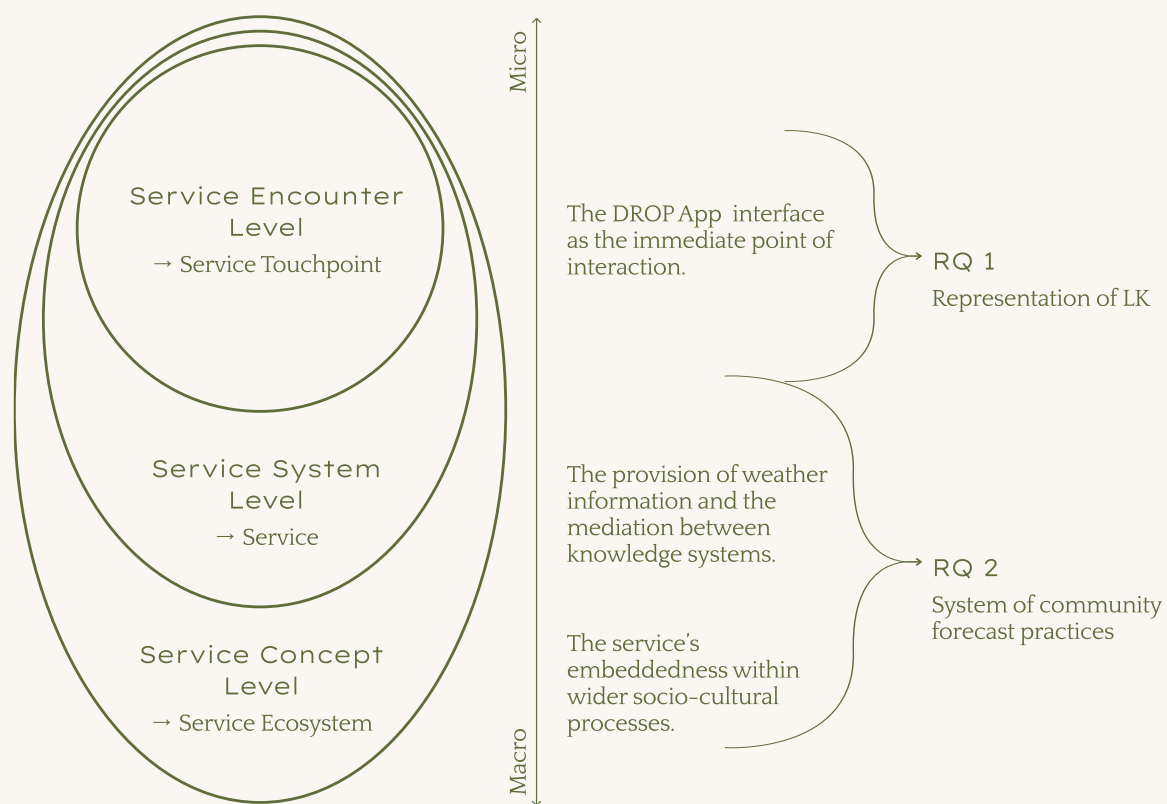


Figure 9: MSD framework by Patrício et al. (2011) (left side, own elaboration) applied to the case of the DROP app to situate the research questions of this thesis (right side)

3.2 Systems Thinking

Systems thinking, as introduced by Meadows (2008), refers to a holistic approach that examines structures, feedback mechanisms, and interactions within a system in order to analyze its dynamics. According to Meadows (2008), systems consist of elements, stocks, and flows, which connect and interact with each other (Figure 10). In detail, elements describe all

human and non-human components of the system. Further, stocks represent the different accumulations in a system, such as materials, information, or other resources, that increase or decrease over time through various processes, termed flows. These interactions create their own behavior through feedback loops that affect other parts of the system. For example, negative feedback loops stabilize and reinforce the system, while positive feedback loops may produce change (Figure 10). Those feedback loops explain the resilience of a system and its malleability to change. External events, such as an intervention or action (Figure 10), can initiate behavior change of the system when they target specific points within that system (Meadows, 1999). Meadows (1999) refers to them as *leverage points*. Because such changes can occur with delays, Meadows (2008) emphasizes the importance of foresight and scenario exploration regarding possible future behavioral shifts.

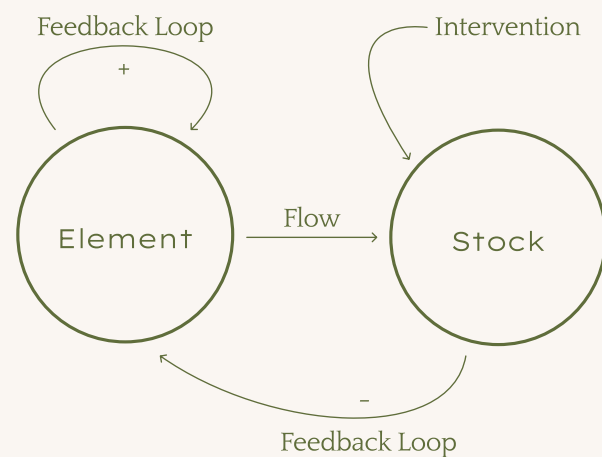


Figure 10: Components of a system (own elaboration based on Meadows, 1999, 2008)

The thesis incorporates a systems thinking perspective by recognizing weather forecasting through community knowledge practices as a dynamic system. It investigates the current structure and behavior of this system by analyzing its elements, stocks, flows, and feedback mechanisms relevant for forecasting and knowledge practices. Moreover, the thesis considers the introduction of the DROP app as a targeted intervention that may affect different parts of the system. Therefore, the thesis seeks to identify and locate some of these potential shifts and changes within the system.

3.3 Soft Impacts of Design and Technology

The thesis is based on the understanding that design and technological innovations produce both hard and soft impacts on individuals who interact with them (Swierstra, 2015). Hard impacts refer to the measurable consequences of technologies, such as their material, economic, or environmental effects (Swierstra, 2015). However, Swierstra (2015) argues that focusing only on these tangible outcomes ignores the dimensions of soft impacts. Soft impacts refer to the social and cultural shifts that occur and slowly stabilize when people

integrate technologies into their everyday lives. Consequently, they reshape norms, meanings, and values in an unnoticeable manner (Swierstra, 2015). Swierstra (2015, p.13) describes soft impacts as “subtle changes in practices” that alter the perception and actions of individuals in the world. Other scholars, such as Jacobs and Huldgtgren (2018), align with this perspective and stress the importance of examining the broader long-term societal effects and transformations initiated through design outcomes. However, Swierstra (2015) emphasizes that these effects allow for no clear classification of positive or negative impacts. Rather, they remain morally ambiguous (Swierstra, 2015).

When applying Swierstra’s (2015) concept to the use case of this thesis, the DROP app appears as a design intervention that may shape social practices and meaning. Its immediate hard impact involves delivering weather forecasts and climate information to farmers. Yet, understanding its soft impacts remains a complex task and the central focus point of the thesis, without assigning them a positive or negative value.

3.4 Connecting the Concepts

When setting these concepts and scholars in relation to each other, differences as well as a clear connection emerge (Figure 11). Distinctions appear through different domain backgrounds (Figure 11 – horizontal axis), different goal orientations, and different levels of abstractions (Figure 11 – vertical axis). For instance, Patrício et al.’s (2011) and Vink et al.’s (2021) perspectives originate from the field of service design. Nevertheless, Patrício et al. (2011) approach this field with an engineering lens, and Vink et al. (2021) with an organizational lens. In contrast, Meadows’s (2008) background lies in environmental science and ecology but offers a cross-domain methodology. Lastly, Swierstra (2015) derives from the field of philosophy, technology, and ethics. These backgrounds influence the goal orientation of the scholar. For example, the MSD framework (Patrício et al., 2011) aims for improved service experience design across multiple levels. Similarly, Vink et al. (2021) seek to increase the impact of services and create long-term organizational transformation by considering the role of social structures. Meadows (1999, 2008), in contrast, focuses on understanding systems, their interdependencies, and reproduction to intervene effectively for systemic change. Swierstra (2015) aims to surface societal shifts in values and norms, introduced by new technologies, to encourage critical reflection. Further, the levels of abstraction of these concepts differ (Figure 11 – vertical axis). Patrício et al. (2011) provide the most concrete and practice-oriented framework. Vink et al. (2021), while similarly grounded in the practice of service design, adopt a more theoretical, systemic, and reflexive stance. In comparison, Meadows (2008, 1999) operates on an abstract, theoretical level while, at the same time,

identifies practical intervention points within systems. Lastly, Swierstra (2015) offers the most philosophical lens by focusing on the ethically and morally ambiguous consequences of design and technology.

Despite their diverse backgrounds, goals, and levels of abstraction, all four scholars recognize systems as multi-layered, interconnected, and mutually influential. They emphasize that changes within a system impact other elements, levels, and processes. This shared understanding forms the theoretical foundation of this thesis and informs the methodological approach. For instance, systems thinking guides data collection and analysis by drawing attention toward connections, relationships, and dynamic processes within community forecasting practices. Further, the differences of the concepts, such as varying levels of abstractions, offer multiple entry points for the analysis of this research and are reflected in the research questions.

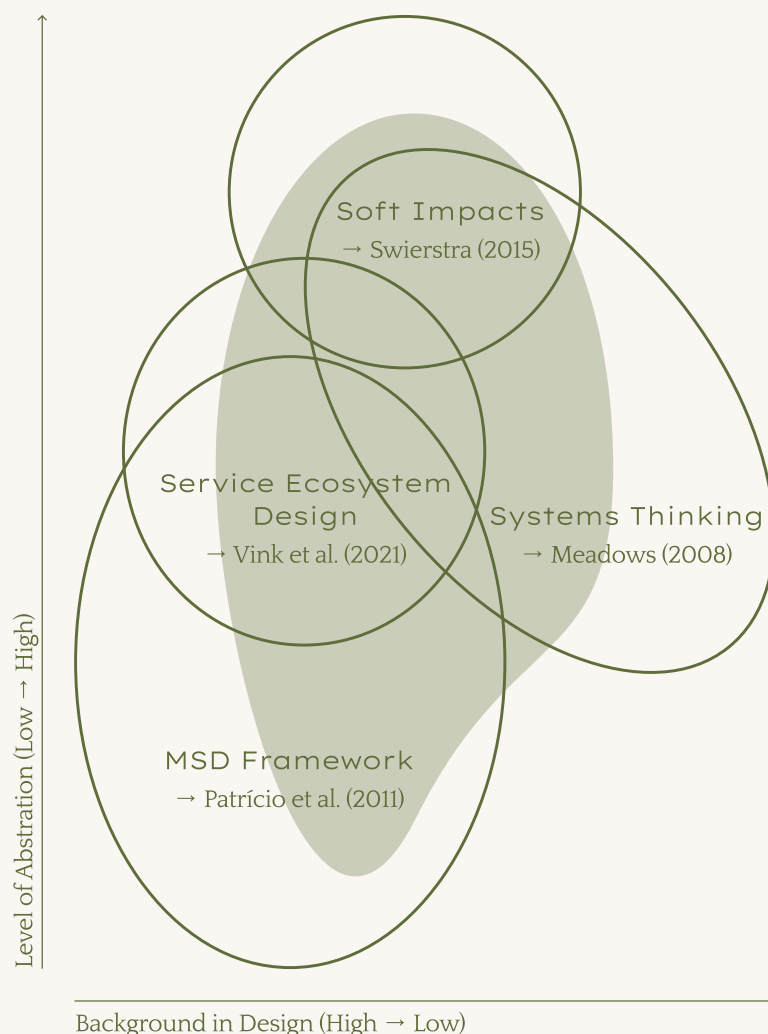


Figure 11: Comparison of the theoretical frameworks based on their background in design (horizontal axis) and level of abstraction (vertical axis) (own elaboration)

4 Methodology

The thesis adopts a qualitative, exploratory methodology with a participatory research approach. Qualitative research focuses on detailed and contextual data for a comprehensive understanding of a topic in relation to the lived experiences of individuals (Ugwu & Eze, 2023). Hence, data collection follows an open-ended process that allows participants to express their thoughts in their own terms and to introduce relevant topics that may not have been anticipated by the researcher (Ugwu & Eze, 2023). To further center the experiences of the participants, the thesis employs a participatory research approach. Participatory research actively includes participants in the process through interactive and visual activities (Sanders & Stappers, 2014). This approach proves especially valuable for revealing behaviors and patterns that influence existing practices (Vink & Koskela-Huotari, 2022). Accordingly, the choice of participatory research aligns with the objective of this thesis, which aims to identify transformations within the system of local knowledge and weather forecasting practices.

Nevertheless, scholars criticize qualitative methods for the involvement of the researcher, who ultimately influences the research process and shapes the findings (Ugwu & Eze, 2023). Rather than seeking generalizability and objectivity, this research aims for depth and immersion in the topic of local forecasting knowledge and practices to generate insights. Thus, the chosen methodology and methods are intentionally adjusted to the specific context of local farming communities in Ghana and to the use case of the DROP app. While acknowledging the subjectivity involved in the research, the thesis combines multiple qualitative methods (Figure 12) to anticipate triangulation and thereby support the credibility of the findings (De Fina, 2015). The following chapters present these methods in a detailed and transparent manner.

4.1 Data Collection Methods

I collected data for this thesis in two phases. First, I applied a desk-based approach, which involves secondary research and qualitative benchmarking through service safari of current climate service weather apps, including the DROP app. Second, I adopted a participatory approach that encompasses semi-structured interviews with the DROP app research team

in combination with generative tools, a workshop with farmers in Ghana, and community-based fieldwork through narrative interviewing. Together, these different methods complement and inform each other (Figure 12). I reflect on the limitations of each method and on their implementation processes in the next chapters.

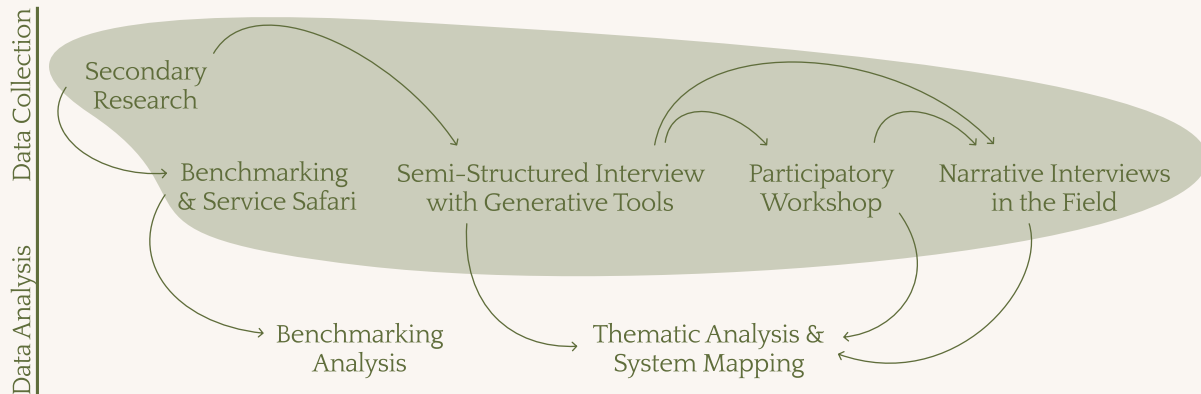


Figure 12: Overview of methods and their interplay during the research process

4.1.1 Secondary Research

Qualitative, secondary research enables an initial understanding of a specific topic by building on existing published studies (Ugwu & Eze, 2023). While these studies are easily accessible and available, critical reflection remains essential because it contains already interpreted and potentially outdated data (Ugwu & Eze, 2023). Furthermore, insight from published studies may not apply to the specific context of this thesis and the farming communities in focus. For these reasons, I utilized secondary research mainly to gain an initial theoretical and conceptual understanding of key terms such as climate adaptation, climate services, local knowledge, local forecasting indicators, and community knowledge practices. Guided by these topics, and with a particular focus on Ghana and weather forecasting, I searched and accessed academic literature and existing studies through Google Scholar. The insights from the literature review informed the earlier chapters of this thesis, as well as the development of the participatory methods.

4.1.2 Qualitative Benchmarking and Service Safari

Qualitative benchmarking and service safari represent methods from the practice of design. Drawing on methods from the practice of design provides a design-oriented approach for exploring the material and functional characteristics of the DROP app as a designed artefact and its placement within a broader ecosystem of climate services. Benchmarking exemplifies a comparative analysis between multiple entities to analyze their differences

and similarities (10x Sheets, n.d.). In contrast to interviews, which capture the perspectives that individuals articulate, I regard benchmarking as a relevant method to examine the DROP app itself, its operation in practice, and its relation to other services. Further, I applied benchmarking to understand the current landscape of CS in the field of weather apps and the representation of LK in these apps. Nevertheless, benchmarking carries limitations, such as potential biases in the selection of entities and variables, interpretation issues due to inconsistent data quality and different data sources (10x Sheets, n.d.), or temporal constraints as compared elements, especially apps, might change through updates. To mitigate some of these limitations and to realize consistency in data sourcing and analysis, I adopted service safari as the data collection method. Service safari represents an exploratory engagement with a specific service, while documenting the process and experiences (Design Council, 2015). The implementation process of benchmarking and safari service included multiple steps:

1 – Selection of Weather Forecasting Apps

Initially, my online search for the selection of relevant apps prioritized those that were developed for the Ghanaian agricultural context and included LK to ensure relevance. Apart from the DROP app, I found one other project, SIW – Smart Indigenous Weather (Artificial Intelligence for Sustainable Development, 2024) (Figure 13). Therefore, I expanded the search to the entire African continent and identified one additional project with LK inclusion, ITIKI (n.d.). However, ITIKI focuses on drought forecasting in Kenya, Mozambique, and South Africa and appears inactive, with the last project update in 2020, the last app update in 2021, no valid contact email addresses, and no compatibility with current Android versions. I conducted the online search exclusively in English, which may have influenced the identification of potential apps. As external validation, I asked participants during semi-structured interviews (chapter 4.1.3) whether they were aware of any additional relevant apps, but only the Uliza-Wi Chatbot, also developed under the SAFE4ALL project, was mentioned. However, the Uliza-Wi Chatbot supports farmers with scientific forecasts and advisories, and does not include LK yet (Weather Impact, n.d.).

Variable	DROP app	SIW app
Target Country	Ghana (+ Zimbabwe, Kenya, Bangladesh)	Ghana
Project	SAFE4ALL EU Horizon Project	AI4SD Project (AI for Sustainable Development)
Status	Development since 2019 App available since June 2022	Development since 2022 App available since June 2025
App Access	Google Play Store	Google Play Store

Figure 13: Overview of identified relevant weather apps as a basis for the benchmarking and service safari

2 - Data Collection through Service Safari

I directly engaged with the DROP app and the SIW app through the method of service safari to collect the data for the benchmarking (Figure 14). I downloaded both apps, created a user profile, and explored the interface. The interface exploration consisted of a general orientation, followed by investigating the process of submitting a local weather forecast prediction and accessing forecast information. I noted down initial observations and documented each UI screen through screenshots.

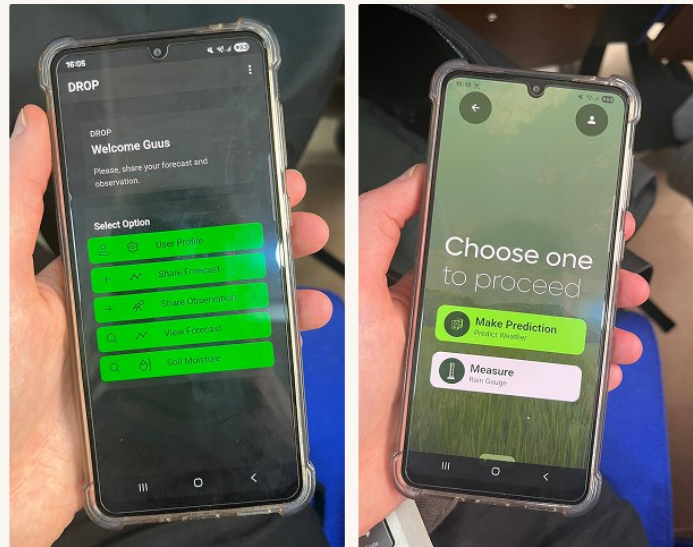


Figure 14: Conducting the service safari of the DROP app (left) and SIW app (right)

Thereafter, I imported all screenshots into Figma and reconstructed the user flows to prepare for the benchmarking analysis (see Chapter 4.2.1).

4.1.3 Semi-Structured Interviews with Generative Tools

Semi-structured interviews combine open- and closed-ended questions and thus allow for unanticipated topics to emerge (Adams, 2015). Interviewers then explore these topics through follow-up questions. Thereby, the method facilitates an in-depth exploration of experiences and perspectives of individuals as questions and topics remain flexible (Adams, 2015). In conventional interview-based methods, the role of the researcher is criticized for dominating the interaction and for framing questions in ways that may embed assumptions (De Fina, 2015). While this concern remains relevant for semi-structured interviews, it applies to a lesser extent. Still, the relationship between the participants and researcher, as well as the connected perceptions of one another, influences data collection. This affects the level of detail, framing, and content of questions and responses, and can result in, for example, participants remaining at the surface level or emphasizing foundational information to ensure shared understanding. Further, in this thesis, interviews were conducted in English, which does not represent the native language of either the participants or me, and may affect nuances in meaning.

To mitigate some of these constraints and to include participants as active collaborators in the research process, I integrated *generative tools* in addition to semi-structured interview questions. Generative tools illustrate an elicitation technique that surfaces tacit knowledge of participants through creative engagement and the process of making (Johnson & Weller, 2002; Sanders & Stappers, 2014). In detail, participants interact with these tools and create collages, maps, stories, or other artifacts to express their experiences (Sanders & Davanta, 1999). Nevertheless, it is relevant to note that the application of generative tools may initially appear unfamiliar or evaluative and may influence participant engagement. In addition, outputs represent highly context-specific and abstract data, which may pose challenges for systematic comparison and analysis across interviews. Yet, the combination of semi-structured interviews with generative tools represents a suitable method to uncover in-depth perspectives, both verbally and visually. Aspects of knowledge practices connected to community roles and relationships may be easier to express through mappings than through words alone. Therefore, this approach supports a layered understanding of local knowledge representation and transformation within the context of the DROP app. The implementation of these methods included a multiple-step process:

1 - Participant Selection and Invitation

I selected participants from the DROP app research and development team. They offer relevant perspectives on the development process of the DROP app since they are responsible for the decision-making on the practical integration and representation of local and scientific knowledge. Further, through close collaboration and fieldwork during the development process, they have accumulated invaluable expertise and insights into the complexities that this integration involves.

After reaching out to the research and development team of the DROP app, I received a list of contacts relevant to my research. I invited all suggested contacts via email for a 45 to 60-minute online interview. In detail, participants included three researchers from the DROP app development team from Wageningen University, Netherlands, and two researchers from the DROP app development team from the University of Development Studies, Ghana. By asking interviewees to suggest further relevant contacts, I identified two additional participants. Both previously completed their thesis in connection with the DROP app, but no longer form part of the research team. As a result, participants differed in terms of their institutional affiliation and location, role within the DROP app team, period of involvement, and level of fieldwork engagement (Figure 15).

Participant	Project Role	Involvement	Level of Fieldwork Engagement
Professor at Wageningen University, Netherlands	Co-Developer	2019 - present	Annual visits; Indirect engagement through student supervision; Close collaboration with UDS colleagues
Professor at Wageningen University, Netherlands	Development of soil moisture module	2021 - present	No direct community engagement; Indirect engagement through student supervision
Professor at University of Development Studies, Ghana	Co-Developer	2019 - present	Annual community visits; Supervises students conducting fieldwork
PhD student at Wageningen University, Netherlands; Master graduate from University of Development Studies, Ghana	Msc Thesis research, Fieldwork support	2019 - 2022	Extensive direct community engagement during thesis work and fieldwork support
Master graduate from Wageningen University, Netherlands	Msc Thesis research, Internship at UDS	2022	Frequent direct community engagement during internship
Professor at University of Development Studies, Ghana	Applied Research Coordinator	2024 - present	Annual visits
Master graduate from University of Development Studies, Ghana	Msc Thesis research, Fieldwork Support	2022 - 2024	Frequent direct community engagement during thesis work and fieldwork support

Figure 15: Overview of interview participants detailing institutional affiliation and location, role within the DROP app team, period of involvement, and level of fieldwork engagement

2 - Development of Interview Guide and Generative Tools

For the development of the interview guide and generative tools, I first outlined different topics that I want to address. These included: *background of interviewees, development process of the DROP app, challenges and future improvements, LK inclusion and representation, community roles and dynamics, LK practices, decision-making, and observed or perceived shifts in these aspects*. Based on this outline, I decided which topics could be explored through interview questions alone and which were better facilitated through generative tools. Following, I developed a mix of open and closed questions in tentative order and simultaneously created the generative tools. The generative tools were largely informed by secondary research and included three activities: First, I applied a **card-sorting activity** to elicit perspectives on key aspects and qualities that best capture local forecasting knowledge (LFK). Participants were then asked to relate these aspects to the DROP app by indicating whether they perceived them as explicitly represented, translated or modified, not represented, or newly emerging through the app. Second, I employed **mapping activities** to explore and visualize community roles and relationships, local knowledge practices, and elements that influence decision-making in farming activities. Third, I utilized **story elicitation** to capture observed or anticipated changes in community

dynamics, forecasting practices, or farming decisions, after the introduction of the DROP app. Overall, the development of the interview guide and generative tools represented an iterative process. I refined multiple versions based on peer feedback, specifically regarding order, content, and framing of questions and activities. Further, I conducted a pilot interview with a PhD student researching CS in Europe, which helped me improve the clarity and comprehensibility of the activities, as well as test the recording process and access to the Figma board. After additional minor modifications, I finalized the interview guide and the generative activities (Appendix 1).

4 - Practical Ethical Considerations

I sent a consent form to each participant a few days in advance of the interview and asked them to sign the document if no questions arose. The consent form consisted of information regarding the background of the research, participants' contribution, recording procedures, data handling, anonymity, and the voluntary nature of participation, including the option to withdraw. While I anonymized the research team members throughout this thesis, I explicitly name the DROP app and the SAFE4ALL project to credit their work and input, and to capture the context that renders the case meaningful for the broader field. At the same time, this decision carries responsibility. My intention with this research is not to evaluate the project but to learn from and reflect on its process, decisions, and challenges.

6 - Conducting the Interview with Generative Tools

I conducted the interviews online and recorded them via Microsoft Teams. The semi-structured interview approach proved beneficial because it allowed me to adjust the order and questions based on the background of the participants, their areas of expertise, and their degree of engagement with farmers. During the interview, I shared the link for the Figma board that presents the generative tools. However, when participants appeared less comfortable with direct engagement in Figma, I shared my screen and navigated the board on their behalf, while inviting them to guide the actions verbally. This proved to be a beneficial alternative for implementing the activities. Furthermore, I observed that participants often approached the topic from a predominantly technical perspective. Aspects related to social or cultural dimensions proved more challenging to elicit, possibly because they did not previously constitute a central focus of their work. Additionally, when social or cultural changes were addressed, some responses from participants revealed contradictions. This underlines the importance of contextual sensitivity when analyzing data and of complementing interviews with additional methods that include farmers and their experiences.

4.1.4 Participatory Workshop in Ghana

Workshops engage participants directly and facilitate an in-depth exploration of experiences (Hyysalo, 2025). They bring multiple participants together, create interactions and discussions, and facilitate collective sense-making. Workshops are highly flexible in their implementation and therefore require adaptation to the specific research objectives, context, and participant group (Hyysalo, 2025). This flexibility makes workshops a suitable method for exploring local knowledge practices and social dynamics. It allows me to create a shared space for participants, namely farmers, to reflect on, challenge, and build upon each other's perspectives. However, effective workshop implementation needs careful planning (Hyysalo, 2025). This includes logistical and strategic considerations such as participant selection, invitations, location, and framing of the session, as well as intermediate design and collaborative work. Intermediate design work refers to the development of materials, tasks, and tools that engage workshop participants with the research topic and each other. Collaborative work represents the facilitation of the workshop by the researcher, such as providing instructions, managing time, guiding group processes, and documenting. Group dynamics, social norms, and the individual personalities of participants may influence the workshop and data collection. Consequently, the role of the researcher and their competencies in facilitation is crucial. The implementation process of the workshop included multiple steps:

1 - Fieldwork Preparation

I conducted the workshop in Ghana in collaboration with the SAFE4ALL project team. As part of their research activities, they regularly conduct field visits to engage with local stakeholders and communities. Fieldwork preparation, therefore, included logistical planning, coordination, and alignment of research goals between my thesis and the broader SAFE4ALL project. The workshop for this thesis was integrated as one session within a multi-day stakeholder engagement program carried out at the University of Development Studies in Tamale, Ghana.

2 - Participant Selection and Invitation

My primary aim was to include a diverse range of farmers who have experience with the DROP app. As key stakeholders, farmers possess valuable direct insights into local knowledge practices, community dynamics and roles, and decision-making processes. Accordingly, the SAFE4ALL project team invited six farmers, who previously attended one of their workshops, on my behalf.

3 - Development of Workshop Activities and Tools

The objective of this workshop was to validate and contextualize insights generated through semi-structured interviews, and, more importantly, to generate new insights based on the lived experiences of farmers. I developed the workshop concept iteratively. While prior interviews informed my understanding, sensitized certain thematic areas, and supported the development of workshop activities, I focused on minimizing framing and designing workshop components that balance collective activities with individual tasks to allow participants to reflect on their own perspectives independently before engaging in group exchange. Subsequently, I brainstormed various activities, reflected on strengths and weaknesses, and mapped them to the research question. After iterating on initial ideas, I discussed the preliminary workshop plan with both of my thesis advisors. Feedback included the need for flexibility in facilitation, clear and concise instructions, and strategies to balance open-ended tasks with specific prompting. Thereafter, I refined the workshop concept and schedule (Figure 16) and designed all workshop materials (Appendix 2), including worksheets, cards, posters, and a slide deck that guides participants through the activities. Finally, I organized a test run with a peer to assess the clarity of instructions.

Time (90min)	Activity	Content
5-10 min	Introduction	Research, Participants, Consent, DROP app
5 min	Warm-Up	Positioning Poster & Optional Extension Activity (DROP app)
15 min	Mapping Activity	My Forecasting Circle
45 min	Main Activity	Card Game
5-10 min	Optional Activity	Sign-Card
5 min	Wrap-Up	Wish for the Future

Figure 16: Workshop schedule and activities

The final workshop concept included the following activities: First, a **positioning poster serves as a warm-up activity** to understand participants' background and experience with local forecasting knowledge and the DROP app. Second, a **mapping exercise of participants' forecasting circles** aims to surface community dynamics, relationships, roles, and changes in forecasting networks. The third and **main activity centers around a card game**, which displays various statements or questions that touch upon tensions connected to decision-making processes, trust between knowledge systems, gaps in LK representation, or social dynamics to spark discussions. Further, a **sign card exercise** seeks to generate a rich, in-depth understanding of specific indicators. Lastly, a **"wish for the future"- poster serves as a wrap-up activity** that elicits participants' hopes and aspirations. Detailed descriptions alongside pictures of each activity are placed in Appendix 2.

4 - Practical Ethical Considerations

Before the start of the workshop, I verbally presented the research information to participants, which consisted of information regarding the background of the research, participants' contribution, documentation process, data handling, anonymity, and the voluntary nature of participation, including the option to withdraw. Translating this information to Dagbani, the local language of the participants, played an important role in ensuring that participants could genuinely understand the research and ask questions before participating in the workshop. Aware that the research context is inevitably shaped by power relations, my goal was to clearly communicate to participants that their knowledge, perspectives, and experiences are central to this research.

5 - Facilitating the Workshop

All six invited farmers attended the two-hour workshop (Figure 17). Further, one translator and two members from the SAFE4ALL project were present, who supported with documentation and assisted participants in carrying out the workshop tasks (Figure 18 – left photo). The workshop was audio-recorded, and photos were taken throughout. During the workshop, the slide deck proved useful for explaining the participatory activities to participants. The flexible workshop schedule allowed me to adjust tasks based on participants' pace and needs. For example, the mapping activity took longer than expected, and thus, I decided to discuss the final part of this exercise verbally. Similarly, I had prepared a large set of cards for the game activity (Figure 18 – right photo) and selected the most relevant cards during the workshop itself, based on topics already mentioned by participants.

Participant	Community	Gender
1	Nakpanzoo	Male
2	Nakpanzoo	Male
3	Chahi Yapalsi	Male
4	Chahi Yapalsi	Male
5	Gbullung	Male
6	Gbullung	Female

Figure 17: Overview of workshop participants

Several challenges arose during the facilitation of the workshop. First, the level of English proficiency varied across participants, and translation was required throughout the workshop, which may resulted in meaning being lost or condensed during this process. Second, engagement levels differed among participants, which could be traced back to gender dynamics or age-related hierarchies. I tried to balance these unequal participation patterns by regularly directing questions towards less active individuals. Further, the workshop had to be divided into two parts with a lunch break in between due to a later starting time, which resulted in lower engagement levels overall in the second half.

Nevertheless, the participatory workshop represented a valuable opportunity to directly learn from the experiences of farmers, touched upon a wide range of relevant topics and provided me with an initial understanding of the local perspectives. This initial overview served as a starting point for the subsequent fieldwork research component, as it helped me to identify aspects to focus on in the upcoming narrative interviews, including topics that remained open, unclear, or particularly relevant.



Figure 18: Translator assisting participants in carrying out the positioning poster tasks (left photo) and participants engaging in card game activity (right photo) – photo credits: Guus Wiersma

4.1.5 Community-Based Fieldwork through Narrative Interviewing

Fieldwork investigates contextual complexities of social life by engaging with people in their own environments (Phillips & Johns, 2024). This approach understands field data as relational, situated, and co-produced between researchers and participants. In this thesis, I implemented fieldwork through visits to farming communities and engagement with their members. In contrast to participatory workshops (Chapter 4.1.4), which constrain and limit attendance by available time, travel distance, and other practical barriers, community-based fieldwork allows me to interact with a broader and more diverse range of participants. It enables contextually grounded exploration of experiences and perspectives, which is essential for understanding local knowledge practices in relation to the DROP app and situated in their daily lives.

Fieldwork requires methodological considerations, as the relationship between the researcher and participants shapes the generated data. My presence within farming communities influences interactions, expectations, and relational dynamics. In addition, my background and association with the SAFE4ALL project places me in a position of power. These power relations became evident when a SAFE4ALL researcher mentioned that

community members may provide socially desirable responses to "please" researchers. I am aware of these limitations, and I aim to mitigate them, to the extent possible, by implementing narrative interviewing techniques (Phillips & Johns, 2024). Narrative interviewing invites participants to tell stories related to past experiences with minimal prompting from the researcher. Nevertheless, I adapted this approach to balance openness with the need to address specific topics relevant to my research questions. In detail, I combined narrative elicitation with semi-structured thematic probing to guide conversations. The implementation process of the fieldwork included multiple steps:

1 - Community Selection and Participant Considerations

Communities were selected based on prior engagement during the DROP app implementation phase. The SAFE4ALL team contacted two communities, Yapalsi and Nakpanzoo (Figure 19), in advance to request permission and confirm participation. I intended to engage with a diverse representation of individuals during the narrative interviews. Insights from the interviews with the DROP app research team highlighted the importance of engaging different social groups, such as women and men, or younger and older community members, separately, because power dynamics, hierarchies, or authority structures may hinder interaction. Based on this, I decided to conduct the interviews individually or in pairs with the following groups to include diverse perspectives: female farmers, young (male) farmers, experienced farmers and rainmakers. Within these groups, I sought to engage both individuals who use and those who do not use the DROP app.

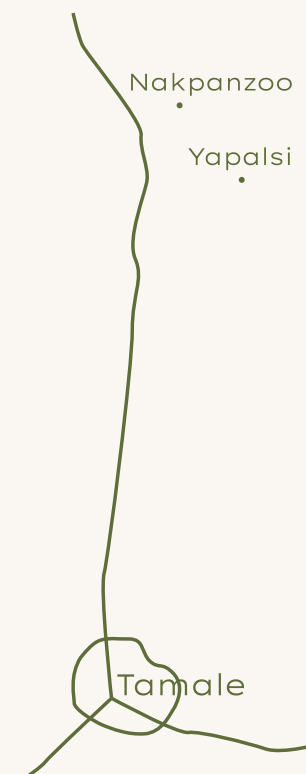


Figure 19: Location of selected communities for the fieldwork (own elaboration)

2 - Development of the Interview Guide

The development of the narrative interview guide represents an iterative process. I focused on simple questions that prompt people to narrate and tell stories and, therefore, planned to begin each fieldwork encounter by asking participants: "Could you walk me through your day?". This question helped me to understand participants' typical activities, daily interactions with others, or moments when weather became relevant, while allowing them to describe their experiences in their own terms. To ensure that I still cover all necessary topics to answer both research questions, I developed a set of thematic prompts, which I would introduce if participants had not mentioned them already. These covered four main topics: *local knowledge practices and weather forecasting*, *DROP app*

experience, experienced changes in community practices, and future concerns and wishes.

To maintain a narrative orientation, I would, for example, initiate the topic of weather forecasting with "Where does weather matter in your day?" or introduce the topic of experiences with the DROP app by asking "Does the DROP app appear in your day? Where?". In addition to narrative interview questions, I planned to shift toward a more material and observational way of engagement when the DROP app was mentioned in the conversation. In detail, I would invite them to show me how they use it in their daily lives. These walkthroughs allow me to observe their interaction with the DROP app and facilitate grounded, tangible discussions on the representation of local indicators. The final interview guide (Appendix 3) reflects a flexible, multi-layered approach. It includes broader narrative prompts to open conversation, specific thematic topics to address if not mentioned, and detailed sub-questions in case participants need more probing. I aimed to remain responsive to participants' engagement and adjust questions based on their background and experiences.

3 - Practical Ethical Considerations

SAFE4ALL project members contacted the selected communities in advance to acquire permission for field visits, set a time, explain objectives and confirm participation. Before conducting each interview, I first introduced myself and my background. Then, I presented the research information, including the background of the research, participants' contributions, the documentation process, data handling, anonymity, and the voluntary nature of participation, verbally. Thereafter, the information was translated into Dagbani, and I asked participants for consent by introducing themselves on the recording. I chose verbal instead of written consent in recognition of the possibly varying literacy levels of participants. Further, through mutual introduction, I aimed, to the extent possible, to create a feeling of trust and convey genuine interest in the personal experiences and perspectives of participants.

4 - Conducting the Narrative Interviews

After arriving in the community, SAFE4ALL project members supported me in selecting participants. In each location, I conducted three interviews, carried out either individually or in pairs based on availability (Figure 20). Each interview lasted between 25 and 45 minutes, and was documented through notes, audio recordings and photos. All interviews required translation between English and the local language, Dagbani. I acknowledge that translation filters or alters the meaning of the responses of participants and thereby represents a limitation in fully understanding intended meanings.

Interview	Location	Participant(s)	Gender	Age Group	Smartphone	DROP app Access
1	Yapalsi	Farmer	Male	Young-middle	Newly bought	Next season
2	Yapalsi	Farmer	Male	Middle-old	No	Indirect, through others
		Farmer	Male	Middle-old	No	No
3	Yapalsi	Farmer	Female	Middle	Yes	Yes
		Non-Farmer	Female	Young	Yes	Not applicable
4	Nakpanzoo	Farmer	Male	Young	Yes	(Yes) Different app, Next season
		Farmer	Male	Middle	Yes	(Yes) Different app, Next season
5	Nakpanzoo	Farmer	Female	Middle	No	Indirect, through others (possibly different app)
		Farmer	Female	Middle	No	Indirect, through others (possibly different app)
6	Nakpanzoo	Rainmaker	Male	Old	No	Indirect, through others (possibly different app)

Figure 21: Overview of fieldwork participants based on location, gender, age group, smartphone ownership and access to the DROP app



Figure 20: Interviewing two farmers in Yapalsi (left photo) and interviewing the rainmaker in Nakpanzoo (right photo) – photo credits: Kwame Dogbey (left photo), Guus Wiersma (right photo)

After each encounter (Figure 21), I briefly reflected on emerging insights to inform subsequent interviews and to adjust my interview guide. For example, initially I planned to begin the interview by asking participants to walk me through a typical farming day. However, the first two interviews revealed that this prompt made participants reflect on farming activities of the entire season instead of their daily routines. Therefore, I switched the opening question to "Where does weather matter in your day?", which proved more beneficial in focusing the conversation on weather forecasting practices. Further, the topic

of *experiences with the DROP app* was less central to the interviews than originally planned, as I was only able to engage with a small number of individuals who directly use the app. This was a result of multiple factors, such as limited smartphone access, especially among women, and reduced availability of participants in the first community at the time of the visit. Additionally, two male farmers in the second community who had been identified as app users were found, in the course of the interview, to have been referring to a different app. Despite these challenges, I was able to engage with a diverse range of participants, including youth, middle-aged farmers, older community members, a rainmaker, and women. The flexibility of the interview guide allowed me to focus the conversation on local knowledge practices and their role in the community, observed changes, and participants' outlook on the future, as well as how weather information and forecasts are received and shared. Therefore, all interviews offered a rich range of perspectives and remained highly valuable.

4.2 Data Analysis Methods

Through the different methods described in Chapter 4.1, I collected diverse qualitative data, which consists of secondary data, app screenshots and notes from the service safari, interview transcripts and visual materials, workshop outputs and notes. They form the basis of the data analysis. I applied three analysis methods, namely benchmarking analysis, thematic analysis, and system mapping, to answer the proposed research questions. I reflect on the limitations of each method as well as their implementation process in this chapter.

4.2.1 Benchmarking Analysis

The benchmarking analysis constitutes the final step in the implementation of the benchmarking method, outlined in Chapter 4.1.2. The objective was to examine the operation of the DROP app in practice and its relation to other services in order to investigate the representation of LK in weather forecasting apps for farmers in Ghana. The analysis followed an iterative process, consisting of multiple steps:

1 – Data Preparation

I prepared the data from the service safari by uploading all screenshots and notes to Figma and recreating the app flow for submitting a local weather forecast prediction and accessing forecast information. Additionally, I added screenshots of the proposed local knowledge

indicator layer of the DROP app (Figure 22). Although these local indicators are not yet selectable in the current published version of the app, I decided to include the screenshots in my analysis for multiple reasons. First, the mock-up already exists and clearly shows the intended implementation. Second, and more importantly, the possible representation of local forecasting knowledge is directly relevant to my research question. Analyzing the proposed design allows me to examine not only what has been implemented, but also what the app *intends* to represent and how the developers *imagine* the integration of LK in the app. Lastly, it is particularly valuable to analyze this feature before implementation, as it can directly inform the design.



Figure 22: Mock-up of the DROP app with indicator selection (screenshots provided by the DROP app developer)

2 – Initial Annotation and Variable Selection

After the data preparation, I started annotating the screenshots of both apps separately through the lens of multiple variables (Figure 23; Figure 24). First, I applied a deductive approach by drawing on predefined variables informed by the research objectives. These included project *background variables* (country, project origin, project status) and *general*

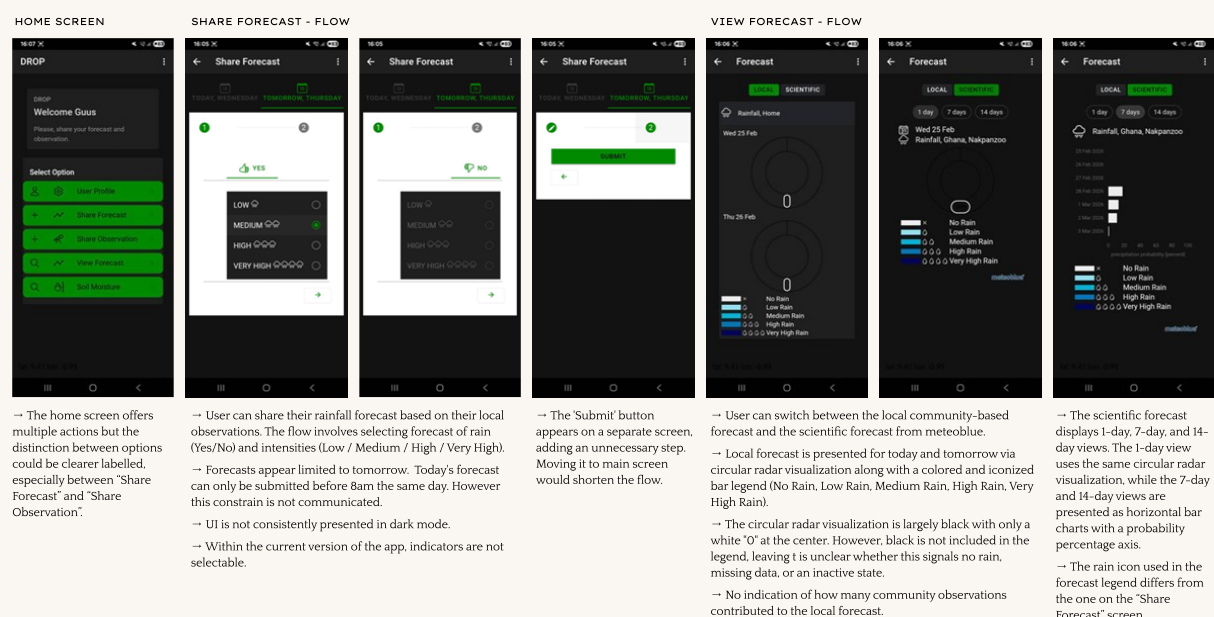


Figure 23: Annotated forecast submission flow of the DROP app (screenshots from the service safari)

information variables (target user, language, types of forecasts). Second, I adopted an inductive approach based on direct interaction with the selected apps. These identified additional variables relate specifically to the *representation of LK*, such as the number of LFI, types of LFI, type of representation (e.g., visual, textual), the style and level of detail (e.g., in descriptions), timeframe for prediction, probability for prediction, prediction for intensity, and presence of special features (e.g., audio elements). Lastly, I included general observations regarding the *user interface (UI)* of the apps. The combination of a deductive and inductive approach facilitated a guided analysis while simultaneously leaving space for additional relevant variables to emerge.

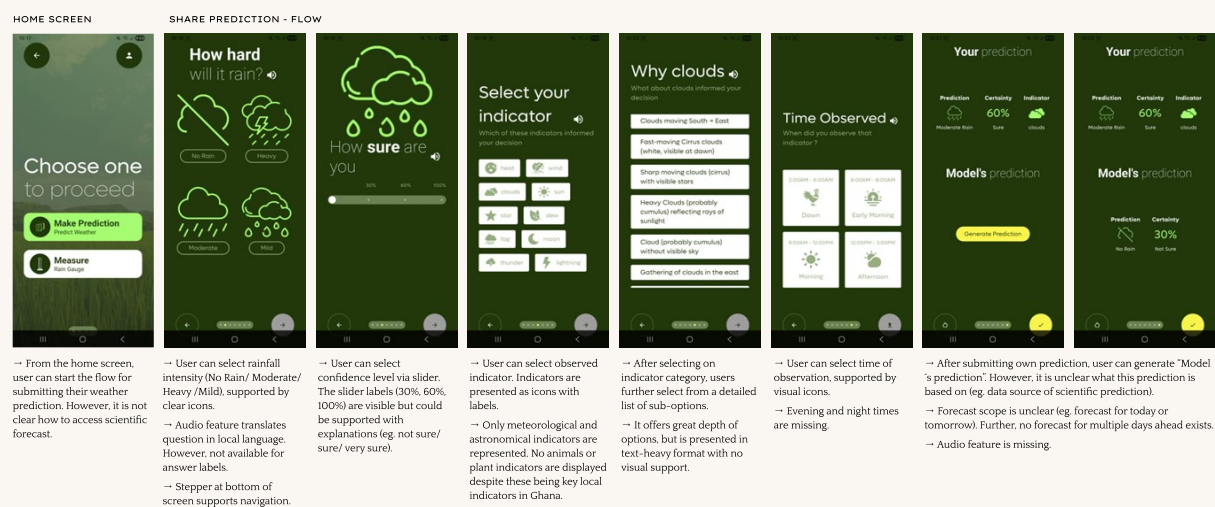


Figure 24: Annotated forecast submission flow of the SIW app (screenshots from the service safari)

3 - Data Organization and App Comparison

I continued the annotation of screenshots iteratively and started to organize them in a systematic and comparative manner. Therefore, I created a table to present the results of the benchmarking (Figure 25). Later, in Chapter 6, I integrate key elements and aspects derived from this table into the findings to answer the research questions.

Variable	DROP app	SIW app
Country	Ghana (+ Zimbabwe, Kenya, Bangladesh)	Ghana
Project	SAFE4ALL EU Horizon Project	AI4SD Project (AI for Sustainable Development)
Status	Development since 2019 App available since June 2022	Development since 2022 App Available since June 2025
Target user	Farmer	Farmer
Language	English	English +Limited audio translation in Twi
Type of forecast	Local-indicator based, scientific	Local-indicator based, scientific
LFI inclusion	yes	yes
Number of LFI	14	10 + Detailed subcategories
Types of LFI	Animal, plant, meteorological, astronomical and other indicators, No seasonal indicators	Only meteorological and astronomical indicators
Type of representation	Visual (icon/ pictogram) +Textual	Visual (icon) +Textual
Style & Level of detail	Conceptual, generic, illustrative, heterogenous style + one word description	Conceptual, generic, homogenous style + one word description (category) + short sentence (sub-categories)
Timeframe for prediction	today and tomorrow (local forecast) 1, 7 and 14 days (scientific forecast)	tomorrow only (local and scientific forecast)
Probability for prediction	no	yes (33%, 67% or 100%)
Prediction for intensity	yes (low, medium, high, very high)	yes (no rain/ mild/ moderate/ heavy)
Special Features	Aggregated local user prediction	Audio feature (limited)

*blue text refers to the proposed local knowledge indicator layer, for which a mock-up already shows the indented implementation.

Figure 25: Results of the benchmarking analysis

4.2.2 Thematic Analysis

Thematic analysis constitutes a qualitative analysis method that synthesizes gathered data into themes (Braun & Clarke, 2006; Braun et al., 2019). It allows for systematic integration and examination of different data sources and materials collected during the research process and therefore represents a suitable method for this thesis. To acknowledge the active role and involvement of the researcher in the analysis process, I adopted a *reflexive thematic analysis* approach (Braun et al., 2019). Reflexive thematic analysis recognizes subjectivity

within analysis as a needed resource for the creation of meaning. Meaning is contextual and therefore constructed through the engagement of the researcher with the data.

I conducted the reflexive thematic analysis in two phases. This separation reflects the temporal structure of the research process, as I implemented the interviews in December, while the workshop and fieldwork took place two months later. The first phase focused on data from semi-structured interviews and generative tools. It consisted of the following steps: data preparation and familiarization, generation of codes, generation of themes, and revision of themes in relation to the research questions. The second phase included additional data from the participatory workshop in Ghana and the community-based narrative interviews. It encompassed the same steps as the first phase, with an additional process of combining and revising the themes from both phases. Overall, the thematic analysis followed an iterative process, and I placed the detailed description of each step in Appendix 4. The final set of themes and sub-themes (Figure 26) forms the basis for the construction of a coherent argumentation in response to the research questions.

RQ1	LK within the DROP app → Direct representation → Indirect representation → Visual style & feedback LFI Variety and Depth → Behavioral detail & simplification → Timeframe & intensity prediction → Reliability differences → Types and range LFK practices → Ceremonial rain-calling & other rituals → Astrological and calendrical calculation → Perceptions Dynamic Nature of LK → Contextuality & relationality → Climate change impacts Uneven Knowledge Holdings and Participation → Mismatch between experienced knowledge holders and app users → LF input in the DROP app Co-production of Knowledge → Knowledge contribution & ownership	RQ2	On-Demand Access and Information Sourcing → Continuously available information → Text-based format → Access & barriers Reconfigured Forecasting Network → Process of information sharing and receiving → Intergenerational dynamic & social roles → Gender → Reach and scale of info-sharing Comparison between SF and LF → Trust and interest in SF → Accuracy, access & availability Future of Forecasting Practices → Role of LK and rainmaker → Expectations & dependencies → Hybrid forecast
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Figure 26: Final set of themes and sub-themes of thematic analysis

4.2.3 System Mapping

System mapping draws upon the systems thinking perspective by Meadows (2008, 1999), outlined in Chapter 3.2. The creation of maps serves as an analytical method to visualize actors, relationships, and processes within a system. This thesis applies system mapping as a complementary tool to analyze and synthesize the collected data.

Throughout the analysis process, I developed multiple system maps and iteratively refined them based on new generated data from each implemented research method. While I acknowledge that system maps may be incomplete and biased (Jones & Bowes, 2017) as they reflect personal interpretations of reality, they constitute valuable design tools for sense-making of collected materials and informed my subsequent research process. Created maps varied in their purposes, level of detail, and temporality. I placed all iterations, along with a description of the process, in Appendix 5. The final set of maps does not aim to visualize the entire system of weather forecasting and community knowledge practices in relation to the DROP app, but rather supports the presentation of individual findings. Each map is tailored to a specific key insight and thereby only communicates the detailed relationship between actors or processes that underpin that argumentation.

5 Findings

The thesis explores how the integration of local and scientific weather knowledge in climate services, such as the DROP app, reshapes the form, meaning, and social organization around forecasting practices in Ghanaian farming communities. This chapter outlines the main research findings connected to this objective by answering the two research questions. The findings are supported by quotes to ground my interpretations, and, most importantly, to let participants speak for themselves. Quotes represent statements made by the DROP app research team and farmers in English, as well as from the translator, who voices farmers' experiences expressed in Dagbani.

5.1 Research Question One

RQ 1: How is local forecasting knowledge represented in the climate service DROP app, and what aspects of this knowledge are not reflected in the integration?

This first research question focuses mainly on the DROP app itself. Specifically, it examines the transformation of local forecasting knowledge when integrated into the app interface to identify aspects of local knowledge that are represented, as well as those that are not fully captured or simplified in terms of meaning, interpretation, or contextual richness. This does not necessarily imply that missing aspects should be implemented. Rather, the aim is to understand what inclusion of LK encompasses when operationalized within climate services such as the DROP app. Following, I present six insights to answer this research question:

Insight 1

Indirect representation of LK reduces the local knowledge to mere data outputs and detaches it from the context and source that give it meaning and credibility.

The service safari indicated that the current DROP app version allows users to submit a forecast prediction by selecting an expected rainfall amount (Figure 27 - left). In practice, this prediction is based on the environmental observations of the user, such as frog croaking, ants carrying eggs, or cloud formations. However, these indicators themselves are not recorded in the app, and the local knowledge behind the forecast prediction remains outside

the interface. Thus, the current published version of the DROP app integrates local forecasting knowledge only indirectly. In contrast, the research team proposes a possible future implementation that directly includes LFK as part of the submitted data, as illustrated in a mock-up (Figure 27 – right). The mock-up of the DROP app presents local environmental indicators as pictorial images with labels. Users would select the specific indicator category on which their rainfall prediction is based. Thereby, the DROP app would not only record a local forecast, which is the case for the current published version, but additionally documents the local knowledge behind it. Workshop and interview participants consistently responded positively to this mock-up and the visual representation of LFK. One farmer immediately opened the DROP app and asked where this feature would appear, indicating their interest. In fact, farmers named multiple reasons for the importance of the visual representation of LK. First, it would make the app more graphical and easier to navigate for non-literate users:

They are really appreciative of all the indicator symbols. I am certain they all want that. It would make things easier for them. Because some of our farmers can't even read. But when they see the symbols, they can understand. The level of illiteracy is high. Not all of them are able to read. But with symbols, they can.¹

– Translator during workshop with farmers

Second, it would serve as a reference point for the source of the forecast, and additionally support the reflection on the reliability of specific indicators:

He would like to select the specific indicator. The reason is that those signs will serve as a reference for them to know where their forecasting is coming from, which particular indicator it is based on. And by them entering it, others will also be able to see their inputs and see if it is accurate or not. It will also help make weather knowledge better.

– Translator during interview 1 with a young-middle-aged male farmer

Third, it would support collective learning:

If it is done in that format, people will be able to share this information and knowledge among themselves. Because if he proves what indicator he has seen, and someone else also shares what he has seen, it is kind of like learning from each other. It is something that will help. And he agrees as well.

– Translator during interview 4 with middle-aged and young male farmers

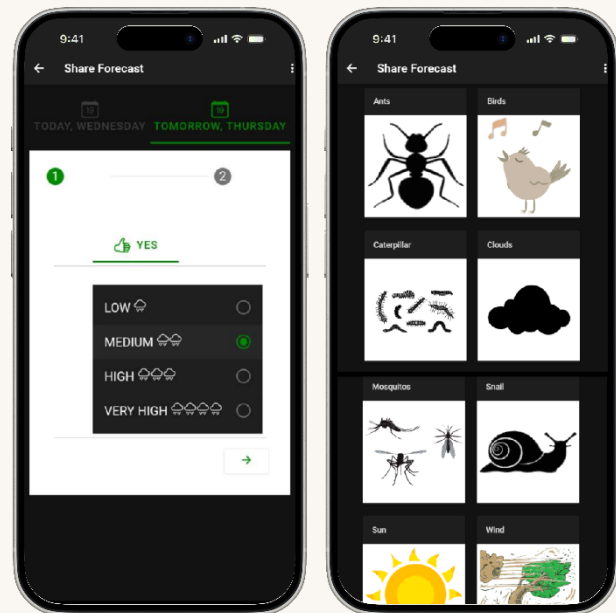


Figure 27: Submission of a local forecast based on the rainfall prediction in the current published version of the DROP app (left) and selection of LFI in the proposed future mock-up (right)

¹ Because of language barriers during the fieldwork, this and most subsequent quotes in this thesis are from the translator, who acted as a bridge between the farmer and me. Therefore, I want to acknowledge the additional layer of interpretation embedded in the quotes.

Currently, the DROP app assumes that recording the local prediction is enough to capture the local knowledge. This decision signals a deliberate effort to keep the development of the tool feasible and simple. Nevertheless, the findings illustrate the importance of presenting the knowledge behind the forecast (Figure 28).

Essentially, LFK represents a process of reading the

environment and interpreting its meaning. Detaching this relationship reduces LK to merely data outputs and does not reflect farmers' realities. Most importantly, the inclusion of selectable pictorial indicators centers the LK and creates credibility, transparency, and accessibility of the LF (Figure 28). Further, documenting farmers' interpretations of LFI produces a feedback loop where community members can learn from each other's observations (Figure 28 – right side). Thus, rather than solely presenting an aggregated prediction, the LK behind the local forecast should remain legible and traceable and thereby could contribute to a shared system of observed knowledge.

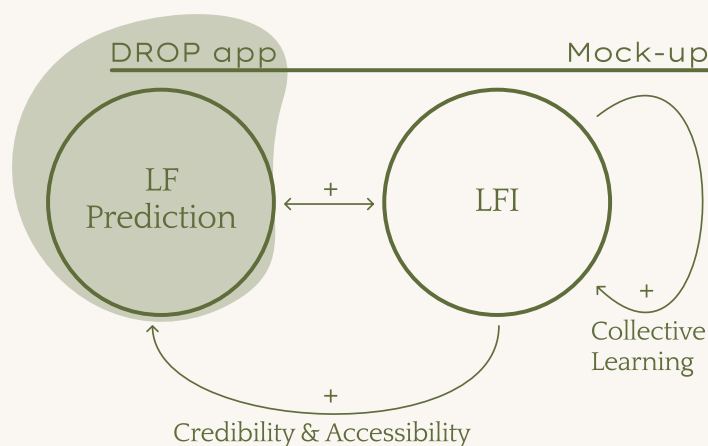


Figure 28: System map illustrating the importance of presenting the local forecast indicator (LFI) behind the local forecast prediction

Insight 2

The degree of simplification of LK determines whether its key aspects are captured within the system and influences perceptions on how intentionally the integration appears.

The DROP app mock-up presents forecast indicators as broad categories, for example, an “ant” as an image with a short label. This representation does not capture the nuanced knowledge of specific behaviors or variations within indicators. The research team acknowledges this simplification:

We are trying to translate the local knowledge in the app in a much more simplified way: Select this indicator and tell me if it's going to rain or not. ... But it's about what they see and what moves they see. They see clouds, but there is a lot of variation in clouds that could mean different things.

– Professor 1 from the project team

We need to always be careful with the interpretations of the local forecast because, for example, in the symbol, we just say ants. But what activities the ants do can lead to different forecast opinions.

– Professor 2 from the project team

In comparison, the SIW app offers multiple specific sub-variations after selecting the broad category of, for instance, “clouds” (Figure 29). The user can choose between different cloud formations, colors, or movement directions. This subcategory structure, which reflects nuanced knowledge on variations, is not represented in the DROP app. However, the

nuances and specific behaviors of indicators are relevant because they carry different forecasting meanings for rainfall onset and intensity. For example, not the ant itself indicates a weather prediction, but what the ant is doing – carrying their eggs – and where the ant is carrying them to – what height and how far:

He is also talking about the actual behavior of the ants. How they behave tells you the intensity of the rain and the duration before the rain comes. Moving to a higher branch indicates heavy rain. If you see ants carrying eggs over a longer distance, it tells you that rain is coming, but it will take some number of hours or days. If it is a very short distance, the rain is very close, just a few hours. Depending on the distance, it can be within 24 hours, the next day, or even a week.

– Translator during interview 1 with a young-middle-aged male farmer

This quote illustrates that *prediction timeframes* vary depending on the nuanced observation of the LFI. However, the DROP app only allows forecast submissions for the same or the next day. Further, farmers connect indicators to *different reliabilities*. For instance, ants and frogs are considered highly accurate indicators, whereas sheep and mosquitoes are ranked low:

He said there is a reliability variation among those indicators. The ant indicator has high accuracy. Another indicator with high accuracy is frogs croaking, where rain will come within a 24-hour period. ... There are two indicators where he would not rank the accuracy high. There is the sheep And then there is also the mosquito bite: if there are intense mosquito bites in the household at night, that is an indicator. But he would not rank it high.

– Translator during interview 1 with a young-middle-aged male farmer

This knowledge regarding differences in the reliability of indicators is not reflected in the DROP app, unlike the SIW app, which allows users to submit their level of certainty for a prediction (Figure 30). During the workshop, farmers expressed interest in including this feature:

We want it to be part of the app. So that when we are making a prediction today, we can indicate how certain we are.

– Farmer during workshop

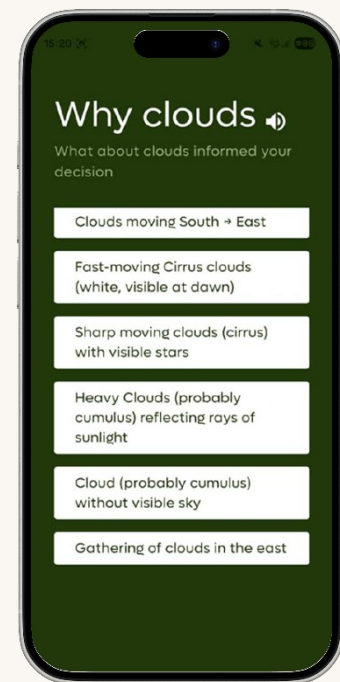


Figure 29: The SIW app offers multiple sub-variations within one LFI category

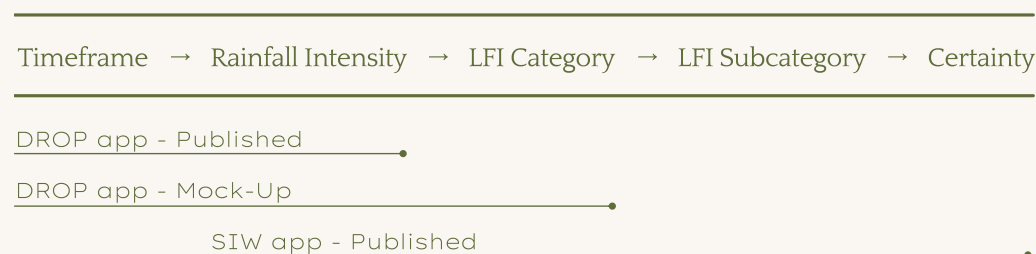


Figure 30: Comparison of LK input flows between the DROP app and the SIW app – The SIW app allows for more precise input in terms of LFI subcategory and certainty selection

Overall, these findings illustrate that the predictive value of an indicator is not reflected in its presence but in its nuanced behavior, which carries information regarding rainfall intensity, onset, and certainty. Reading this behavior embodies the local knowledge. When digital climate service tools oversimplify LK, for instance, into broad categories, key aspects of local forecast predictions are not captured and thus cannot enter the system.

Apart from the missing behavioral detail, a further aspect of the simplification concerns the *visual style of the indicator*. The DROP app mock-up does not illustrate any specificity in terms of locally specific species of animal indicators. For example, the ant indicator is presented as a flat black icon. Despite workshop participants emphasizing that they rely on a distinct type of ant for weather prediction, they largely agreed to the proposed visual representation of LFI in the DROP app mock-up:

How you designed it and described it is okay. The person who didn't go to school, if they see this ant, they are able to understand that this is an ant. It is the same for other things. You can understand it.

- Farmer during workshop

Similarly, one researcher noted that farmers had pointed out that a depicted frog did not match the look of the frog from their region. Another researcher, however, countered that the species is trivial:

The local farmer said that this is not the frog in their region. You know, they are very, very specific about the symbol.

- Professor 2 from the project team

It doesn't need to be a specific frog, because what is important is that the frogs make noise at night.

- Professor 3 from the project team

Overall, this illustrates that the simplification of the specificity of LFI poses no barrier to understanding the representation. A generalized icon conveys the concept, but it simultaneously impacts the perception of whether the place-based, situated nature of that knowledge is recognized and valued in the design process. The visual language carries meaning beyond aesthetics, as it influences how credible and intentional LK inclusion appears in the app. When translating nuanced LK into digital climate services, simplification is unavoidable and represents a balancing act between feasible, practical functionality and detailed representation. Design decisions regarding what aspects to simplify and what meanings to preserve require awareness, reflection, and transparency. Most importantly, it needs collaboration with farmers to navigate this translation process (Figure 31). LK integration within climate services, therefore, illustrates a co-production process, where farmers contribute LK and feedback during development (Figure 31 – left), while the research team introduces technical infrastructure and practical boundaries (Figure 31 – right). Practical boundaries include, for instance, the limited number of selectable indicators, the English language interface, or the requirement to submit a prediction for the same day before 9 am. These boundaries need to be negotiated based on the input of communities to

create meaningful climate service tools, as they not only determine usability but also the extent to which local knowledge enters the system.

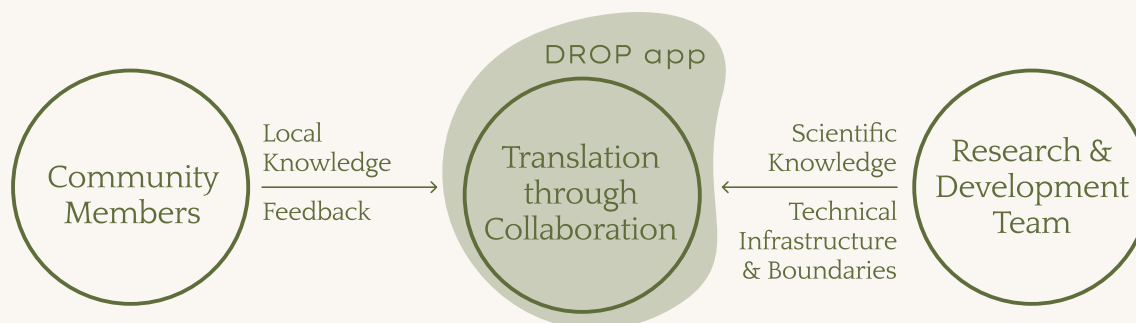


Figure 31: Knowledge co-production process between community members (left) and the research team (right)

Insight 3

Project scope and accuracy perception constrain the breadth of LK representation and leave the broader local knowledge system invisible.

Community members draw on a wider range of indicators for rainfall prediction than the selection currently depicted in the DROP app mock-up. During field interviews, participants mentioned, for example, bats, snails, specific houseflies, or sheep as indicators, which are missing in the current DROP app mock-up. The DROP app research team acknowledges this:

After the first implementation, we identified other indicators that could be included in the app.

– Former project team member 1

The reason behind the limited selection lies in the time-intensive collection of these indicators. Given the contextual, highly localized nature of LK, more indicators continue to emerge throughout the process. To partially address this gap, the mock-up interface includes an "other" category option to allow inputs beyond the listed indicators. Further, the DROP app only represents forecast indicators connected to rainfall prediction, which constitutes the main focus point of the project. However, other prediction purposes of LFI exist as well (Figure 32). For instance, farmers employ environmental observations for wind, storm, and drought prediction:

During the night, if you see a ring around the moon, and if the color of the ring is brown or red, it means that before it rains, there is some wind or storm.

– Translator during interview 5 with middle-aged female farmers

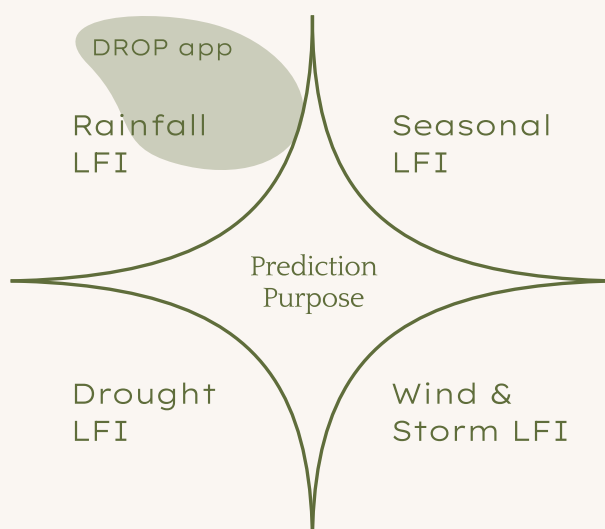


Figure 32: Overview of LFI and its various prediction purposes

He also said that for the dry spell, there is a tree they use to forecast it. If that tree starts to have flowers during the rainy season, then it shows that very soon you will start to experience a dry spell.

- Translator during interview 4 with young and middle-aged male farmers

This finding illustrates that the project scope affects LK representation. By focusing on specific LFI for rainfall prediction, the DROP app implements only one part of a broader knowledge system applied by local communities to forecast weather changes.

Further, farmers utilize seasonal indicators once per season to predict the onset and intensity of the upcoming rainy and farming season:

He gave an example of one local tree, a local grape tree. He is saying that when that local plant has fruits, by that time they will know that the season starts, the farming season, or the rainy season onset.

- Translator during interview 1 with a young-middle-aged male farmer

The research team is aware of these seasonal indicators, but excludes them in the DROP app due to accuracy concerns:

There are indicators that can predict when the rainy season will start. And then there are indicators that can show when it will end. So farmers use this one for the long-term prediction, for the entire season. The seasonal indicators are also the indicators that are not reliable. ... The indicators that predict the season are not yet in the app.

- Former project team member 1

While farmers during the workshop mostly shared those accuracy concerns, another farmer during the narrative interview expressed interest in including them in the DROP app:

We have them, the seasonal indicators, but we are likely to fail. If you are going to predict a week, you can do your best. But when the season begins, the indicators or symbols are no longer as effective as they used to be.

- Farmer during workshop

He would like to share what he has seen this season to be and also hear from others what they have seen.

- Translator during interview 4 with young and middle-aged male farmers

The dynamic indicates a conflict between farmers' trust in traditional practice and their awareness of its decreasing reliability, especially for longer prediction timeframes such as seasonal forecasts. This conflict also manifests itself in the design of climate services, which must balance the breadth of LK representation with the provision of accurate information. The differing opinions across community members on which LFI to include in the DROP app illustrate that accuracy is not a shared understanding, but a negotiated value. Overall, the findings demonstrate that the definitions of accuracy, as well as the collection process of LK and the scope of the research project, impact LK representation in climate services tools, such as the DROP app. To create transparency and acknowledge the true breadth of local knowledge systems applied by communities to forecast weather changes, it remains essential to precisely articulate which types and to what extent local forecast indicators are included or excluded within a specific CS tool.

Insight 4

Epistemological differences between scientific and local knowledge cannot be resolved. Whether spiritual and ceremonial LK inclusion is treated as an act of recognition or as a scientific equivalence determines which forms of knowledge enter the system.

Participants describe rainmakers as important community figures. They represent active, experienced, and trusted LK holders, and thus an invaluable source of weather information for other community members:

The rainmakers in the community are important for weather forecasting. They have one rainmaker in the community who forecasts for them, and the number of times it comes true is far greater than the times he himself was able to forecast.

- Translator during interview 1 with a young-middle-aged male farmer

Especially in cases where you can see clouds forming, but later the rain does not come. When that happens, they can go to the rainmaker and ask. They tell him what they saw and what happened, and he shares whatever information he has with them.

- Translator during interview 5 with middle-aged female farmers

They are saying they don't go to the rainmaker. He is the one who normally comes out to tell them whenever he sees something regarding rain.

- Translator during interview 2 with middle- to old-aged male farmers

In addition to forecasting rainfall through environmental indicators, rainmakers practice LFK in the form of ceremonial rain-calling, rituals, and astrological and calendrical calculations for the community (Figure 33 - middle):

And then also with the rain calling, he said that one is different from just using the signs. He can call for the rain maybe once in a year, or sometimes twice, or up to three times. ... depending on how problematic the rain becomes that year. But he is saying that there is another form of it, ... like a ritual they can perform. You can come and have it placed at your farm, so that even during a dry spell, when other people are experiencing drought, whenever there are rain clouds, they will follow onto your farm. ... They are also looking at something like astrology; they also have that kind of system. That is how he normally predicts the rainy season and dry spells.

- Translator during interview 6 with the rainmaker

Ceremonial and calendrical local forecasting knowledge is not represented in the current app or proposed implementation, despite all members of the DROP app research team being aware of this dimension of LK. It is seen as a continuing practice outside of the DROP app:

The rainmaker performs a ritual to call the rain. So this part is not embedded in an app in any way. The app can tell them that the rains are not going to come at all, or it's going to come, but the rituals give them the room to create the rain. The app cannot do that in any way. So, I don't think they're going to let go of the rituals in the future. They're still going to continue it.

- Former project team member 1

The question of whether and how this form of LK could be included in the DROP app revealed uncertainty from workshop participants. One participant suggested it could be recorded as an event, but not submitted in advance as part of the prediction:

When the rainmaker calls, and it rains, ... we can add it to the DROP app. But not before, not as a prediction, because that is about the traditional system. How would you input it as a prediction?

- Farmer during workshop

Whenever the rainmaker calls for rain or whenever there is a ceremony, they would be able to put it there. And then also, if it actually happens, they will be able to compare afterwards. That is what they are saying.

- Translator during workshop

This uncertainty may stem from differing opinions on these practices. Some participants described a growing dependency on the rainmaker and his ceremonies because of increasing dry spells and unpredictable rainfall, while other community members questioned the accuracy (Figure 33 – right):

He is saying that in the future, rainmakers will be equally or more important in this society in relation to climate change. Because now the rain is becoming less and the dry spells have become more frequent. And that is the reason why they normally depend on the rainmaker to call the rain.

– Translator during interview 1 with a young-middle-aged male farmer

The ceremony he normally performs is not 100% accurate. They say it is a 50-50 kind of forecast.

– Translator during interview 3 with female non-farmer and farmer

Ceremonial LK represents a fundamentally different practice compared to LFI. LFI, as observable natural phenomena, are to some extent scientifically grounded:

When this bird is flying very low, you know that the storm is coming. But it's because you have some air depression, and then you have all the insects that are flying low at this moment, so it's perfect for them to hunt.

– Former project team member 2

In contrast, ceremonial LK is rooted in spirituality. The practice of rain-calling, for instance, represents an intervention in the weather, not a prediction. However, because it anticipates rain within a certain timeframe, it could be translated into the logic of the DROP app. Therefore, the design decision for the exclusion points towards epistemological limits within CS development, where spiritual and observational LFK are not treated as equal knowledge sources. However, precisely because of these epistemological differences, an inclusion can signal an interest in not only avoiding instrumentalizing of LK but also in offering a way to record and preserve it. Essentially, inclusion should not aim to resolve epistemological differences but to make them visible and acknowledge that various ways of knowing exist and hold value for communities beyond the immediate purpose of accurate weather prediction.

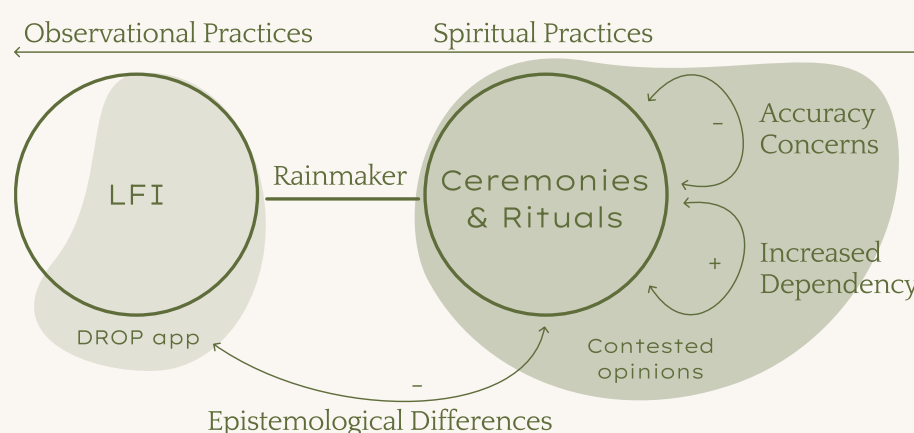


Figure 33: System map illustrating the epistemological differences between observational, indicator-based practices included in the DROP app (left) and ceremonial practices currently excluded (right), alongside contested opinions on the future role of ceremonial practices

Insight 5

Static, fixed interfaces cannot capture the contextual, changing nature of LK.

The DROP app documents LFI based on field research conducted in 2020. Therefore, the LFI within the app represents a selection of indicators that were in use in a specific community, by selected community members, at a specific point in time. While the research team interviewed a diverse range of community members during the LFI collection process, they placed a special focus on adequate representation of elderly men because they are recognized as primary knowledge holders:

Most of the people who hold information for forecasting are men. You have in the community specific people who are only interested in the weather, like rainmakers. They are mostly men, old men, because they have enough experience and so on.

– Professor 3 from the project team

During the interviews, we made sure that we had a good number of old people from the community.

– Former project team member 1

Overall, LK is highly heterogeneous. Different community members hold different knowledge, based on their experience, location, role within the community, and gender:

For example, they were saying, if this specific bird is doing that at night, it means this, but then you discuss with someone else who is saying something else. ... Also, you could see that women had different local knowledge than the men. ... For example, worms in the ground that were indicating something, and I never heard that from the man before. This is one of the examples, but it happened a few times.

– Former project team member 2

Especially gender differences within LK became evident during the fieldwork of this thesis. While male participants referred almost exclusively to animal indicators, women named celestial indicators for rainfall prediction. At the same time, however, data indicated that women hold less LFK compared to men:

They are saying that for them, the only indicator they know is clouds. When they see clouds forming, they will know that it is going to rain or is about to rain, and also if they don't see the sun.

– Translator during interview 3 with female non-farmer and farmer

Elders advising other people in the community, but mainly women, because they have lesser experience with the local system.

– Translator during interview 2 with middle- to old-aged male farmers

This variability in LK distribution makes it highly contextual and relational and may explain the continuous discovery of additional indicators that are not yet represented in the DROP app (see insight 3). Most significantly, local knowledge is changing as a result of climate change. Researchers are aware of these changes, but cannot reflect them in the DROP app:

We know that we may lose indicators because of climate change. You cannot really see it in the apps, but it's happening.

– Professor 3 from the project team

What if the animal or plant is no longer there because of the changes in the environment? ...The research was done some years back. ...I expect some change in the community. So if we go in and do an evaluation ... whether they lost some indicators, we may find it.

– Former project team member 1

During the fieldwork, every farmer consistently described the reduction in the accuracy of indicators as the main climate change impact on LK:

The indicators are no longer as effective as they used to be. Some years ago, if some small clouds formed, it would rain. But now the clouds form in much greater numbers and then pass without rain. That means there are changes, the accuracy or probability has become lower.

- Farmer during workshop

He is saying that even just recently, he could tell that a particular date in the coming month was going to have rain. He used to be able to do that, and it would happen. If not in this community, then in the nearby communities. But right now, it is not like that anymore.

- Translator during interview 6 with the rainmaker

Beyond accuracy, some indicators disappeared. For example, earthworms and snails cannot be observed around the community anymore, possibly linked to rising temperatures:

He is saying that the indicators are reducing and also becoming scarce. He is using earthworms and snails as an example. He is saying that if you take a look at Tamale or the communities closer to Tamale, it would be difficult to find snails around there. So those who were using snails or earthworms as indicators. Now, it will be hard to find them in that particular area. They were also using ducks as an indicator. ... He is also saying that now, in the whole community, nobody has ducks anymore. So that indicator has just disappeared.

- Translator during interview 2 with middle- to old-aged male farmers

Further, one member of the research team pointed to the impact of agrochemicals on certain indicators. Farmers did not mention this aspect themselves:

With the advent of modern technology in farming practices, farmers are now using agrochemicals. And these agrochemicals, I'm talking about herbicides, weedicides, these ones have had adverse effects on these ants and these frogs. ... They are killing or they are destroying some of these insects and birds in the communities.

- Professor 3 from the project team

Overall, these findings illustrate the changing and contextual nature of LK shaped by age, gender, the role of community members, as well as external factors such as climate change or household and farming decisions (Figure 34 - bottom). In contrast, the DROP app, with its fixed interface and static set of indicators, cannot register any of these changes or dynamics. It preserves knowledge at the moment of collection (Figure 34 - middle). For example, indicators that disappeared continue as possible selections inside the app. This reveals a tension between LK as a living practice and LK as a documented artifact when integrated into the DROP app. However, the digital format of such climate service tools could reflect this living nature of LK if designed in a way that allows for editability and updatability through its users.

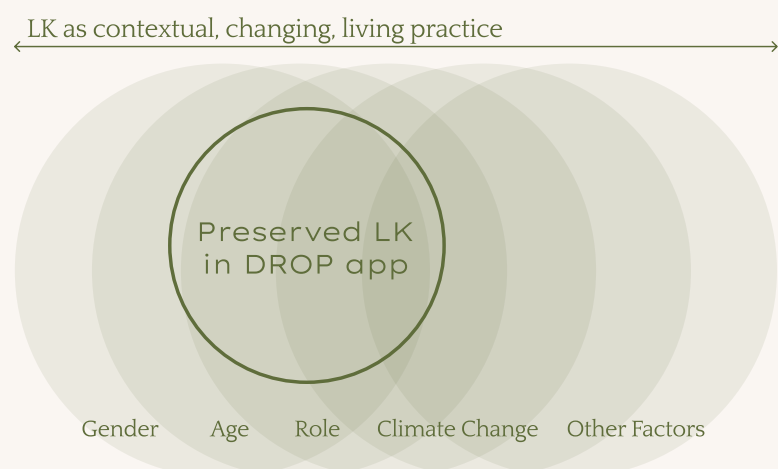


Figure 34: Tension between LK as a contextual, living practice and LK as a preserved, documented artifact within CS

Insight 6

Structural barriers prevent LK inputs from knowledge experts such as elders, which limits the actual representation of the community's forecasting expertise in the DROP app.

Traditional forecasting represents a consultation-based system with community members consulting skilled forecasters (Figure 35). Especially, elders or rainmakers are recognized as reliable local forecast experts within the community. They accumulated experience over decades, know a wide range of indicators, and are trusted by community members. At the same time, they are excluded from using technologies like the DROP app due to barriers such as low literacy and smartphone access (Figure 36 – right). The community members who do use the DROP app are younger, more educated, and tech-savvy, but less experienced with LFK (Figure 36 – left):

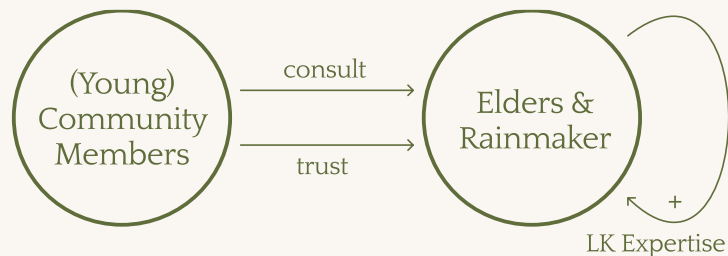


Figure 35: Traditional consultation-based forecasting system with community members (left) consulting elders with LK expertise (right)

He is saying it would be as a result of the elders not being familiar with the phone. They are not tech-savvy due to their lack of education. The youth are much more experienced with mobile phones than the older generation.

– Translator during interview 4 with young and middle-aged male farmers

During the training, we saw that those with a basic or higher education understood the app more easily, and reading it was simpler for them than for those who didn't go to school.

– Farmer during workshop

As a result, a structural mismatch evolves, which shapes local forecast inputs in the DROP app and their level of accuracy (Figure 36 – top middle). LK inputs derive from a small group of technologically literate smartphone owners, and not necessarily from the knowledge experts in the community, such as elders and rainmakers. The research team recognizes these uneven skill and accuracy levels within the data of the DROP app inputs:

We need to find farmers who are skilled in giving this local forecast, because some cannot. I don't know, maybe they just want to play around or maybe they make a prediction, but are not experienced enough.

– Professor 2 from the project team

Further, the lack of experience with LK among the DROP app users led to the app being accessed primarily to receive scientific forecasts:

Definitely, I would go to collect the information. I will not use my own idea. If I open the DROP app, I will go directly to pick the information from the DROP app.

– Farmer during workshop

She was mainly focused on viewing the forecast. ... Once she got the forecast information, she wouldn't even bother about sharing her own.

– Translator during interview 3 with female non-farmer and farmer

Overall, these dynamics suggest that the actual LK representation within the DROP app remains sparse in practice and does not reflect the community's local forecasting expertise.

While the design of the DROP app in theory encourages all community members to input their observations, in practice, knowledge holders, especially elderly, face access barriers. These findings illustrate that openness as a design principle within CS development does not guarantee participation as an outcome, as not everyone is equally positioned to contribute. As a result, LFK continues to be the most active and present outside the app, for instance, in personal observations or in consultations with the elders.

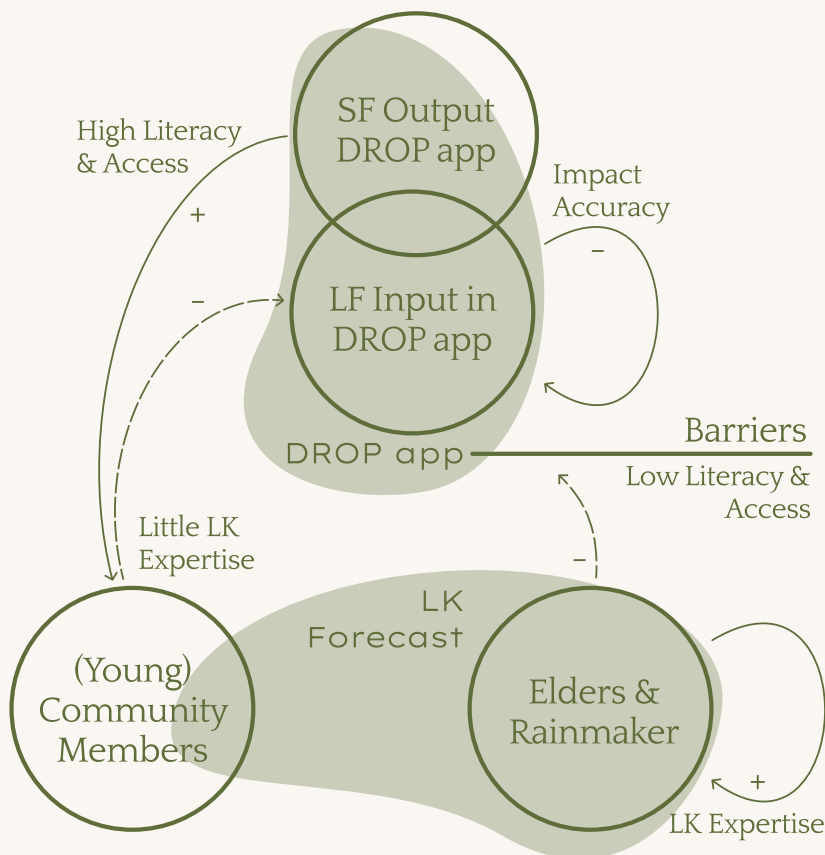


Figure 36: System map illustrating the structural mismatch between knowledge holders (elders and rainmakers - right) and app users (youth - left)

5.2 Research Question Two

RQ 2: How may the introduction of the climate service DROP app reconfigure the system of community forecasting practices?

This second research question positions the DROP app within other processes and systems. In detail, it explores the connection between the DROP app and community norms, institutions, and existing weather forecasting practices by investigating potential shifts in these relationships. Following, I present five insights to answer this research question:

Insight 1

The DROP app bridges the gap between availability and need for weather information and, at the same time, reveals temporal boundaries between the app's logic and farmers' forecasting practice.

Traditional forecasting practice is based on environmental observations. Farmers look out for signs consistently throughout the day (Figure 37 – left side):

He was saying that normally they look for the signs early in the morning and also later in the evening, because those are the critical points of the day that determine what particular activity to carry out. ... This man is also saying that all parts of the day, morning, afternoon, and evening, are important. Maybe you didn't get any sign in the morning, but maybe a few hours later, you will see the weather changing. Also, during the rainy season, rain can come at any point in time. Because of that, the weather is important to them at any point of the day.

– Translator during interview 2 with middle-old-aged male farmers

However, most participants observe indicators only once a week (Figure 37 – right side):

I look out for signs multiple times a day, but see a sign around once per week.

– Farmer during workshop

It's once in a while. The frequency of the signs is hard to say. Like, you can see one this week and then not see another one for a whole week.

– Translator during interview 2 with middle- to old-aged male farmers

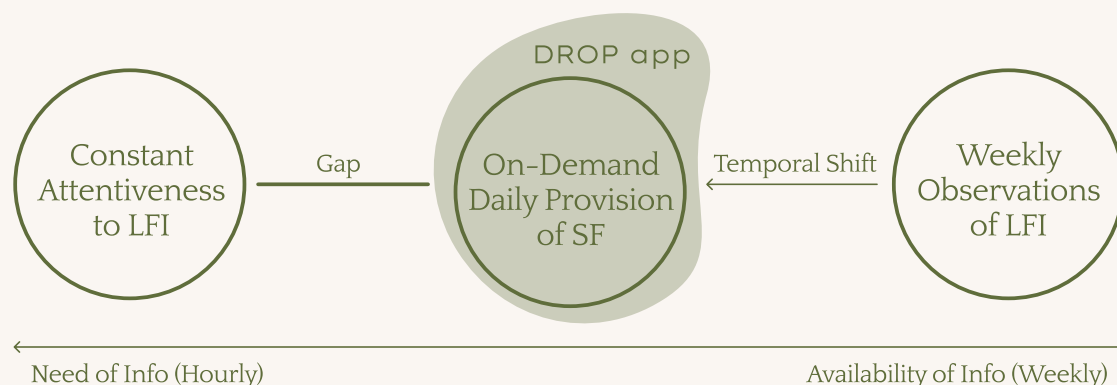


Figure 37: The DROP app bridges the gap between the actual presence of LFI for weather forecasting and the need for consistently available weather information

This indicates a gap between the actual presence of local indicators for weather forecasting and the need for consistently available weather information. The DROP app, with its scientific forecast, closes this gap by providing a daily weather forecast (Figure 37 – middle). In contrast to TV or radio, which similarly offer daily SF to farmers, the DROP app introduces a temporal shift through its constant availability and on-demand access to weather. Farmers describe checking the DROP app multiple times a day:

Sometimes the local indicators, it can happen, and you didn't see it. ... But the scientific forecast is different. If you go, it will give you information about the rain.

– Farmer during workshop

It is normally whenever they are to do some work in their household or maybe on their farm, that they check it in the morning and before they go out.

– Translator during interview 3 with female non-farmer and farmer

The practice of retrieving the forecast frequently within a day reflects the importance of being aware of spontaneous weather changes. As mentioned earlier, farmers constantly read their environment, because even quick weather changes carry direct relevance to their farming activities, decisions, and livelihood. Hence, farmers during the workshop expressed a wish for implementing an hourly forecast within the DROP app, in addition to the daily forecast, to better match their needs:

It may determine that today it will rain, but the time will not show. We want it hourly.

– Farmer during workshop

Further, these findings surface different perceptions between researchers and farmers on what counts as a forecast. For the research team, a forecast only qualifies as one if it is made sufficiently in advance. Thus, the DROP app limits LF inputs to 9 am the same day and does not allow for spontaneous prediction submissions. For farmers, however, observing clouds forming rapidly and estimating rain an hour away constitutes an actionable, relevant prediction. Thus, the desire of farmers for a greater temporal resolution illustrates a deeper asymmetry between their practice of forecasting and that of the research and development team. This example indicates that assumptions are implicitly embedded in the design of CS but need to be surfaced to ensure such tools reflect the realities of farmers.

Insight 2

The DROP app introduces a shift within the landscape of weather information channels toward a text-based, digital format, which creates new barriers to participation.

Farmers access weather information through traditional and technological sources. On one side, farmers apply their LK and additionally receive local forecast information from elders and rainmakers:

They are saying there is a rainmaker in the community and they do receive information from him.

– Translator during interview 4 with young and middle-aged male farmers

On the other side, farmers draw on different technologies, such as radio, TV, SMS, hotlines, and most recently apps like Facebook, Telegram, or the DROP app (Figure 38):

They receive weather information from radio stations, radio broadcasts, and TV stations. There was also an organization that came and registered their numbers. Sometimes they call them, like a hotline, and tell them on which dates it is going to rain.

– Translator during interview 5 with middle-aged female farmers

Radio, TV, Apps such as Drop app, Facebook, WhatsApp, Google, Utilize-WI Telegram Chatbot.

– Network Mapping activity during workshop

This suggests that the DROP app enters a landscape already characterized by a broad engagement with multiple information channels and growing access and interest in scientific forecasts from digital tools. One research team member notes:

They go to many other channels of communication because there are also other projects, other apps. And for example, I told you last time I went there to introduce the DROP app. Someone was saying, I'm using the DROP app but then showed me a different app.

– Professor 1 from the project team

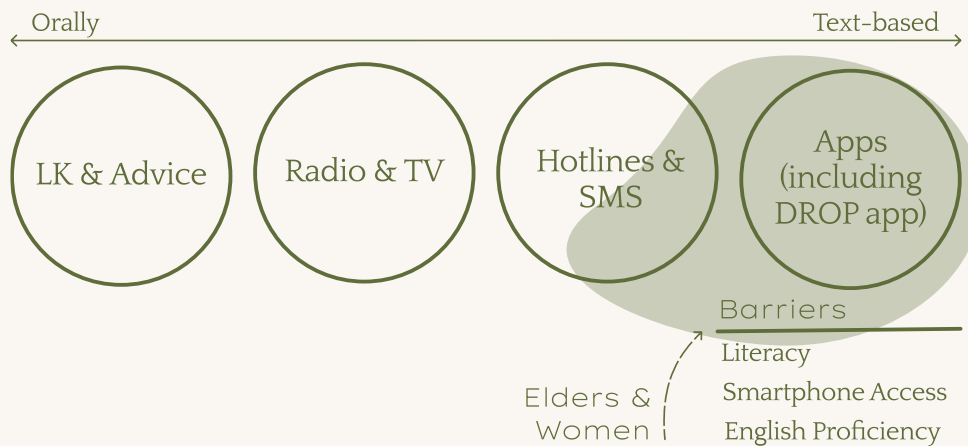


Figure 38: Overview of weather information sources ranging from oral (left) to text-based formats (right) and the barriers that digital tools introduce

However, the new tools that farmers gain, such as the DROP app, differ in their dissemination format of weather information. Most existing information sources, such as elder advice, radio or TV, operate orally (Figure 38 – left). In contrast, digital apps are text-based and require technological literacy, smartphone ownership, internet connectivity, and English language proficiency (Figure 38 – right). This shift mediates access, which the research team is aware of:

My interaction with the farmers shows that they are interested. They want to use the app, but then there are some constraints that we have also identified from the field. Many of them are illiterate, so they don't have the knowledge to use it. If the elder is illiterate and cannot read meaning into the DROP app, that becomes a constraint for the elder rather than a resistance. So the ability to use the app is the problem. Also, many of them don't have a smartphone.

– Professor 4 from the project team

This quote illustrates that not the resistance to technological development but rather the barriers to participation hinder the adoption of tools like the DROP app within farming communities. Most significantly, these barriers apply to a specific social group, namely, elders and women, who have less access to smartphones and lower levels of formal education. To resolve these barriers, farmers propose potential solutions, such as implementing voice input for local forecast submissions and voice output in Dagbani or SMS-based delivery:

They are suggesting making a WhatsApp version of the system. If DROP could also be made in that form, it would also help.

– Translator during interview 3 with female non-farmer and farmer

He is saying that if they added a place where you can press to get the output read out loud in audio and in Dagbani would be helpful. And also for the part where they have to input what they think about the situation, like how they do on WhatsApp, if they could press and say what they have to share and it would go into the app. He is also saying that if they could provide an option where you can receive the forecast through SMS from the DROP app for those who don't have smartphones, that would also be helpful.

– Translator during interview 4 with young and middle-aged male farmers

These features would essentially reestablish an oral transmission format and lower the threshold for accessing digital climate service tools. Overall, these findings illustrate that design decisions on how information is presented also determine who is excluded from the service itself. Ultimately, specifically for the DROP app, this shapes whose knowledge and local forecast predictions enter the system.

Insight 3

Access to the DROP app's scientific forecast creates new social positions within existing forecasting networks, especially for the youth and women.

Traditionally, rainmakers and elders are recognized as trusted experts in local weather forecasting and consulted by community members with lesser local knowledge, particularly by women and young farmers, for advice:

Elders advise other people in the community, but mainly women, because they have less experience.

– Translator during interview 2 with middle- to old-aged male farmers

This reflects a social structure of forecast authority, defined by experience and age. However, with climate change impacting local forecast accuracy and with the advent of new technologies, including the DROP app, the dynamic is shifting. Rainmakers, experienced elders, farmers, and other community members increasingly consult smartphone users for scientific forecast information (Figure 39 – right):

He said that he himself doesn't use it, but he asks other people. He gets information from them, but he doesn't know which apps they are using.

– Translator during interview 6 with the rainmaker

He receives DROP app information from somebody in the community. He normally goes to that person to ask about weather information.

– Translator during interview 2 with middle- to old-aged male farmers

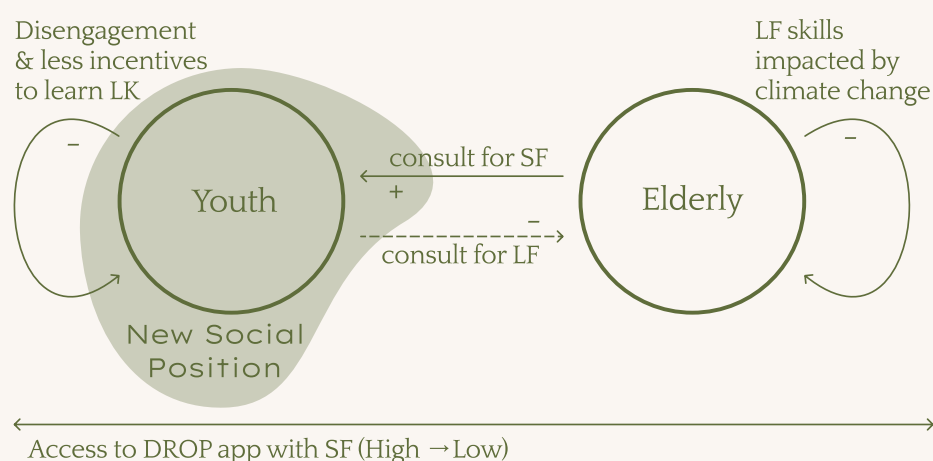


Figure 39: System map illustrating shifts in intergenerational forecasting knowledge exchange

Due to access barriers, smartphone users largely represent young, educated, male farmers. Findings suggest that they inhabit a new social position within the forecasting networks

(Figure 39 – left). This creates a layer of social capital, which is built upon digital literacy rather than environmental experience. The research team describes this emerging *intergenerational dynamic* as a bidirectional exchange, where elders contribute local expertise and younger generations contribute technological skills:

Young people see the scientific forecast and explain it to the oldest, and the oldest pass on the local knowledge.

– Professor 1 from the project team

However, the reality of this bi-directional exchange in practice and its continuation in the future is contested. Participants name a *generational disengagement from LK*, in which the younger generation is described as less interested in learning local indicator-based forecasting practices because the availability of SF through digital apps reduces the incentives to develop LF skills (Figure 39 – top left):

This man is saying that for the coming years, he sees that local forecasting knowledge will become lesser in the system, because of the lack of interest from the youth. ... The youth prefer applications over traditional knowledge. ... Even if they could have ten of these applications, they would prefer to have all of them. ... He does not feel happy about the youth abandoning the local knowledge. He would have preferred them to use both. Because now, it is kind of like the knowledge is being lost, ... because the youth are not paying attention to learning it.

– Translator during interview 4 with middle-aged male farmer

The research team expresses the same concern regarding this generational shift:

I'm afraid when they depend on it and in the end they also forget about their local indigenous practices. Especially the young generations, you know, because the young generation is always very open to technology, always likes to use the phone more often. ... in the future, when all the farmers are like the young generation, then it might happen.

– Professor 2 from the project team

However, during the interview, a young farmer disagreed with this framing. Rather than the lack of interest, he notes that applying LK represents a skill that needs time to learn and one he will acquire at a later age:

This other man is saying that for them, they feel like they haven't yet reached the right age. So it is kind of like, he says, they have to get to a certain age before they also start using the traditional system.

– Translator during interview 4 with young male farmers

The findings do not show a clear outlook for the future of LK practice and transmission. Nevertheless, this intergenerational dynamic, defined by shifts in social positions, significantly impacts whether local knowledge practices will continue to be learned. This applies not only to LFI but also to traditional ceremonial knowledge practices.

Apart from intergenerational dynamics, the reorganization of social positions within forecasting networks has an equally relevant gender dimension, which acts in two directions. On the one hand, women remain largely excluded from participating in forecasting practices (Figure 40 – middle). They hold different LK, feel less confident in their own LK and further have lower smartphone ownership compared to men. Thereby, existing inequalities are reproduced.

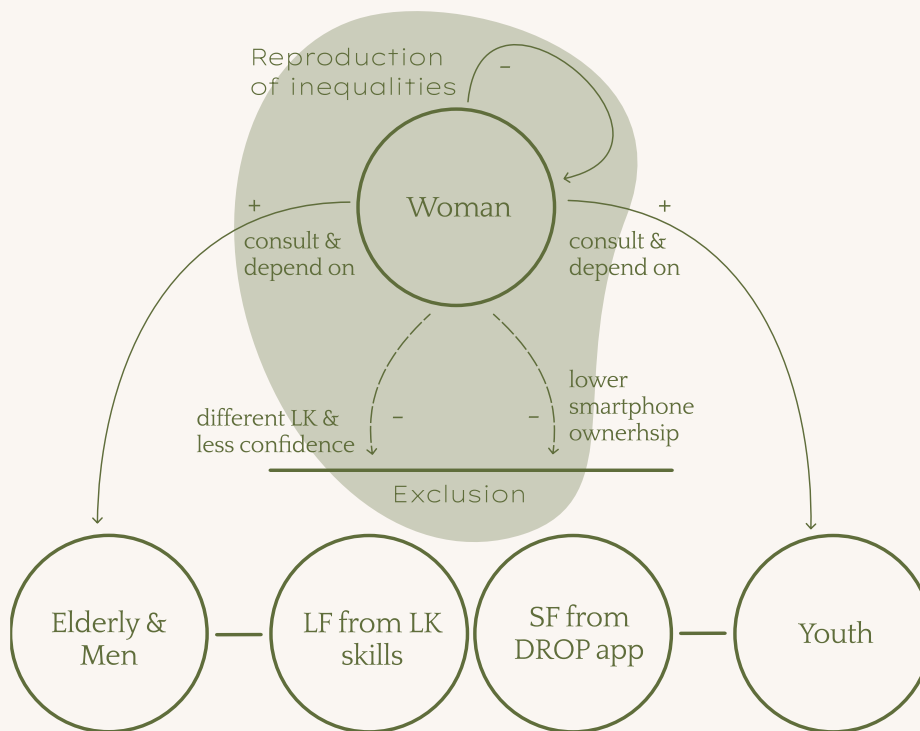


Figure 40: System map illustrating the exclusion of women from participating in forecasting practices

Women depend on other community members for receiving forecast information (Figure 40), both local and scientific information, and describe using their own LK merely as a fallback:

For the women, those who have smartphones are very few. It is mainly the men who have them. ... The women ask those who have the apps for the weather information. ... Sometimes, when the people they normally ask for information are not around, because maybe they have gone out early in the morning, they also use the knowledge they have. So their local knowledge is used mainly when they don't have anyone to ask.

– Translator during interview 5 with middle-aged female farmers

However, on the other side, in cases where women do have access, roles shift. Women become active sharers of forecast information within their household and community. Similarly to the youth, women start to inhabit a new social position as respected knowledge holders (Figure 41).

For the household, even some of the men living within their household ask her about the weather information every day, and she will tell them. ... Because some community members knew she was part of the training when the DROP app was being introduced, they would come and ask her about the information on the app.

– Translator during interview 3 with female non-farmer and farmer

There are patriarchal families where the man is the first in the hierarchy at home. But now you see women who have access to technology that they could bring this new information to the household.

– Professor 1 from the project team

Overall, these dynamics portray the DROP app as a social intervention that influences relationships between community members. The effects on social and cultural structures,

introduced by climate service tools, may not be immediately visible, but are central to understanding the impact and future implications for communities.

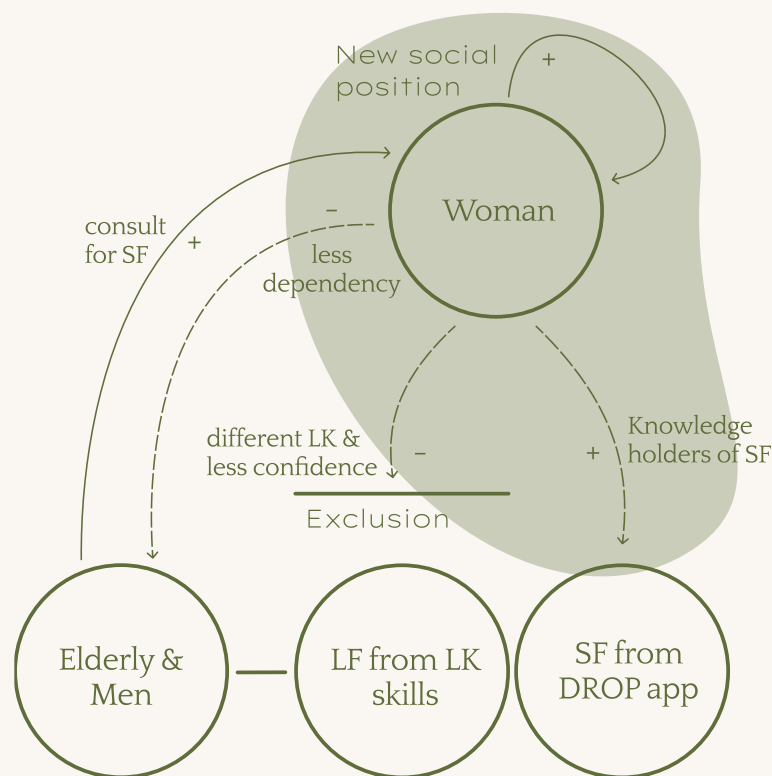


Figure 41: System map illustrating women with access to the DROP app as knowledge holders of the SF

Insight 4

Growing interest and trust in the SF represent a pragmatic strategy to minimize financial loss under the impact of climate change.

Farmers consistently indicated growing interest, use, and trust in the SF from the DROP app. One reason stems from the continuous availability of scientific weather information (Figure 42 – left):

They rely more on the scientific forecast than the local one. With the traditional system, you have to see a sign before you have a prediction. With the scientific forecast, every day it gives you some information.

– Translator during interview 2 with middle- to old-aged male farmers

A second reason is the perceived higher accuracy compared to other scientific forecast services, such as TV (Figure 42 – middle):

I mostly use the DROP app because I trust this one. But the other apps, it can give information, but it won't work. Even when the climate is changing, the DROP app knows there is no rain again. Also TV is not so reliable.

– Farmer during workshop

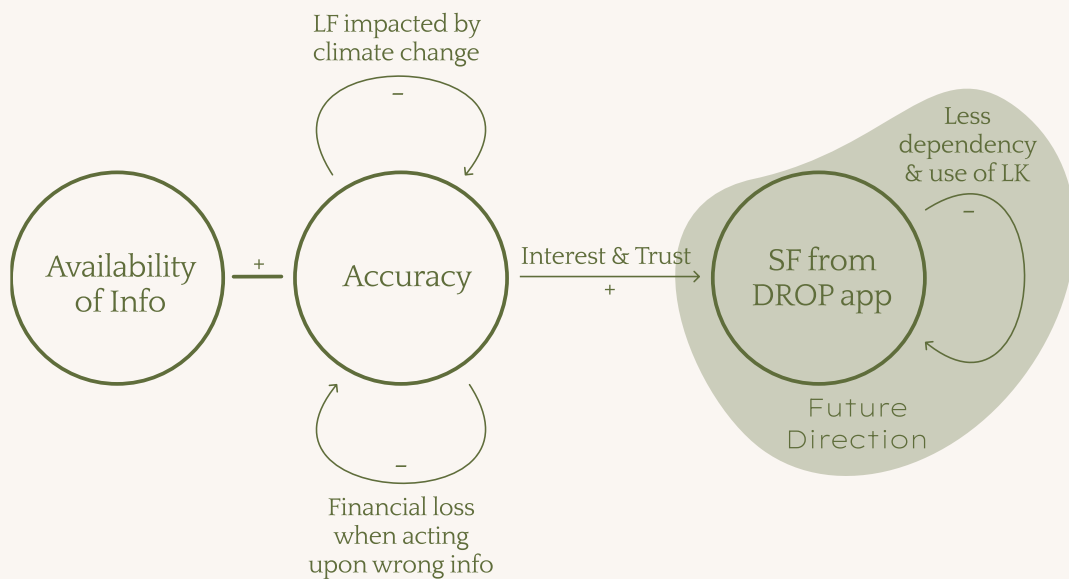


Figure 42: System map illustrating the dynamics behind the growing interest and trust in the SF

The main driver, however, links to the decrease in accuracy of the local knowledge forecast under climate change (Figure 42 – top). All farmers described a future in which the dependency on LFI continues to decline, because of increasing environmental pressures and unpredictable weather patterns (Figure 42 – right):

He is saying that based on his experience, in the coming years, the scientific forecast is going to overtake their local tradition. Because of the changes in the weather and climate change. He is saying that even just recently, he could tell that a particular date in the coming month was going to have rain. He used to be able to do that, and it would happen. ... But right now it is not like that anymore. Sometimes they can even see the signs and it will not happen.

– Translator during interview 6 with the rainmaker

They are saying that their dependence on local knowledge is something they see becoming a thing of the past in the coming years, because of climate change. Looking at how temperatures have become higher and how the seasons have changed, it is kind of affecting the accuracy of the traditional knowledge forecast. So they are saying that in the coming years, it is going to be of no use, or maybe of lesser impact.

– Translator during interview 5 with middle-aged female farmers

Even elders and rainmakers, the traditional knowledge holders, have begun integrating SF into their own practice, which signals the future direction of forecasting:

Even he himself sometimes benefits from it. The scientific forecast is able to tell which particular side of the country the rain is going to fall on.

– Translator during interview 6 with the rainmaker

Even the elders and the rainmaker prefer scientific knowledge over the signs they were using. Some of them still use both. They use TV stations where the Ghana Meteorological Agency normally does weather forecasts on television. Sometimes when they see it on television, and it says it will rain, they will also wait. They won't go to their farms. It is kind of like they now also trust or have more trust in scientific forecasts than in the local system.

– Translator during interview 1 with a young-middle-aged male farmer

The consequences of acting upon wrong forecast information impact farmers' livelihoods substantially and lead to material consequences and financial losses (Figure 42 – bottom):

They do rainfed agriculture. ... So, meaning if a farmer misses the ability to predict the beginning of the season. ... That means he's going to lose some money, because if you cultivate and it never rains again,

the seed may rot in the ground, and then he has to buy seeds again. So there is a lot of financial loss, and sometimes the farmer may not be able to even buy seeds again.

- Former project team member 1

So they had mispredicted. They planted their seeds, and there was not enough moisture, so germination was very poor. Some crops did not germinate at all. So they had to plough again to resow. A lot of farmers were there, and they also saw what was happening. So it triggered them: next time, I need to have accurate information before I act. There was a lot of financial loss.

- Farmer during workshop

In essence, the growing interest and trust in the SF can be seen as pragmatic and outcome-based rather than an ideological shift. Farmers turn to the information source that best supports their survival. In times of climate change and decreasing accuracy of their LK, technological adoption represents an appreciated resource to navigate uncertainty, as one workshop participant explained:

As a farmer, rain is everything to us. When the season comes, every farmer is expecting to hear information about that. So any app, any DROP app you use that can indicate forecasting information, when you are gathered at a meeting like this and discussing that issue with farmer groups, everybody is listening.

- Farmer during workshop

Nonetheless, LK remains practiced and actively compared against the scientific forecast (Figure 43 – top), because of its predictive value, but also its embeddedness in the social and cultural life of communities. Farmers navigate different forecast sources and need to make decisions when they contradict each other. This decision-making process includes strategies such as asking others or picking the source that predicts rain:

Sometimes she will look at the scientific aspects, and sometimes she will look at the local side. When there is a contradiction, she would choose the one that shows rain. But she normally depends on the scientific one.

- Translator during interview 3 with female non-farmer and farmer

If there is a difference, you need to sit down and do more research, maybe ask others, so that you can know where the discrepancy is coming from. ...So we just compare everything we know.

- Farmer during workshop

In the future, the research team plans to implement a hybrid forecast that directly combines the local with the scientific forecast (Figure 43 – bottom middle). Within this hybrid model, the local forecast functions as a means to improve the accuracy and localization of the scientific one:

The hybrid result actually predicts the weather better than if you only had maybe satellite information, and also better if you only had your local knowledge. So we have realized that the integration of the two is actually going to give you better accuracy and that is where we were working on. ... The advantage is that it will give you very local information for the community.

- Professor 3 from the project team

Ultimately, the quality of local forecast inputs remains vital for the accuracy of the hybrid forecast. The issue of accuracy does not merely represent a technical challenge of implementing improved algorithms. Rather, it requires addressing the challenges presented in previous insights, such as the gap between knowledge holders and the app users, barriers to participation, the low number of LF submissions, or the possible generational disengagement from learning LFI, to create the conditions for LK to enter the DROP app. Overall, the findings point towards the importance of applying a systemic perspective to

understand the interconnectedness of technical design, social dynamics, and cultural practices, in order to support the adoption and improvements of climate services.

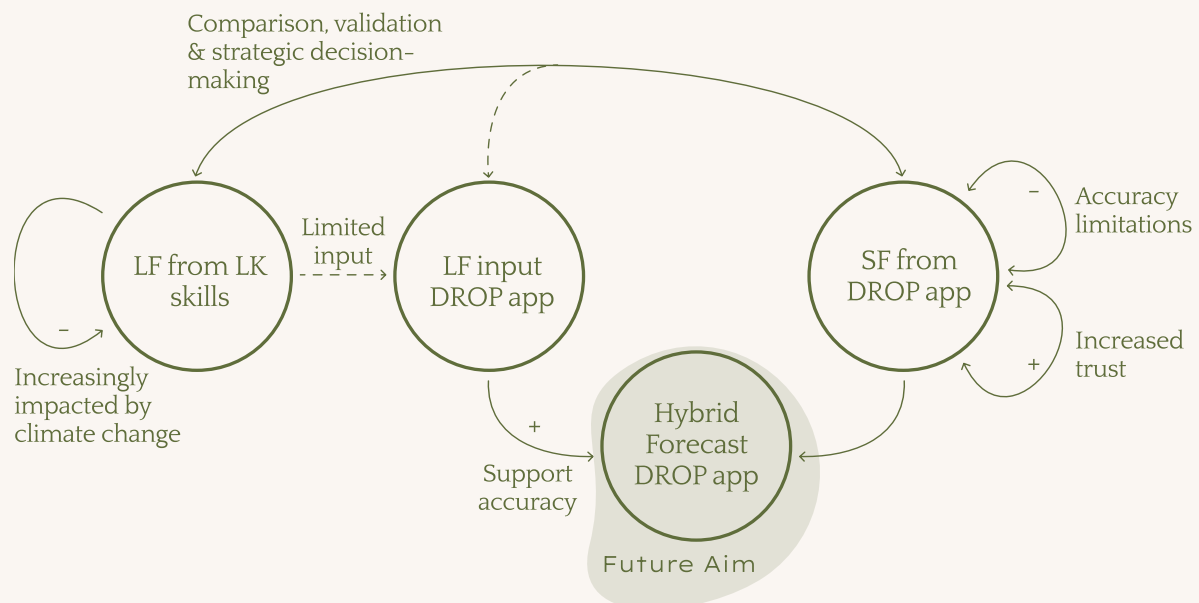


Figure 43: System map illustrating the decision-making process of farmers between the LF and the SF, and the future aim of the research team to implement a hybrid forecast

Insight 5

Strong informal community networks carry weather information, primarily designed for farmers, to community members with a wide range of other occupations and needs.

The primary goal of the DROP app is to support farmers in decision-making by providing reliable weather information (Figure 44 – left). However, farmers during the workshop mentioned that they share the forecast with a wide range of community members, such as traders, shea nut pickers, teachers, or pastoralists (Figure 44 – right):

I will share the weather information with farmers, traders, shea pickers, and teachers. So that they will know how to plan their time in advance. Teachers, when they go to school, they can better plan to arrive at the school without being beaten by rain. – Farmer during workshop

It is not only for farming activities. If someone wants to travel, people don't want to get sick, so they normally ask about weather information. – Translator during interview 4 with young and middle-aged male farmers

Additionally, women adopted the forecast information for a wide range of household and market-related:

They also use it to advise themselves on domestic household work and their responsibilities in the household, and also when they are going to the market. As she mentioned, for example, when they are to dry flour, the millet, they dry it under the sun. – Translator during interview 3 with female non-farmer and farmer

Sister: uses for cooking and washing. My wife: because she has to dry things. – Network Mapping activity during workshop

The quotes indicate that forecast information supports a variety of activities based on profession and role. A wide variety of decisions in daily life are weather-dependent and impacted by changing weather patterns in times of climate change. This sharing of information is only possible because of strong informal communication networks (Figure 44 – bottom). Researchers observed a high degree of discussion on the DROP app and weather forecasts, even beyond communities:

They tell us that they are not using it only for themselves. ... They share information even with people who are not in their communities. But most of these exchanges are on an individual level, informal passing of information from one person to another.

– Professor 4 from the project team

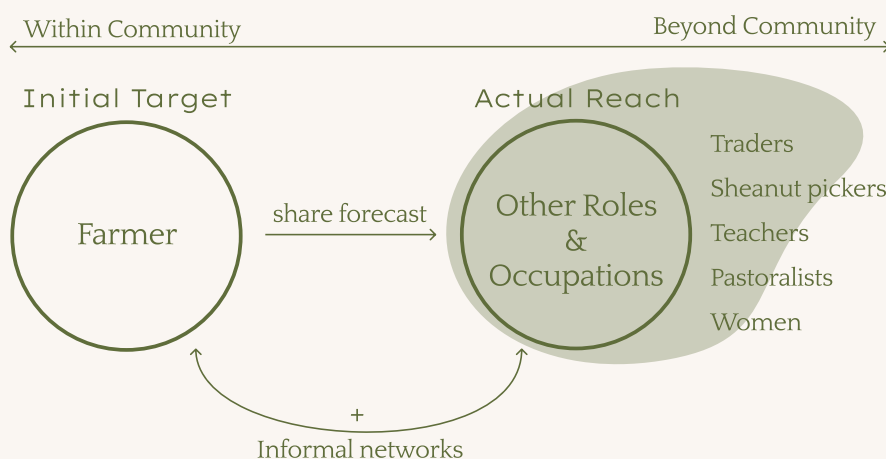


Figure 44: System map illustration initial target (left) and actual reach (right) of weather information from the DROP app

These findings suggest that the scale and reach of the DROP app exceed the initial anticipation. Most importantly, it illustrates that the value of a service is always co-created between actors. Rather than understanding users as merely passive recipients, they represent active agents who generate value for themselves and others through their use and adaptation of a service to their needs. Thereby, they influence the actual impact of a service in practice. This points towards a relevant implication for the evaluation of climate services. For instance, measuring impact by tracking forecast accuracy and farming outcomes alone may not capture the full reach of a service. It requires rethinking how to measure impact to account for the broader, often unanticipated outcomes of a service beyond the intended objective.

6 Discussion on Implications

This thesis is situated in the intersection between climate services and service design, using the DROP app as a practical case. The findings from the previous chapter open up discussions on implications for the practice of these fields. Therefore, this chapter contextualizes those findings across three dimensions: implications for the design of the DROP app, implications for the practice of climate services, and implications for the practice of service design. Finally, this chapter presents five design principles for the future development of climate services tools that aim to integrate LK.

6.1 Design of the DROP app

The design implications for the DROP app summarize observations from the research process rather than function as final recommendations. Further assessment and evaluation of their feasibility and potential impact would be needed, but remain out of scope for this thesis. Nevertheless, they represent starting points for future design iterations of the DROP app. The design implications span across four areas: technical functionalities, UI/UX design, LK features, and design speculations (Figure 45).

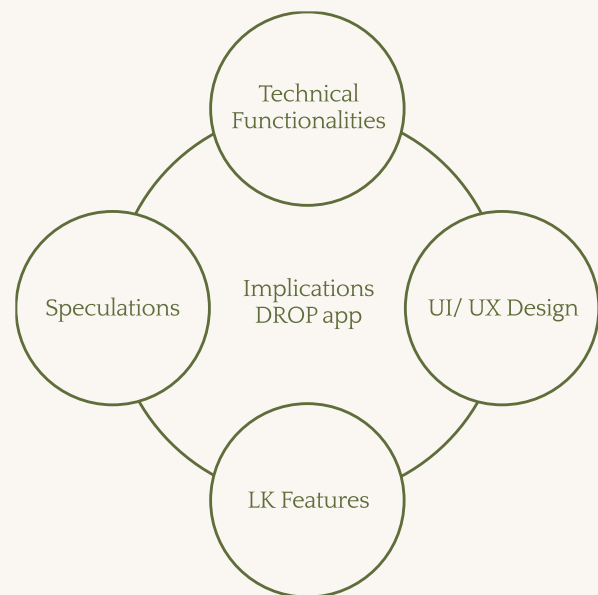


Figure 45: The four areas of implications for the design of the DROP app

Technical Functionalities

Implications for the technical functionalities refer to the app's underlying infrastructure and capabilities. The following opportunities emerged from the interactions with farmers:

Hourly Forecast: Farmers expressed a need for an hourly weather forecast as weather matters at any hour of their day.

WhatsApp or SMS Version: To increase accessibility of DROP app services, farmers proposed switching to SMS-based delivery for those without smartphones or implementing a WhatsApp version for easier navigation.

Audio-Input-Output: Farmers suggested an audio feature, which supports verbal submission of the local forecast observations and audio delivery of the scientific forecast.

Local Language: Farmers wished for the app to operate in their local language, because language remains a central barrier for the adoption of the app.

Forecast Accuracy: Farmers reported low accuracy, particularly for the low rainfall forecast in the DROP app.

Push Notifications: Fieldwork indicated that the actual LF input in the DROP app appears low. Push notifications could serve as a tool to communicate the value of LF submissions and, at the same time, remind farmers to submit a LF.

UI/UX Design

Implications for the UI/UX design concern the interaction with the interface of the DROP app. Overall, farmers expressed the wish for a more graphical and simpler experience. While I did not perform a full UI/UX audit, the service safari surfaced some currently existing friction points within the interface design:

Labelling and Terminology: The home screen offers multiple actions, but the distinction between options could be clearer labelled, especially between “Share Forecast” and “Share Observation”. Further, the forecast graphics are presented as a circular radar visualization. During the service safari, the forecast state appeared largely black with only a white “0” at the center. However, black is not included in the legend. It may seem unclear whether this indicates no rain, missing data, or an inactive status.

Communication of Constraints: Local forecast predictions for the same day need to be submitted before 9 am. This temporal constraint remains invisible to users. When encountered, they may assume an error. Therefore, it is essential to clearly communicate this limitation.

Visual Design and Consistency: Across screens, inconsistencies in the visual language exist. For example, the UI is not consistently presented in dark mode, or the rain icon used in the forecast legend differs from the one on the “Share Forecast” screen. Further, the scientific forecast displays 1-day, 7-day, and 14-day views. The 1-day view applies a circular radar

visualization along with a colored and iconized bar legend, while the 7-day and 14-day views present horizontal bar charts with a probability percentage axis. As a result, users need to reorient themselves across views. All forecast information could be displayed in a similar manner to support understanding. Further, other approaches to illustrate information in less technical formats may be relevant to explore.

Simplification of User Flow: The 'Submit' button for the LF appears on a separate screen. This adds an unnecessary step. Moving it to the main screen would shorten the flow. Additionally, fieldwork indicated that the screen for the account creation presented a major friction point and required external guidance. Therefore, simplifying this step may lower the entry barrier for new users.

LK Features

Implications for LK features relate to practical elements that may enrich current LK representation in the DROP app. The research revealed some of these opportunities:

Visual Indicators: During the fieldwork, farmers consistently responded positively to the visual indicator mock-up. The implementation of this layer may generate an increase in LK inputs as the submission flow would appear more graphical, intuitive, and tangible.

Visual Consistency and Localization of LFI: In the current mock-up, some indicators are presented as flat black icons, such as ants or clouds, while others are colorful and illustrative, such as the bird or dew-covered leaf. This inconsistent visual style carries meaning beyond the lack of coherence or aesthetics, as it communicates how LK is perceived and influences how credible and intentional its inclusion appears in the app. Further, the visual representation of LFI in the DROP app mock-up illustrates no specificity in terms of locally specific species of animals. Despite workshop participants mentioning that they rely on a distinct type of ant for weather prediction, they were largely accepting the presented visual representation in the DROP app mock-up. Nevertheless, adapting visual representation to local realities would enrich the LK component.

Behavioral Depth and Subcategories of LFI: Only a one-word label is presented as a description for the indicators. Therefore, LFI lack behavioral details and nuances that carry the specific forecasting meaning. Adding subcategories may introduce more detail or granularity regarding the depth and variety within broad categories of LFI.

LF Scope: The current DROP app version limits LF inputs to today or tomorrow. However, certain indicators predict the weather multiple days in advance. Extending the range may better reflect the actual scope of LFI.

Transparency and Quality of LF: Farmers connect different LFI with different reliabilities. Therefore, adding a feature where farmers can select their certainty level may increase the accuracy of aggregated LF. Further, the inclusion of a feature that allows farmers to edit or submit multiple inputs may support data accuracy and user autonomy. To add an additional layer of transparency, the DROP app could indicate the number of LF submissions that underpin the aggregated LF.

Design Speculations

Design speculations reflect on the deeper dynamics and underlying tensions encountered throughout the research process. They represent provocations that question the status quo and address certain points in the system rather than proposing solutions. Speculations aim to open new directions that may be further explored through experimentation. The following design speculations rethink and re-imagine the role that apps, such as the DROP app, could play for local communities in the future. They are grouped into four categories:

1 - Sustaining LK documentation and transmission

What if the DROP app became a learning tool for LK? – A present tension concerns the unevenly held local knowledge across generations and the uncertain continuity of LK transmission. While elders accumulated extensive LK expertise, they regard youth as disconnected from the practice because technologies lower the incentives to learn LK skills. Currently, the DROP app targets users who already have local forecast expertise. Reframing climate service apps as a learning tool could respond to this intergenerational dynamic. For example, the integration of short explanations, contextual guidance, and specific information on how to read the environment to arrive at the forecast within the app interface in the form of audio, video, image, or text may increase youth participation, support the development of forecasting skills, sustain LK transmission and increase the reach of the service.

What if the DROP app became a tool that supports the collection of LK? – A current tension reflects the highly contextual, localized, and nuanced nature of LK, which complicates the collection and therefore impacts the representation of LK within the DROP app. Currently, researchers first document LFI and subsequently translate them into the DROP app. Reimagining climate service apps as tools for the collection and documentation of LK could offer a new perspective on this process, but would require a new, flexible, and open submission format. For instance, rather than selecting from fixed categories, farmers could describe their observations and detailed interpretations through voice recordings in their own language. This format connects back to another tension: while knowledge transmission is traditionally oral, the DROP app mock-up presents LK textually and visually. Therefore, an

audio-based input would better reflect traditional transmission. Over time, climate services tools may build a shared knowledge base and evolve into a living LK archive, which is directly produced by the community itself.

2 - Expanding the scope

What if the DROP app became a tool that extends prediction purposes beyond daily rainfall? – Apart from daily rainfall prediction, farmers apply indicators to forecast wind, storm droughts, and seasonal patterns. Currently, the DROP app only focuses on rainfall indicators. Extending this forecasting scope to a wider variety of indicators could better reflect the actual breadth of LK. Further, it would signal recognition for these other forecasting practices, possibly strengthen their use, and support the documentation of a broader range of LFI.

What if the DROP app became a real-time community warning and sharing platform for LF? – The weather matters to farmers at any time of the day. They constantly look out for signs, as even a one-hour window remains relevant for determining farming decisions. However, the DROP app imposes a temporal constraint that limits same-day forecast inputs to before 9 am. This constraint is rooted in the understanding that same-day observations do not constitute a prediction. Rethinking this perspective could transform climate service apps into a dynamic, real-time, community-driven warning and sharing platform. For instance, by allowing community members to share spontaneous observations, they could quickly warn each other of incoming rain and collectively adapt to the changing weather.

3 - Creating transparency and credibility

What if the DROP app makes knowledge holders visible? – Elders and rainmakers are recognized as LK experts and, therefore, actively consulted and trusted. Rather than reflecting these hierarchical structures, the DROP app aggregates all forecast inputs equally into a single LF. Individual contributions remain invisible. One approach to rethinking climate service apps in order to reflect existing knowledge structures of communities involves making knowledge holders visible within the app's interface. For instance, during the profile creation or the submission flow, community members could indicate their experience level, and this information would later be included in the aggregated forecast. This may create transparency, credibility of LF, and connect back to the social meaning that community members attach to forecasting expertise.

4 - Challenging epistemological dominance

What if the DROP app became a tool that reflects the ceremonial and cultural dimension of LK? – Apart from local indicators-based forecasting, communities, especially rainmakers,

employ ceremonial and ritual practices as part of their forecasting tradition. This dimension remains excluded in the DROP app. Reimagining climate service apps to reflect the ceremonial and ritual dimension of LK may better reflect the diversity of LK, signal recognition for these practices, and most importantly, may contribute to long-term cultural sustainability.

6.2 Practice of Climate Services

Observations from the DROP app use case, as a practical example for the integration of LK, directly inform the broader field of climate services. These design implications for the practice of CS span across four areas: decoding LK, reflective and iterative learning, interdisciplinarity, and measuring impact (Figure 46):

Decoding LK

In recent years, the emphasis on LK integration within the development of climate services for climate adaptation has grown (Nyadzi et al., 2021a; Warner et al., 2022; Kom et al., 2022; Filho et al., 2022a; Makate, 2020).

However, LK seems to be treated as a singular element and its integration as a straightforward, simple process. The use case of the DROP app challenges this assumption and illustrates that LK contains many dimensions, is highly localized, nuanced, culturally embodied, and applied for different purposes. The research showed that not all dimensions of LK are reflected in the DROP app. For instance, only a limited range of indicators for rainfall prediction are represented in the DROP app, while community members apply LFI indicators for wind, storm, drought, or seasonal prediction (see Insight 3 – Chapter 5.1 Research Question One), and further draw on ceremonial and calendrical local forecasting knowledge (see Insight 4 – Chapter 5.1 Research Question One). Additionally, LK is held unevenly across roles, gender, and generations and transmitted orally and through practice. For instance, women hold different knowledge than men, and elders are seen as knowledge holders with vast expertise (see Insights 5 and 6 – Chapter 5.1 Research Question One). As a result, the integration of LK within CS requires transparency about which dimension of LK is

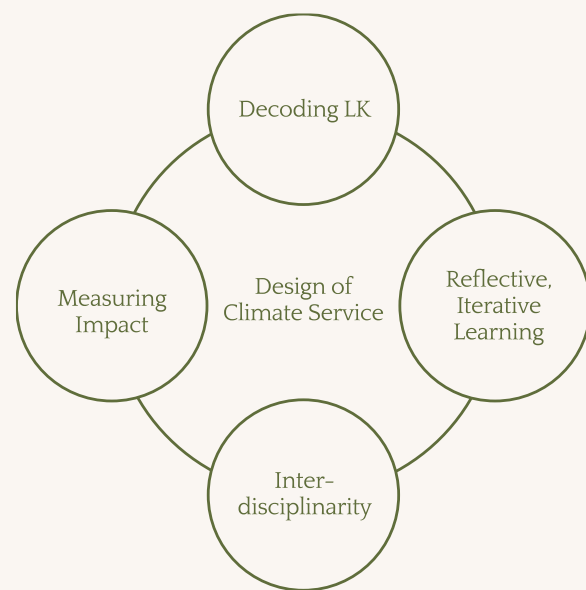


Figure 46: The four areas of implications for the practice of climate services

represented, whose LK is included and, subsequently, what aspects remain outside of the scope. While arguing for the integration remains a meaningful starting point for the future practice of CS, we simultaneously need to decode LK and be aware of all these facets. Therefore, the case of the DROP app invites practitioners to reflect more rigorously on the meaning of LK integration. Future climate service design should move beyond broad theoretical claims and toward a precise articulation of the specific dimensions, manner, and extent to which LK is addressed in a specific CS.

Reflective, Iterative Learning

Despite the growing emphasis on LK integration, practical examples within digital climate services remain limited (Filho et al., 2025; Gbangou, 2021; Nyadzi et al., 2021a). The DROP app represents one of possibly many approaches for the inclusion of LK, as one member from the research team expressed: “We are still at the very beginning.” In fact, the implementation process involves various challenges that the broader field of CS needs to address, such as the involvement of power dynamics, assumptions regarding different knowledge systems, or uncertainties of how to translate LK into CS. As findings illustrate, simplification is unavoidable when nuanced LK is integrated into digital climate services (see Insight 2 – Chapter 5.1 Research Question One). Navigating what aspects to simplify and what meanings to preserve demands ongoing negotiation and a reflective, iterative practice. Some tensions only surface after the implementation of a certain feature or structure. For example, findings illustrate that the DROP app’s current indirect representation of LK detaches LK from the context and source that give it meaning and credibility and, thus, requires further iteration (see Insight 1 – Chapter 5.1 Research Question One). Additionally, experimenting with alternative dissemination formats within the DROP app through voice inputs and outputs, as proposed by farmers, could lower current barriers that hinder participation (see Insight 6 – Chapter 5.1 Research Question One). Similarly, exploring editable interfaces could better capture the contextual, changing nature of LK (see Insight 5 – Chapter 5.1 Research Question One). Therefore, revisiting past design decisions and iterating on them, with transparency and in collaboration with users and non-users, constitutes a fundamental part of the process. The future practice of CS should explore, develop, test, and iterate more practical approaches for integrating LK. Most importantly, it should share and reflect on these observations and experiences with other practitioners in the field to collectively learn from the process.

Interdisciplinarity

Climate services as a discipline originates primarily from technical research fields. However, the DROP app use case illustrates that emerging challenges within LK integration, such as the

gap between knowledge holders and the app users (see Insight 6 – Chapter 5.1 Research Question One), the barriers to participation (see Insight 3 – Chapter 5.2 Research Question Two), the possible generational disengagement from learning LFI (see Insight 3 – Chapter 5.2 Research Question Two) or epistemological tensions (see Insight 4 – Chapter 5.1 Research Question One), are not primarily technical in nature. In fact, they represent social and design challenges and therefore require corresponding expertise. While scholars (Chiputwa, 2020; Nyadzi et al., 2022; Filho et al., 2025) from the field of CS increasingly recognize the need for new approaches and methodologies, including the co-creation with local communities and stakeholders, this commitment must deepen in practice. Co-Creation seems to be primarily understood as user feedback, but at its core, it refers to intensive collaborative engagements before, throughout, and after the research and design process (Hyysalo & Johnson, 2024; Sanders & Stappers, 2008; Schuler & Namioka, 1993). CS development can benefit from engaging with these approaches that design disciplines have long established and refined. Therefore, the thesis argues for the involvement of (service) designers in the CS research and development process. Their education and skill set are based on participatory methods, systems thinking, and the facilitation of interdisciplinary collaboration. Further, the application of creative and speculative approaches to problem-solving opens new directions for the role that climate services, such as the DROP app, could play for local communities in the future (see design speculations – Chapter 6.1 Design of the DROP app). As discussed earlier, this could involve rethinking the DROP app as a learning tool or reimagining it as a LK collection platform to address the tensions that this thesis encountered during the research process. Beyond design, CS may benefit from social science perspectives to study the social and cultural impacts these tools may introduce on communities. For instance, the thesis only briefly touched upon gender dynamics or intergenerational shifts (see Insight 3 – Chapter 5.2 Research Question Two). Nevertheless, they impact the uptake of CS and therefore remain relevant to explore further. Consequently, the future practice of climate services should integrate technical, design, and social science expertise to achieve successful co-creation and implementation of CS.

Measuring Impact

The use case of the DROP app illustrates that the impact of a service in the field of CS is predominantly evaluated with technical measures of forecast accuracy relative to actual weather outcomes or observable changes in farming practices. These direct and anticipated effects correspond to what Swierstra (2015) coins as hard impacts. However, Swierstra (2015) points out that technological interventions also produce soft impacts, which refer to the social and cultural shifts in norms, meaning, and values. The findings of this thesis indicate that climate service tools create precisely these kinds of effects. They influence relationships and dynamics between community members, such as the social positions of the youth or

women within community forecasting networks or intergenerational processes around LK transmission (see Insight 3 – Chapter 5.2 Research Question Two). Further, the scale and reach of a specific CS may exceed the initial target group. In the case of the DROP app, strong informal community networks carried weather information, primarily designed for farmers, to individuals with a wide range of other occupations and needs (see Insight 5 – Chapter 5.2 Research Question Two). Evaluating services in context and in use by understanding their value as co-created between actors helps to surface these soft impacts. The thesis argues for the rethinking of conventional evaluation frameworks for CS to capture a holistic understanding of impact that accounts for broader long-term socio-technical effects and transformations initiated through design outcomes. This involves rethinking goals as well. Instead of designing an accurate information service, the focus could shift to designing an inclusive service by addressing the issue of participation barriers and thus measuring impact by, for instance, the number of people from generation 50+ who engage in the service.

6.3 Practice of Service Design

As mentioned earlier, service design practice could contribute valuable human-centered, participatory, and system-oriented skills to the field of CS. At the same time, however, service designers may face challenges and limitations within their current practice when applying their skills to this space. The following implications for the broader field of service design largely reflect my own observations and experiences as an aspiring service designer working with a digital climate service tool, the DROP app, for this thesis. They span across four areas: frameworks, systems mapping, intercultural collaboration, and reflexive methodological practice (Figure 47):

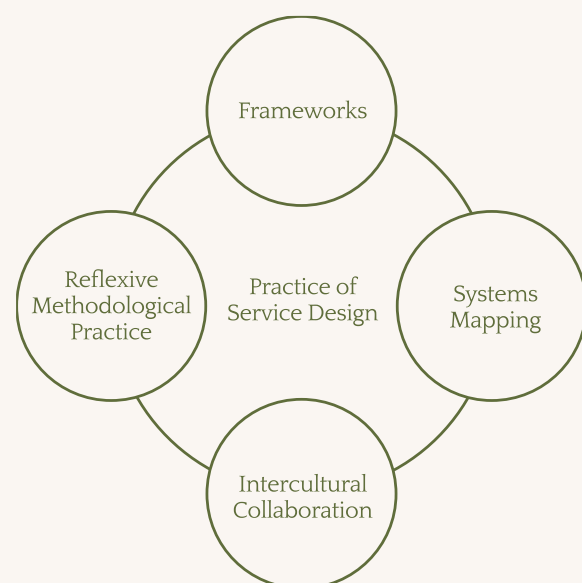


Figure 47: The four areas of implications for the practice of service design

Frameworks

Existing service design frameworks, especially the MSD framework (Patrício et al., 2011) applied in this thesis, provide a structural lens for examining complex service systems. The MSD framework offers different levels of abstraction as entry points for the research and

design of services and proved valuable in this thesis for the framing of the research questions. However, reflection on this framework reveals the relatively conventional service design contexts on which it is based. The MSD framework conceptualizes actors within service systems primarily as neutral and equal participants in a value co-creation process. Hence, it lacks consideration of power dynamics and structural imbalances between involved stakeholders. These power dynamics in this research became evident when researchers from the SAFE4ALL team mentioned that community members may provide pleasing answers (see Chapter 4.1.5 Community-Based Fieldwork through Narrative Interviewing) or during the participatory workshop, where surfacing critical feedback from participants at times proved challenging (see Chapter 4.1.4 Participatory Workshop). Especially in the context of CS development between European institutions and smallholder farming communities, these issues require critical awareness and reflection. Similarly, the MSD framework does not account for the complexities of designing within different knowledge systems. For instance, the DROP app case illustrates that design decisions always reflect whose epistemologies are prioritized, as seen, for example, in the exclusion of ceremonial knowledge practices within the app (see Insights 4 – Chapter 5.1 Research Question One). These choices may differ from the perspectives of the intended user and, therefore, demand careful navigation and new approaches to make different ways of knowing visible. Future service design should rethink service design frameworks to address current limitations and develop tools or concepts that resolve them.

Systems Mapping

Service design practice engages with systems thinking to understand complex dynamics (Jones & Kijima, 2018; Vink et al., 2021) and visualizes these dynamics through system maps (Jones & Bowes, 2017). In this context, system mapping is primarily understood as a communication tool and as polished, visually structured, audience-facing outputs (Jones & Bowes, 2017). However, in this thesis, I applied system mapping primarily as a reflexive practice and as a means of sense-making during the analysis process (see Chapter 4.2.3 System Mapping). I experimentally constructed maps at different levels of scale and focus points to explore the collected data (see Appendix 5) in a relational manner and to work towards the generation of insight. These maps were part of the thinking process itself, and, therefore, researcher-facing without the pressure to be legible to others. After I had constructed the insights, I produced entirely new maps for the findings chapter. They served the purpose of communicating only a focused extract of a specific relationship or process relevant to that insight. Overall, system mapping for the purpose of communicating insights as well as for the purpose of experimental sense-making remains an essential tool for the practice of service design. Nevertheless, this thesis emphasizes its value for inwards-facing analysis, because this role appears underexplored in current discourse.

Intercultural Collaboration

Service design requires collaboration with a diverse range of actors who hold different understandings, values, and goals. The intercultural context of CS development between the Global North and the Global South, in which this research was situated, adds another layer of complexity to this process. The DROP app use case illustrates the various tensions that may evolve from different values, cultural dynamics, and implicitly embedded assumptions. For instance, this thesis surfaced that differing perceptions on accuracy constrained the range of represented LFI in the DROP app (see Insights 3 – Chapter 5.1 Research Question One). Further, different perceptions between researchers and farmers on what counts as a forecast invisibly materialized in the DROP app's temporal boundary for local forecast submissions (see Insight 1 – Chapter 5.2 Research Question Two). In this context, certain design decisions may be made subconsciously without questioning one's own beliefs and implicitly embedded norms. Consequently, collaboration across cultures requires skills, awareness, and reflexivity from all involved parties, and particularly from service designers who may increasingly engage with CS development in the future. Service designers facilitate the collaboration, shape its condition, and navigate the design process. While they should be neutral facilitators, they also enter the space with their own cultural frameworks, and thereby invisibly influence research and co-creation processes. Thus, the future of service design practice requires deepened skills in intercultural design work, where designers reflect on their own position in the design process and acknowledge the influences their presence might impose on the outcomes, as this carries consequences for the experience, impact, and uptake of a service.

Reflexive Methodological Practice

The reflexivity about the designers' role and positionality should extend to the methodological practice, especially during the fieldwork and research process. In CS research that involves rural communities, ethical considerations beyond simple ethical compliance are fundamental. While fieldwork guides (Phillips & Johns, 2024) offer practical examples for an ethical and decolonized practice, each research context carries its own dynamics and requires thoughtful adjustments. In this thesis, this involved ongoing reflection on my positionality within the specific context in which my research is situated (see Preface). It also included awareness of the specific power relations and influences that may be present in my research, despite my commitment to an ethical research practice that recognizes research participants as co-producers of knowledge (Phillips & Johns, 2024). While these power dynamics and influences may never be fully resolved, designers need to develop skills and culturally sensitive methods, adjusted to the specific research context, to mitigate them. Methodological decisions in my research involved, for instance,

considerations on how I collect consent and data in ways that lower hierarchies between me, the researcher, and participants, as well as between participants themselves, for instance, during the participatory workshop (see Chapter 4.1.4 Participatory Workshop). In this context, I developed a card game with the aim of mitigating gender dynamics in group activities. Essentially, methods must be sensitive to social hierarchies connected to culture, age, or gender as well as to different levels of literacy as they influence participation. Therefore, the future practice of service design requires deepened skills in reflexive methodological development to support ethical decolonized fieldwork and meaningful participatory engagement. Ultimately, design education plays an essential role in training these competencies.

6.4 Design Principles

Based on the literature, this thesis defines CS as the provision of relevant climate and weather information to targeted user groups to help them anticipate, prepare for, and respond to environmental changes. However, insights illustrate that CS not only provide information. They also intervene in the social and cultural systems of communities. Thus, meaningful climate adaptation requires learning to design CS that reflect the realities and knowledge systems of the communities they aim to serve. To support this, I translated the insights of this thesis connected to the first research question, which specifically focused on LK representation, into design principles for the future development of CS tools that aim to integrate LK (Figure 48). Design principles support design decisions and give guidance during the design process (Kruse et al., 2016). Thus, they are actionable and generalizable guidelines that designers, developers, scientists, researchers, institutions, and other involved actors can apply across projects.

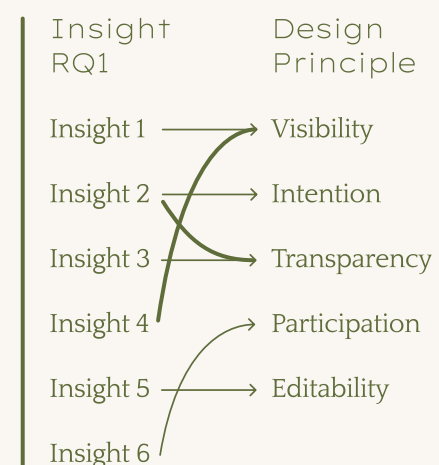


Figure 48: Mapping of insights into design principles

Set of Design Principles for LK Integration within CS

1. **Design for Visibility** – Make different ways of knowing directly visible within CS to treat the integration of LK as a recognition and not as a scientific equivalence.
2. **Design for Intention** – Acknowledge the various dimensions of LK and decide collaboratively and intentionally which meanings to simplify and which to preserve to capture the relevant aspects of LK for a specific CS.

3. **Design for Transparency** – Recognize the various application contexts in which LK exists and communicate precisely and transparently which specific context, aspect, and extent LK is integrated into a specific CS.
4. **Design for Participation, not Openness** – Identify and lower structural barriers of CS that systematically exclude certain social groups and their LK from participating.
5. **Design for Editability** – Capture the contextual, changing, and living nature of LK by allowing users to contribute, edit, and update LK within CS.

This set of design principles (Figure 49) may not be exhaustive, but it serves as a starting point for iteration and expansion through future CS research and design. Hence, I invite practitioners to apply those principles in their own CS projects, reflect, challenge, and refine them based on their experiences and context to collaboratively advance the intersection of design, climate service development, and local knowledge integration.

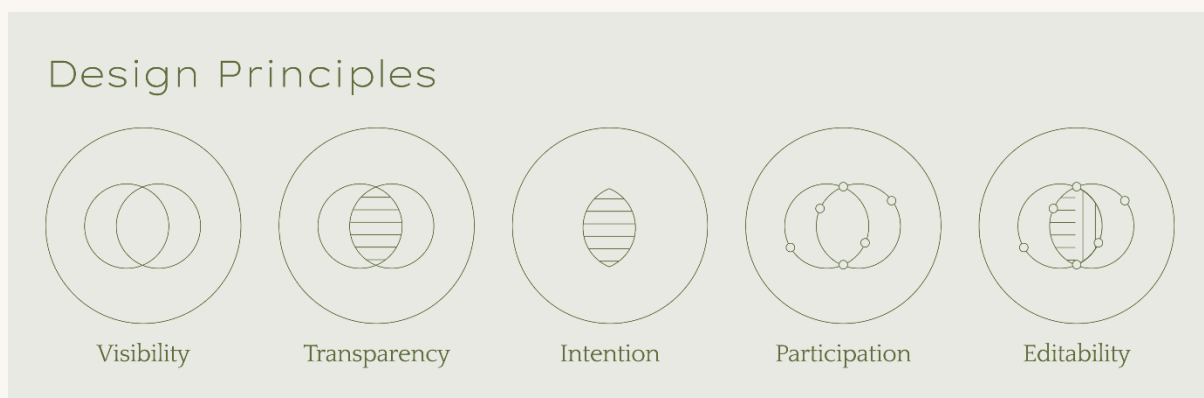


Figure 49: Set of design principles for CS tools that integrate LK

7 Conclusion

This final chapter concludes with an overview of the research process and answers the research questions. Further, it addresses the limitations, outlines avenues for further research, and discusses contributions of the thesis across the research fields of climate services and service design.

7.1 Summary of Research

The thesis explored how the integration of local and scientific weather knowledge in climate services influences community practices in Ghanaian farming communities. In detail, the guiding research questions focused on understanding the representation of LK within CS and explored possible shifts in knowledge practices initiated through CS by drawing upon the case of an existing climate service and its most visible user touchpoint, the DROP app. The DROP app provides weather forecast information to smallholder farmers, and the analysis focused on its current design and deployment in the country of Ghana, where the app was initially developed. The framing of the research questions was based on three theoretical frameworks, namely the multi-level service design concept by Patrício et al. (2011), the systems thinking perspectives by Meadows (2008), and the concept of soft impacts of design and technology by Swierstra (2015).

The thesis adopted a qualitative, exploratory methodology with a participatory research approach. The data collection process consisted of secondary research, benchmarking through service safari of current CS weather apps, semi-structured interviews with the DROP app research team in combination with generative tools, a participatory workshop with farmers and community-based fieldwork in Ghana. Together, these different methods complemented and informed each other. Thereafter, the collected data were analyzed through benchmarking analysis, thematic analysis, and system mapping. The analysis allowed for the generation of insights to answer the proposed research questions and initiated a discussion on the implications for the design of the DROP app, and more broadly for the future practice of climate services and service design. Lastly, this thesis presented five design principles for the future development of CS tools that aim to integrate LK.

7.2 Answers to the Research Questions

This thesis provides answers to the following research questions:

RQ1: How is local forecasting knowledge represented in the climate service DROP app, and what aspects of this knowledge are not reflected in the integration?

In the current published version of the DROP app, farmers submit their local forecast by selecting their predicted rainfall. The LK behind that prediction remains outside the interface. In contrast, the proposed future mock-up of the DROP app includes selectable LFI as part of the submitted data. Field research illustrates the relevance of directly presenting LK in the app as it centers the knowledge, creates credibility and transparency of the LF, and supports collective learning. Thus, rather than presenting LK merely as data outputs, the LK behind the local forecast should remain legible and traceable and thereby could contribute to a shared system of observed knowledge.

Further, findings indicate that the nuanced behavior of an indicator carries forecast information regarding rainfall intensity, onset, and certainty. These nuances are not represented in the DROP app mock-up, which simplifies LFK into selectable icons of broad indicator categories. Further, the DROP app mock-up does not display any specificity of LFI in terms of locally specific species of animals or plants. Overall, simplification remains unavoidable when LK is translated into digital climate services. The translation is a balancing act between feasible, practical functionality and detailed representation. Thus, design decisions regarding what aspects to simplify and what meanings to preserve require more awareness, reflection, and collaboration with farmers, as it influences how credible and intentional LK inclusion appears in the app and determines whether key aspects of the LK are captured within the system.

Community members draw on a wider range of indicators for rainfall prediction than the selection currently depicted in the DROP app mock-up. The reason behind the limited selection lies in the time-intensive collection of these indicators. Further, farmers apply LFI for wind, storm, drought, and seasonal predictions, but these are excluded in the DROP app. Their exclusion is determined by project scope and accuracy concerns, which reflects a broader challenge of balancing the breadth of LK representation with the provision of accurate information in CS development. As a result, the DROP app implements only one part of a broader knowledge system applied by local communities to forecast weather changes. To create transparency and acknowledge the true breadth of LK, it remains essential

to precisely articulate which types and to what extent it is included or excluded within a specific CS tool.

Additionally, ceremonial and calendrical local forecasting knowledge, mainly practiced by rainmakers, is not represented in the current app or proposed mock-up. The question of whether and how this form of LK could be included in the DROP app revealed uncertainty from farmers, which may stem from differing opinions on the value of these practices. The design decision for the exclusion points towards epistemological limits within CS development, where spiritual and observational LFK are not treated as equal knowledge sources. However, inclusion should not aim to resolve epistemological differences but to make them visible and acknowledge that various ways of knowing exist and hold value for communities beyond the immediate purpose of accurate weather prediction.

Findings illustrate the changing and contextual nature of LK, which is shaped by age, gender, and the role of community members, as well as external factors such as climate change or household and farming decisions. In contrast, the DROP app, with its fixed interface and static set of indicators collected from selected community members, in specific communities, at a specific point in time, cannot register any of these changes or dynamics. This reveals a tension between LK as a living practice and LK as a documented, preserved artifact when integrated into digital climate service tools. However, the digital format of such tools could reflect this changing nature of LK if designed in a way that allows for editability and updatability through its users.

Lastly, while the design of the DROP app in theory encourages all community members to input their observations, in practice, knowledge holders, especially elderly, face access barriers. The community members who can access the DROP app are younger, more educated, and tech-savvy, but less experienced with LFK. This structural mismatch affects LF inputs and accuracy and suggests that the actual LK representation within the DROP app remains sparse in practice and does not reflect the community's local forecasting expertise.

RQ 2: How may the introduction of the climate service DROP app reconfigure the system of community forecasting practices?

Findings indicate a gap between the actual presence of local indicators for weather forecasting and the need for consistently available weather information. The DROP app, with its scientific forecast, bridges this gap by providing a daily, on-demand access. Nevertheless, farmers expressed a desire for an hourly forecast resolution within the DROP app. This context surfaces different perceptions between researchers and farmers on what counts as a forecast. For the research team, a forecast only qualifies as one if it is made sufficiently in advance. Thus, the DROP app limits LF submissions to 9 am for the same. For farmers,

however, observing clouds forming rapidly and estimating rain an hour away constitutes an actionable, relevant prediction. This example indicates that assumptions are implicitly embedded in the design of CS but need to be surfaced to ensure such tools reflect the realities of farmers.

The DROP app enters a landscape already characterized by a broad engagement with multiple information channels and growing interest in scientific forecasts from digital tools. However, the new tools that farmers gain, such as the DROP app, differ in the format of weather information. They are text-based and require technological literacy, smartphone ownership, internet connectivity, and English language proficiency, which present barriers and hinder adoption within farming communities. Most significantly, these barriers apply to a specific social group, namely, elders and women, who have less access to smartphones and lower levels of formal education. Overall, these findings illustrate that design decisions on the presentation of information also determine who is excluded from the service itself. Ultimately, specifically for the DROP app, this shapes whose knowledge and local forecast predictions enter the system.

Traditionally, forecast authority is defined by experience and age. However, with climate change impacting local forecast accuracy and with the advent of new technologies, including the DROP app, experienced elders now increasingly consult smartphone users for scientific forecast information. Smartphone users largely represent young, educated, male farmers, who start to inhabit a new social position within the existing forecasting networks. Nevertheless, this intergenerational dynamic, defined by shifts in social positions, significantly impacts whether local knowledge practices will continue to be learned as technology is perceived to reduce incentives for the younger generation to develop LF skills. Further, access to technologies determines the extent to which women remain largely excluded from forecasting practices or, similar to the youth, gain a new social position as respected knowledge holders. Overall, these dynamics portray the DROP app as a social intervention that influences relationships between community members. The effects on social and cultural structures, introduced by digital climate service tools, may not be immediately visible, but are central to understanding the impact and future implications for communities.

Further, findings indicate a growing interest and trust in SF, which represents a pragmatic and outcome-based rather than an ideological shift. In times of climate change and decreasing accuracy of their LK, technological adoption acts as a strategy to support farming decisions and minimize financial losses. Nonetheless, LK remains practiced and actively compared against the scientific forecast. In the future, the research team plans to implement a hybrid forecast, in which the local forecast functions as a means to improve the accuracy and localization of the scientific one. Ultimately, the quality of local forecast inputs remains

vital for the accuracy of the hybrid forecast. The issue of accuracy requires addressing the challenges presented in previous insights, such as the gap between knowledge holders and the app users, barriers to participation, to create the conditions for LK to enter the DROP app. Overall, the findings point towards the importance of applying a systemic perspective to understand the interconnectedness of technical design, social dynamics, and cultural practices, in order to support the adoption and improvements of climate services.

The primary goal of the DROP app is to support farmers in decision-making by providing reliable weather information. However, the findings indicate that strong informal community networks carry weather information to various community members with a wide range of other occupations and needs, as numerous decisions in daily life are weather-dependent and impacted by changing weather patterns in times of climate change. This illustrates that the value of a service is always co-created between actors and, thus, requires rethinking how to measure impact to account for the broader, often unanticipated outcomes of a service beyond the intended objective.

Summary of Implications

The findings are contextualized into concrete implications. First, the thesis discusses implications for the design of the DROP app (Chapter 6.1) across the dimensions of (1) technical functionalities, (2) UI/UX design, (3) LK features, and (4) design speculations. Second, the research surfaced implications for the future practice of climate services, which should (1) decode LK and precisely articulate the specific dimensions, manner, and extent to which LK is addressed, (2) explore, develop, and iterate more practical approaches for integrating LK, (3) increase interdisciplinarity for successful co-creation and implementation of CS, and (4) rethink conventional evaluation frameworks to capture a holistic understanding of impact. Third, the thesis identified implications for the practice of service design (Chapter 6.3), which should (1) rethink existing service design frameworks to address current limitations, (2) recognize the value of system mapping for inwards-facing analysis, (3) deepen skills in intercultural design work, and (4) deepen skills in reflexive methodological development to support ethical decolonized fieldwork. Lastly, this thesis produced five design principles (Chapter 6.4) for the integration of LK in CS tools that practitioners can apply and refine across projects: (1) Design for Visibility, (2) Design for Transparency, (3) Design for Intention, (4) Design for Participation, (5) Design for Editability.

7.3 Limitations

Within this research, several limitations must be acknowledged. The first limitation concerns the framing of the research. The thesis frames the DROP app as a targeted intervention that reshapes forecasting practices in communities. However, when reflecting on this framing, it may implicitly position the DROP app as an active driver and community members as passive recipients of transformation. This implication was not intended. In fact, research revealed a more complex picture than simple technology adoption. Communities are actively navigating the changing landscape of weather forecasting through a process of relational interaction between various interconnected actors and elements.

Further limitations concern the data collection process. First, underlying and implicit power dynamics influenced the research. My position as an outsider researcher may have led participants to perform a specific narrative, hide information, or offer pleasing answers. For instance, during the participatory workshop, surfacing critical feedback from farmers proved challenging at times. Second, participant selection and representation constitute a limitation. I was able to engage only with a small number of individuals who directly use the DROP app during the fieldwork. Further, youth participants remain underrepresented and restrict the identification of patterns for this group. Third, because this research represents a Master's thesis, time and scope were constrained. This reduced the possibility of engaging more deeply with certain aspects, such as gender or intergenerational dynamics. Fourth, all interviews required translation between English and the local language, Dagbani. I acknowledge that translation filters or alters the meaning of the responses of participants and thereby represents a limitation in fully understanding intended meanings. Fifth, the fieldwork only focused on two specific communities in northern Ghana, close to Tamale. However, the DROP app was also introduced in Zimbabwe and Bangladesh. These different contexts may differ fundamentally from the Ghanaian context. Thus, the highly localized nature of this thesis limits the transferability of the findings. Lastly, ethical decolonial research suggests returning insights to the community in an accessible format for clarification and final approval (Phillips & Johns, 2024). This was not possible within the constraints of this thesis. Nevertheless, the research seeks to bring benefit to the communities involved by translating their issues and priorities into implications for the future development of climate services.

7.4 Avenues for Future Research

The limitations discussed above suggest multiple avenues for future research. Future studies could build upon this thesis and expand the research with the DROP app. For instance, they may involve more participants in the process, particularly women, youth, and app users, and deepen the investigation of the relationship between digital climate services and traditional forecasting customs. Further, comparative studies between communities across regions and countries where the DROP app was introduced would support the understanding of how different communities navigate the changing landscape of weather forecasting.

Additionally, the implications for the design of the DROP app represent starting points for further research. Their benefit should be evaluated, tested, and iteratively co-created with local communities. Similarly, implications for the future practice of climate services and service design, as well as the presented design principles, represent a research agenda that invites practitioners to implement, reflect on, and refine them in practice to collaboratively advance the intersection of service design, climate service development, and local knowledge integration.

This thesis focused on LK practices for weather forecasting. However, future research could explore LK in relation to nature observation beyond practical prediction purposes to generate a full picture of LK dimensions or explore the integration and representation of other forms of LK within CS.

7.5 Research Contribution

The contribution of this thesis spans across the field of climate services and the field of service design. Firstly, it contributes to the growing discussion on LK integration within climate services by drawing upon the use case of the DROP app. The thesis surfaces various dimensions of local weather forecasting knowledge in two Ghanaian farming communities and investigates their practical implementation by reflecting on representation and limitations. Further, the thesis contributes to the, in current literature underexplored, discussion on the impacts of CS on social and cultural dynamics by surfacing shifts in community knowledge practices.

Secondly, the research applies service design lenses and methodologies to the discipline of CS and thereby surfaces benefits and limitations of both practices. Subsequently, these reflections are translated into concrete implications for the future practice of climate

services and service design. Thereby, the thesis contributes to and argues for the bridging of these disciplines.

Thirdly, this thesis aims to benefit local farming communities by communicating their experiences and translating them into implications for future tool development. Specifically, findings of this research offer concrete implications for the DROP app to better meet the realities of community members and to contribute to long-term cultural sustainability. Further, this thesis produced five design principles for the integration of LK in CS tools that practitioners can apply and refine across projects.

Postface

I started this thesis with thoughts on climate change and climate adaptation, and with a reflection on the dynamics between Europe and Africa connected to these themes. The use case of this thesis, the DROP app, which is part of the EU-funded research project SAFE4All that develops climate services for stakeholders in Africa, grounded the thesis in the specific context of Ghana and in the complexities that arise when designing with multiple knowledge systems for climate adaptation.

As I mentioned before, African regions currently face greater vulnerability to climate impacts compared to Europe, where the effects feel less visceral. However, the climate will continue to change everywhere, and its consequences will also accelerate in the Global North. Thus, the development of climate adaptation strategies, including the development of CS, will continue to gain more importance in the European context as well. I argue that the question of how we design the tools that carry us through a changing climate and how they interact with the local knowledge, practices, and dynamics will become more relevant in the future for other places too, including Europe.

Reflecting on this perspective, one may argue that Europe largely already lost its connection to its place-based knowledge and traditional practices due to processes such as industrialization, urbanization, and the dominance of technology. If this is true, then this loss should make us even more attentive to the design of climate adaptation tools for regions where this knowledge still exists. It determines whether climate services pose the risk to accelerate the same process of LK displacement in countries like Ghana, or, if mindfully designed, could do the opposite. But have local knowledge practices entirely disappeared from Europe? Indigenous communities, such as the Sámi living in Sápmi (a territory spanning across northern Norway, Sweden, Finland, and Russia), continue to engage with their own knowledge systems. Further, if we look at local knowledge beyond weather forecasting, we might recognize the variety of its applications. LK also includes traditional practices of agricultural irrigation systems, cooling techniques for the built environment, water harvesting, and various other methods that support living with and responding to environmental changes. Nowadays, these traditional practices are starting to receive more attention as they represent sustainable alternatives to address current challenges. In this

context, power asymmetries and epistemological tensions, similar to those surfaced in this thesis, may exist in Europe, for instance, between funders who favor technical solutions and local stakeholders who may recognize the potential of applying traditional practices.

Essentially, there is no single path for climate adaptation. We need multiple strategies and should draw upon all available ways of knowing. As CS development will scale globally and within times where technology is primarily seen as the answer, the risk of overlooking the value of LK in that process is critical. Therefore, I started to wonder: What could local knowledge practices from the Global South offer to Europe? Not in an extractive manner but as a bi-directional exchange of expertise and ways of knowing.

Appendix

Appendix 1: Interview Guide and Generative Tools

Introduction (5min)

- My background and research project
- Consent Form

Background (5min)

- Can you describe your role in the development of the DROP app?
- How long have you been involved in the DROP app project?
- How often and what kind of interactions have you had with end-users or local communities in Ghana?

LK Representation in DROP (10min)

- Can you walk me through the process of deciding how to include LFK in the DROP app?
 - How did you gather LK?
 - What decisions did you make along the way?
 - What challenges did you encounter?
 - What other ways/ ideas for LK integration existed?
- Do you have thoughts on the current state of the app and the integration of LFK?
- Do you see opportunities for improving the integration of local knowledge?

Card Sorting Activity: Attributes & Transformation (15min, Appendix Figure 1)

- Based on your experience, which aspects or qualities best capture LFK?
 - Could you explain why you selected those?
- How do these aspects and qualities relate to the DROP app? Could you explain?

Mapping Activity: Community forecast practices as a system (without DROP app, 15min, Appendix Figure 2)

- How are community roles connected in relation to weather forecasting and farming?
- What examples of LK practices exist in the community?
- What elements are relevant and influence the decision-making on farming activities?

Story Elicitation: Observed or predicted changes (with DROP app, 5min)

- When looking back on your fieldtrips, what kind of observations did you make in the communities in relation to the introduction of the DROP app and changes in community knowledge practices?

- When, where and how was the DROP app used? By whom? What role did it play?
- Did you observe changes in community roles, dynamics, and relationships?
- Did you observe changes in knowledge practices, norms, institutions, and rituals?
- Did you observe new behaviors, shifted behaviors, or changes in decision-making?

→ Imagine a future where the DROP app is integrated into the daily life of the community. It serves as a key tool for weather prediction and adapting farming activities. What changes would you foresee?

Outro (5min)

- Is there anything else you think is important for me to understand about the integration of LK in the DROP app, or its impact on forecasting practices and community dynamics?
- Do you know other CS tools for farmers that integrate LFK? If yes, is it similar?
- Do you recommend any further contacts that I should interview for my research?

Card Sorting

Based on your experience, which aspects or qualities best capture local forecasting knowledge?

The cards below summarize aspects or qualities of local forecasting knowledge identified in secondary research. Please select the cards you consider most relevant. Add or edit cards to reflect aspects that are missing.

Intergenerational	Community-based	Embedded in authority structures	Embedded in collective decision making	Socially negotiated
Environmentally sensitive	Place-specific	Context-specific and nuanced	Adaptive to environmental changes	Sensitive to environmental change
Culturally embedded	Linked to social roles	Structural	Contextually variable	Action-oriented
Interpretative	Meaning centered	Narrative based	Risk-oriented	Anticipatory
Write your own				

How do these aspects and qualities relate to the DROP app?

Based on observations and expected outcomes:

Explicitly represented
Which aspects or qualities are explicitly represented within the DROP app (e.g., through features, structure)?

Translated or modified
Which aspects or qualities are translated or modified from their original social or cultural context?

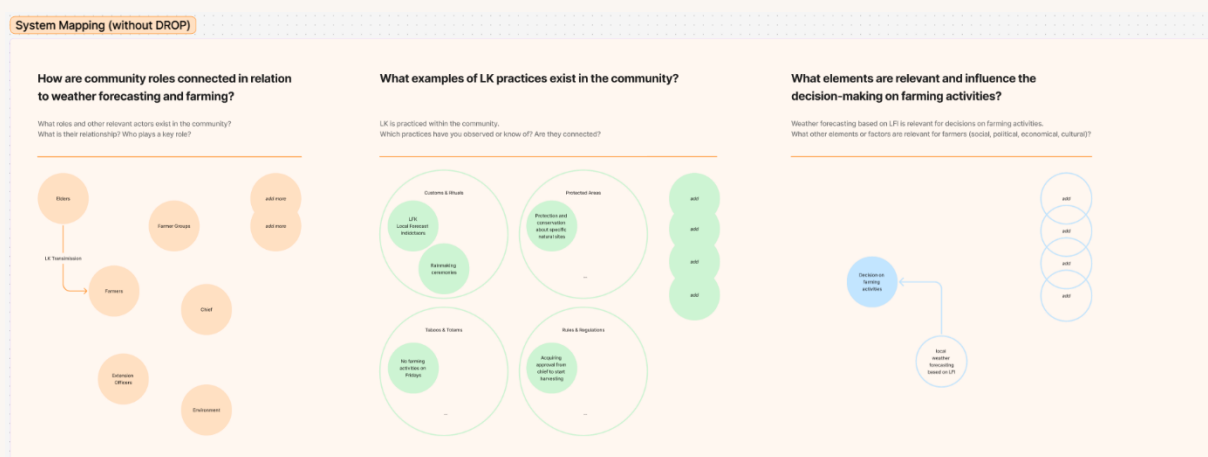
Not represented but continuing outside the app
Which aspects or qualities are not represented by the DROP app, but continue to exist and operate within community knowledge/practices?

Reconfigured in practice
Which aspects or qualities are reconfigured by the DROP app, even if they are not explicitly represented in the app itself?

Emerging or app-induced qualities
Are new aspects or qualities emerging through the use of the DROP app (e.g., individualization, new forms of authority)?

Write your own

Appendix Figure 1: Card sorting activity displayed in Figma



Appendix Figure 2: Mapping activity displayed in Figma

Appendix 2: Participatory Workshop Materials

The final workshop activities included:

Positioning Poster (Warm-Up) - The goal of this activity is to understand participants' background and experience with local forecasting knowledge and the DROP app. Participants position themselves along a spectrum in response to two questions: "How often do you forecast weather through environmental observations?" (ranging from never to multiple times a day) and "When was the last time that you used the DROP app?" (ranging from never to today) (Appendix Figure 3). Further, I prepared an optional extension for this

warm-up activity. Depending on participants' DROP app experience, they pin sticky notes on an additional poster for the question "What is one thing that you like and don't like about the DROP app?" (Appendix Figure 4).



INTRO

How often do you forecast the weather through environmental observation?

never sometimes/ depends once a month once a week daily multiple times a day

When was the last time that you used the DROP app?

never one year ago a few month ago last month last week yesterday/ today

Appendix Figure 3: Positioning poster for warm-up activity



INTRO

What is one thing that you like and don't like about the DROP app?

I like...

I do not like...

Appendix Figure 4: Additional poster activity for warm-up

My Forecasting Circle (Mapping Activity) – This activity aims to understand community dynamics, relationships, roles, and changes in these forecasting networks. Participants work individually on worksheets where they map everyone they talked to about the weather in their everyday life by writing down each person in a circle, placing them close or far based on frequency of contact, drawing connecting lines, and indicating topics or reasons for connection (Appendix Figure 5). To support this task, I provide examples of personas of community members, such as neighbors, elders, or family members. The final part of this exercise centers around the question: "Has anything changed in the last few years?". Here, I remained flexible about whether this would be discussed verbally or implemented as a continued mapping exercise, depending on time and participant engagement.

MY FORECASTING CIRCLE

Who do you talk to about weather? Who do you trust?

Step 1:

Think of everyone you talk to about weather and farming decisions. Write or draw each person* in a circle.

Step 2:

Place them close to you or far from you. Close = you talk often. Far = you talk rarely.

Step 3:

Draw a line between you and each person. Why and about what do you talk to them? Write it on the line.

Step 4:

Do you use any technology for weather (e.g. radio, weather app, WhatsApp...)? Add it.

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graph LR
    ME((ME)) --- P1(( ))
    ME --- P2(( ))
    ME --- P3(( ))

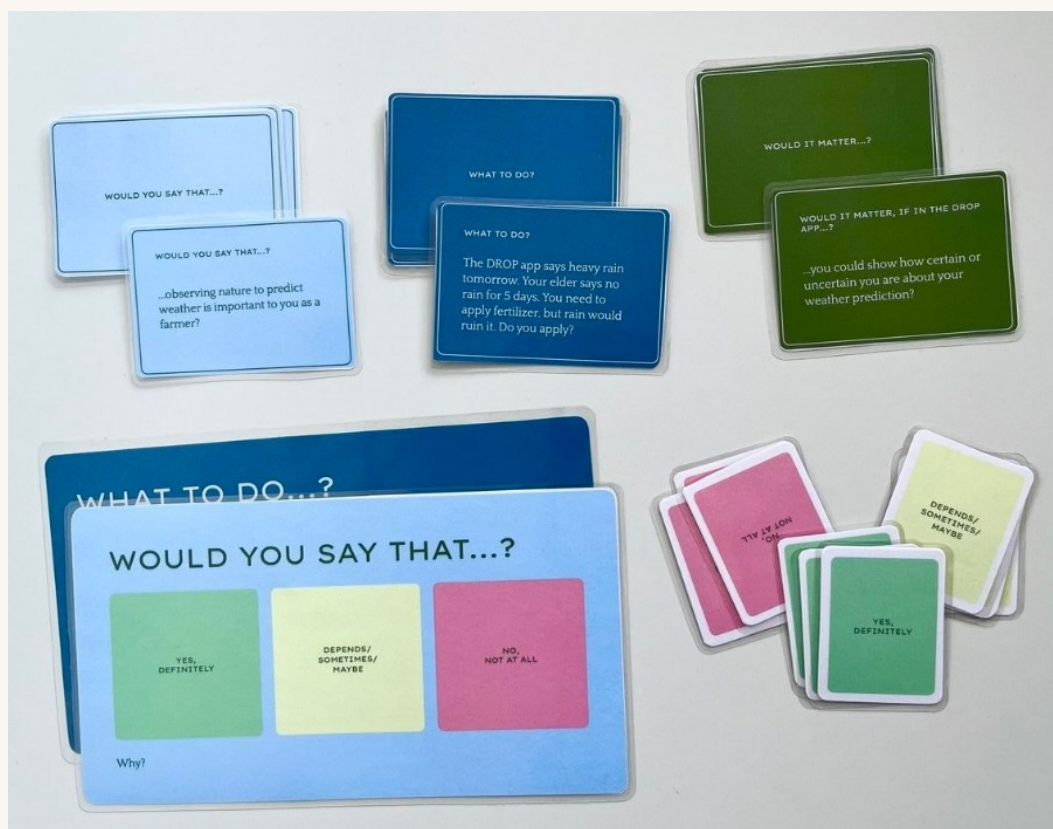
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*examples: wife/husband, brother/sister, son/daughter, father/mother, chief, elder, rainmaker, neighbor, young farmer, extension officers...

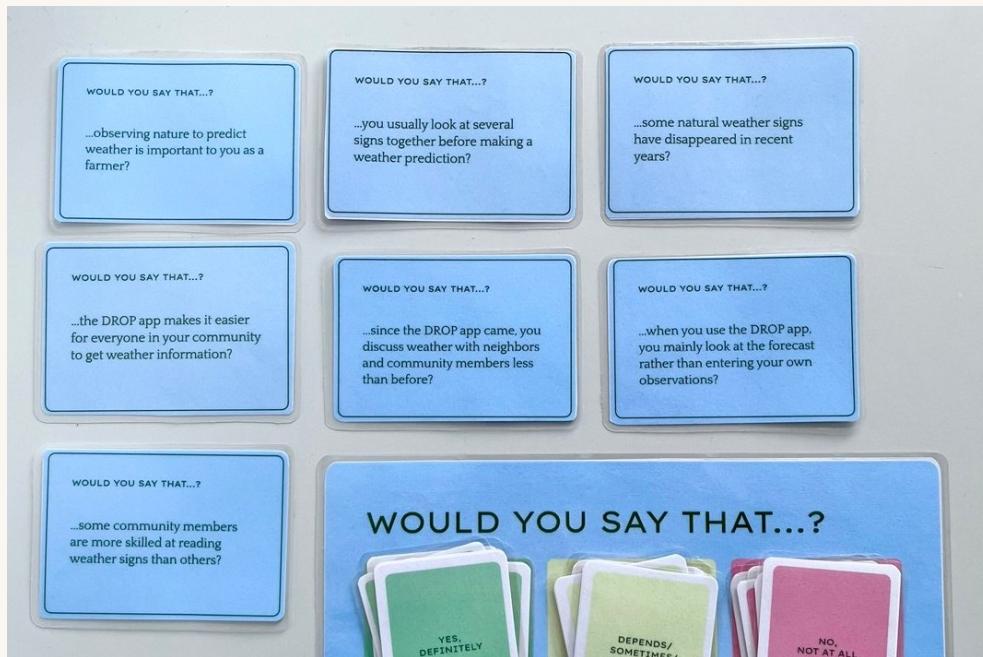
Look at your circle.
Has anything changed in the last few years?

Appendix Figure 5: Forecasting circle worksheet

Card Game (Main activity) – Participants sit around a table for a card-based group activity (Appendix Figure 6). Each round, one participant draws a card from the pile and reads the statement or question aloud. Then, all participants individually select a response card (Yes/ Maybe/ No), place it face down simultaneously, and reveal and discuss. There are three categories of cards. The first category, “Would you say that...?”-cards, presents statements that I derived from interview insights and aim to validate, or open questions that remain from desk research (Appendix Figure 7). Cards display controversial assertions that aim to spark discussion around knowledge practices, DROP app usage, and social dynamics. The second category, “What to do?”-cards, portrays dilemmas, for example, situations where scientific and local forecasts conflict (Appendix Figure 8). The purpose is to understand decision-making processes, especially when tensions emerge between different knowledge systems. The third category, “Would it matter, if...?”-cards, represents potential missing features in the DROP app and aims to surface gaps in the representation of local knowledge (Appendix Figure 9). Here, I purposely chose the more neutral framing “would this matter” instead of “would you like” or “would it be helpful” to avoid socially desirable responses where participants might agree with all suggestions without critical reflection.



Appendix Figure 6: Overview card game materials



Appendix Figure 7: “Would you say that...?” - cards



Appendix Figure 8: “What to do?” - cards



Appendix Figure 9: “Would it matter if...?” - cards

Wish for the Future (Wrap-Up) – This activity serves as a future-oriented conclusion, while also eliciting participants' hopes and aspirations. On a poster, participants are invited to contribute wishes in three categories, namely a wish for the future of local knowledge, for the community, and for the future of weather forecasting.

OUTRO

What is your wish for the future...?

...of local knowledge:

...of the community:

...of weather forecasting:

Appendix Figure 11: Wish for the future – poster as wrap-up activity

Appendix 3: Community-based Fieldwork – Narrative Interview Guide

Introduction & Consent (5 min)

- My background and research project
- Consent

1 - Daily Life Activities (5 min):

Could you walk me through your day?

- How does a typical farming season day from wake-up to sleep look like for you?
- What do you usually do in the morning, midday, and evening?

2 - LK Practices & Weather (5-10 min):

Where does weather matter in your day? // When do you think about or check the weather?

- How are you observing & predicting the weather?
- Who are you talking to? And why? (share/receive info, validate...?)
- What decisions are you making?

Are spiritual, ceremonial practices relevant to the community for weather forecasting?

- When do they happen? // → Who is practicing them? // → What is their role?

3 - DROP app (15 min):

Does the DROP app appear in your day?

When, where, and how (often) and for what do you use it? // Who is using it?

- Do you use it to input your own forecast, or mainly to look at inputs from others, or to check the scientific forecasts?
- Do you discuss the forecast with others? Which one? With whom?
- Do you share information from DROP with other that do not have a phone?
- If DROP contradicts what you observe/ what elders say: What do you trust?

Why?

Can you show me how you check the DROP app normally?

- Is there something annoying or confusing?
- What if you notice a weather sign that is not in the DROP app? Has this happened before?

(3a) App interface – indicators :

How is what you observe different from what is represented?

- What exactly do you observe? When? What context matters (e.g., time)?
 - What does it mean? (e.g., not just "rain" - When? How much? Certainty level?)
 - Do you combine with other indicators or practices?

- Do variations exist? (based on locations/seasons, or “species”)?
- Loss of indicators due to climate change?

Is there something missing? Or something you would like to add here?

→ Seasonal Forecast Indicators, longer time-spans for prediction, visual representation of LK (e.g., specific type of frog), multiple indicators, probabilities, ceremonies

(3b) App induced changes:

Were there new behaviors or practices that you adopted while including DROP in your life?

- Does the app increase or decrease youth interest in local knowledge?
- Do more people have access to forecasts now? Or are barriers overshadowing?

(3c) Without own smartphone:

Where do you get weather information from?

If you had a phone, would you like to use the DROP app?

4 - Changes/a few years ago (5-10 min):

A few years ago, what was different in your daily life? //What is still the same?

- Did your practices (daily tasks, ceremonial practices, traditions, rituals) change?
 - Did your role in the community change?
 - Did relationships and dynamics between community roles/members change?
 - Did your access to (weather) information change?

What initiated these changes?

How do you feel about these changes?

5 - Future + Wishes (5 min):

Based on the changes you have experienced in the past, how do you expect your future daily life to look?

- Is there something that worries you?
- Is there something that will disappear?
- Is there something new emerging?
- What will stay the same/ get strengthened?

Do you have a wish for the future?

→ A wish for local knowledge, a wish for your community, a wish for the future of weather forecasting?

Is there something I missed, something that you want me to understand that I have not asked?

Appendix 4: Reflexive Thematic Analysis Process

The reflexive thematic analysis followed an iterative process consisting of multiple steps:

1 – Data Preparation and Familiarization (Semi-Structured Interviews with Generative Tools)

In the first step, I prepared the data from the interviews, including transcripts, recordings, notes, and visual materials produced through generative tools. I manually reviewed each transcript against the recording to ensure accuracy. Then, I started reading through the dataset and the visual materials. During this process, I made initial annotations by marking interesting, surprising, and generally relevant passages. In a subsequent reading, I revisited these annotations, transferred them onto digital post-it notes on a Figma board and added the visual data. By bringing all collected data together in one shared space, I was able to start with the cross-material analysis.

2 – Generating Codes (Semi-Structured Interviews with Generative Tools)

Coding represents a systematic, iterative process that assigns specific meanings to similar data items. In the first round of coding, I was working with each interview individually and employed a combination of deductive and inductive coding. Deductive codes were informed by the interview guide and included categories such as *interviewee background*, *development process*, *challenges*, or *future improvements*. Simultaneously, inductive codes emerged directly from the engagement with the data, for instance: *growing trust and confidence*, *local knowledge versus scientific forecasting*, *rainmaking practices*, *decision-making over multiple days*, *seasonal indicators*, or *perceived usefulness*. In the second round of coding, I revisited all codes across interviews. I identified overlaps and merged related codes with similar meanings. Additionally, I integrated the visual material into the clusters. At this stage, code clusters included, for instance: *development process*, *future improvements*, *accuracy*, *challenges*, *meanings and types of local knowledge practices*, *seasonal forecasting*, and *new dynamics or behaviors*.

3 – Generating Themes (Semi-Structured Interviews with Generative Tools)

Generating themes involves further combining related codes into central concepts or patterns within the dataset. During this step, I realized that several code clusters were conceptually broad and functioned more effectively as themes rather than codes. I therefore

redefined them as themes and organized the underlying data into sub-themes. For example, the theme *development process* encompassed sub-themes such as: *collection of local forecasting indicators, translation of local knowledge into the DROP app, training of farmers, evaluation and feedback, or hybrid forecast*.

4 – Revising Themes in Relation to the RQ (Semi-Structured Interviews with Generative Tools)

In this stage, I stepped back from the data-driven clustering process and re-engaged explicitly with the research questions. Based on my previous engagement with the data, I drafted preliminary finding statements for each research question. Next, I mapped the previously developed themes and sub-themes and their respective data excerpts onto the finding statements. Where themes did not sufficiently support a finding, I reframed the finding statement or created a new one. Finally, I revisited each finding statement and organized the associated themes and data clusters for a coherent argument. This phase involved additional refinement, merging, and re-articulation of themes. In the end, refined themes related to the first research question included: *depth, types, and representation of local forecasting indicators; lack of spiritual or ceremonial dimensions of local knowledge; static nature of indicator documentation; and shifts from community-owned to co-produced knowledge*. Themes related to the second research question included: *modes of prediction, changes in decision-making practices, increased trust in scientific forecasts, dynamics of participation and expertise, intergenerational knowledge transmission, gendered social change beyond agricultural practices, and other contextual influences on traditional practices*.

At this point, I paused the analysis, turned my focus to the implementation of the participatory workshop and fieldwork, and afterwards continued the analysis process with the newly gathered data as described in the subsequent steps.

5 – Data Preparation and Familiarization (Workshop and Fieldwork)

Similar to the first phase of the earlier analysis, I began with the preparation and organization of the data from the workshop and narrative interviews. I digitized and uploaded all analogue materials, including the filled worksheets from the participants, annotated posters and post-its, and my written notes, to a Figma board. Additionally, I transcribed all workshop and interview recordings manually, as recording quality, background noise, and multiple languages and accents within the recordings made automated transcription largely unusable. Following, I annotated the transcripts and transferred the data to the same Figma board to start coding.

6 – Generating Codes (Workshop and Fieldwork)

First, I coded each interview and the workshop data individually through an inductive approach. Nevertheless, prior experience and frameworks from the first phase of analysis informed my interpretations. Second, I revisited all codes across interviews, identified overlaps, and combined related codes with similar meanings. Final generated codes at this stage included, for instance, *timely relevance of weather forecasts*, *input of local forecast in DROP app*, *seasonal indicators*, *differing probabilities of LFI*, *LFI changes*, *relationship between youth and LK*, *relationship between elders and SF*, *role of the rainmaker*, or *future outlook*.

7 – Generating Themes (Workshop and Fieldwork)

During this step, I clustered similar codes into central themes in an iterative process. For example, I merged codes such as *seasonal indicators* and *differing probabilities* under the theme of *nuances and dimensions of LK*. In other cases, I reorganized existing codes, for instance, *changes in LFI*, as a theme, under which I articulated new codes, such as *accuracy* and *scarce indicators*. Final themes include the *timely relevance of weather forecasts*, *receiving and sharing of weather information*, *forecast information usage*, *nuances and dimensions of LK*, *rainmaker and LK practices*, *LF and SF in the DROP app*, *LFI changes*, and *future outlook*.

8 – Combining and Revising Themes in Relation to the QR

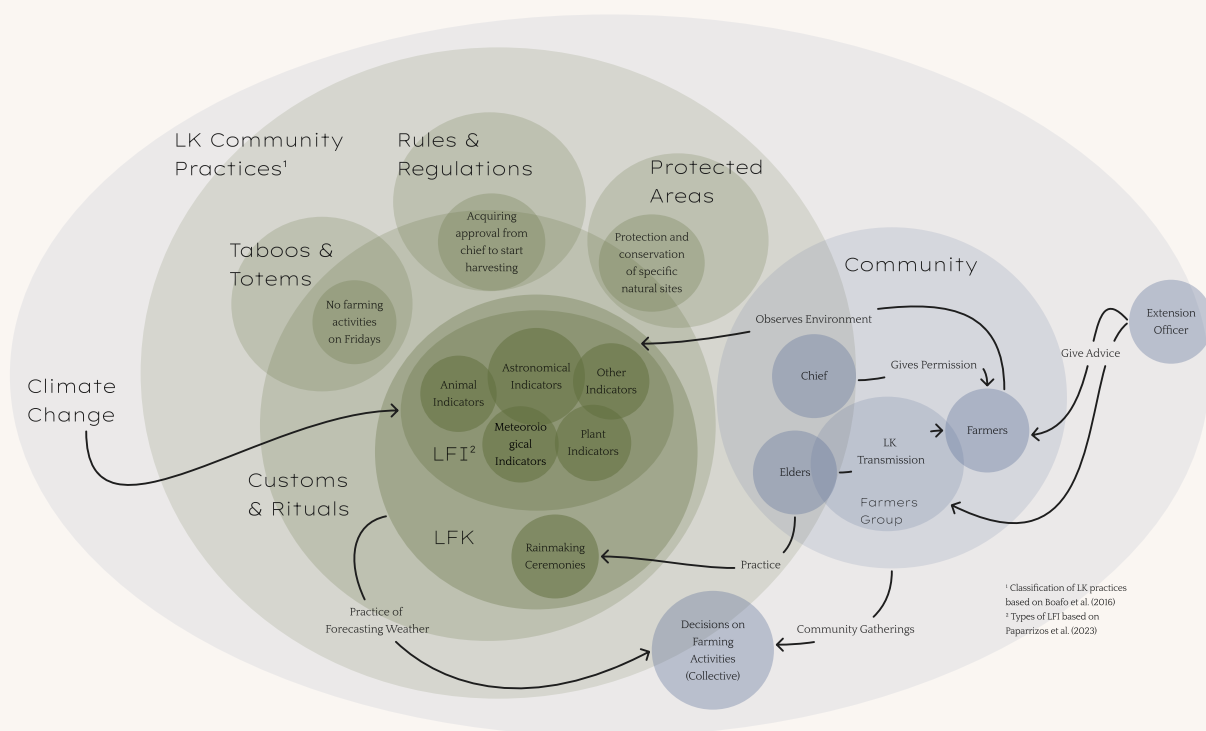
In this final step, I revisited the themes, sub-themes, and coded data excerpts from the first analysis phase of the semi-structured interviews. Subsequently, I compared them with the themes from this second phase of analysis, which was based on the data from the workshop and fieldwork. The review process involved direct engagement with the research questions and revealed differences and overlaps. For example, the theme of *spiritual and ceremonial dimensions of LK* from the first analysis phase reappeared and expanded in the second, now encompassing *astrological and calendrical practices* as well as *farmers' perceptions of these practices*. Ultimately, I reorganized or merged themes and re-coded data clusters iteratively. The final set of themes and sub-themes forms the basis for the construction of a coherent argumentation in response to the research questions.

Appendix 5: System Mapping

Throughout the analysis process, I developed multiple system maps and iteratively refined them. Each iteration reflects the evolving understanding of the system as the application of different methods led to new data and insights:

Iteration 1

For the first version of the system map, I drew exclusively on findings from secondary research and mapped my understanding of local knowledge practices in relation to community dynamics (Appendix Figure 12). Moreover, this version informed the design of the generative tools that I created for the interviews.

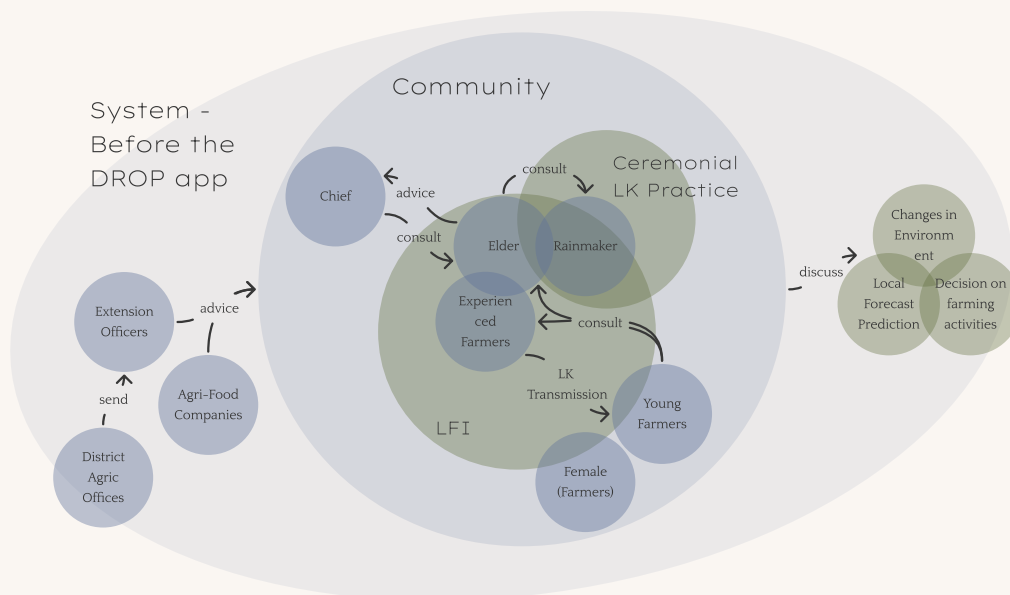


Appendix Figure 12: Iteration 1 based on secondary research

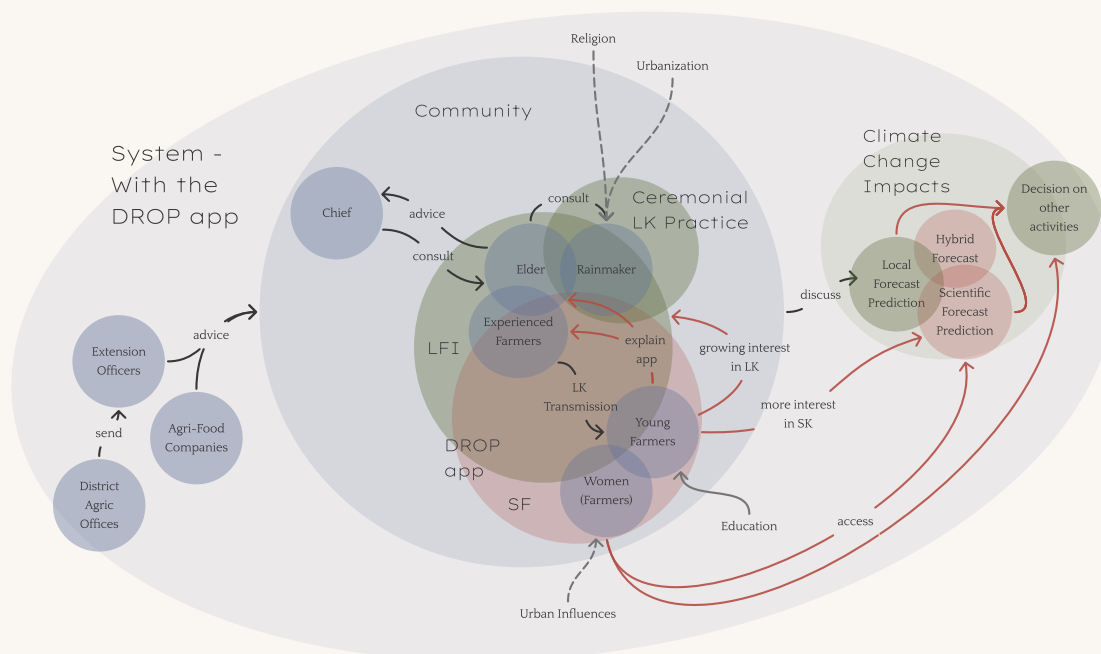
Iteration 2

After the implementation of the semi-structured interviews with generative tools, a substantial amount of mapped data from the secondary research proved not applicable for the respective communities in Ghana. Therefore, I omitted these elements and added new insights. Further, I created two versions of the map, one that shows the system before the DROP app was introduced (Appendix Figure 13) and a second, which illustrates induced

changes through the DROP app as a systemic intervention (Appendix Figure 14). Similar to the first iteration, the map primarily visualizes community roles and dynamics in relation to knowledge practices.



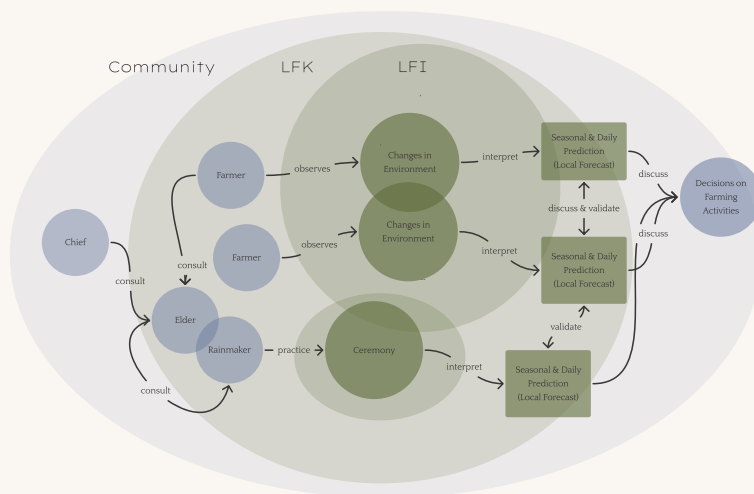
Appendix Figure 13: Iteration 2 based on data from semi-structured interviews illustrating the system before the DROP app was introduced



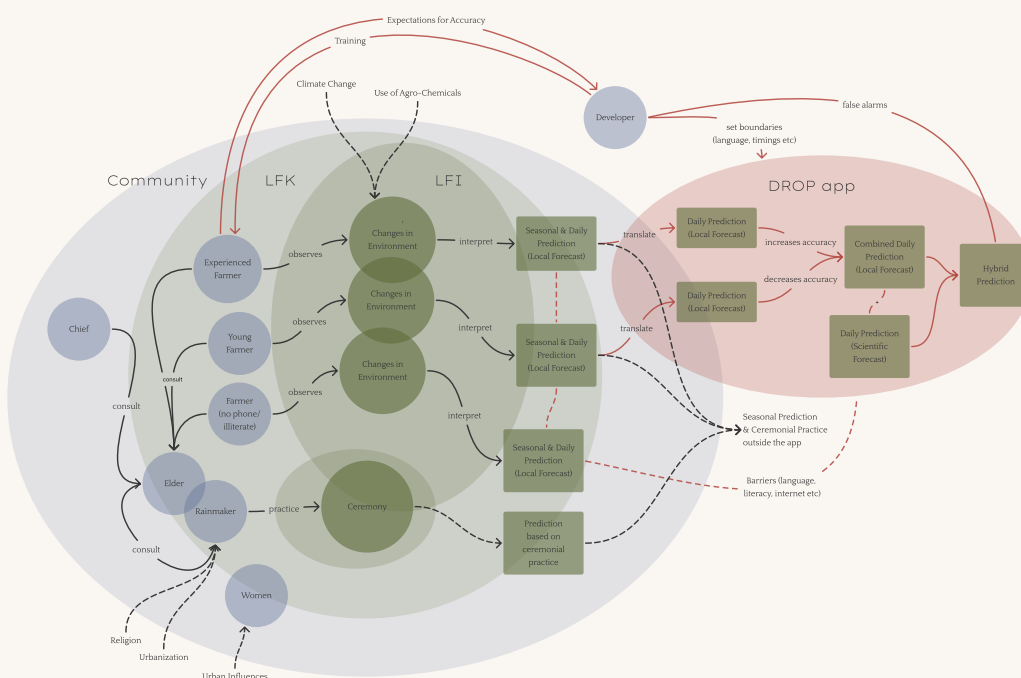
Appendix Figure 14: Iteration 2 based on data from semi-structured interviews illustrating the system after the DROP app was introduced with changes in red color

Iteration 3

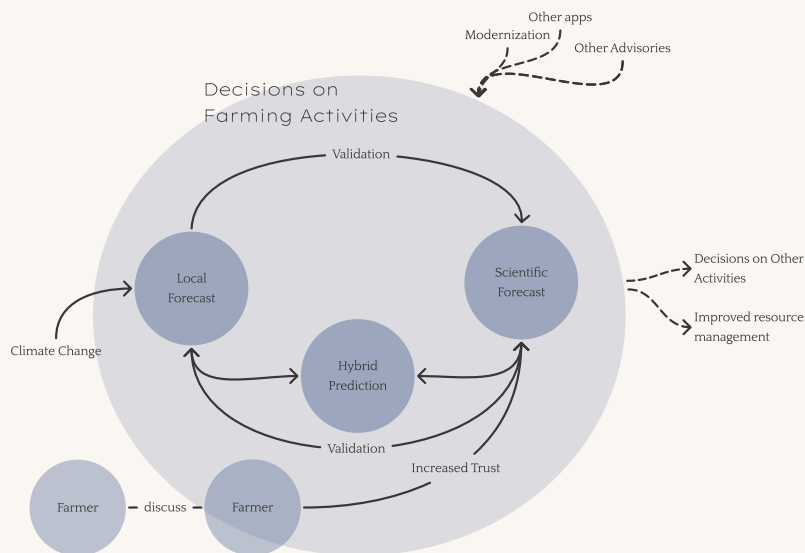
For the third iteration, I was still working with the data from the semi-structured interviews. However, I experimented with a different angle for the system maps. I visualized my understanding of the forecasting process. Specifically, I mapped how different community member derive at their decisions on framing activities (Appendix Figure 15) and how the process may differ after the introduction of the DROP app (Appendix Figure 16). Further, a third map zooms in on the tension between local and scientific forecast information within decision-making processes (Appendix Figure 17).



Appendix Figure 15: Iteration 3 mapping data from semi-structured interviews to illustrate the process from local observations to farming decisions before the DROP app was introduced



Appendix Figure 16: Iteration 3 mapping data from semi-structured interviews to illustrate the process from local observations to farming decisions after the DROP app was introduced with changes in red color



Appendix Figure 17: Iteration 3 mapping data from semi-structured interviews to illustrate tensions between local and scientific forecast information within decision-making processes

Iteration 4

After the implementation of the benchmarking, participatory workshop, and fieldwork, I changed the approach of mapping and concentrated on the visualization of my insights. As a result, I created maps that illustrate a detailed relationship or process and thereby communicate a specific key finding (see Chapter 6 Findings).

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